

3.2. Point groups and crystal classes

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3.2.1. Crystallographic and noncrystallographic point groups

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3.2.1.1. Introduction and definitions

A *point group*¹ is a group of symmetry operations all of which leave at least one point unmoved. Thus, all operations containing translations are excluded. Point groups can be subdivided into crystallographic and noncrystallographic point groups. A *crystallographic* point group is a point group that maps a point lattice onto itself. Consequently, rotations and rotoinversions are restricted to the well known crystallographic cases 1, 2, 3, 4, 6 and $\bar{1}$, $\bar{2} = m$, $\bar{3}$, $\bar{4}$, $\bar{6}$ (cf. Section 1.2.1); matrices for these symmetry operations are listed in Tables 1.2.2.1 and 1.2.2.2. No such restrictions apply to the *noncrystallographic* point groups.

The numbers of the crystallographic point groups are finite: 2 for one dimension, 10 for two dimensions and 32 for three dimensions. The numbers of noncrystallographic point groups for dimensions $n \geq 2$ are infinite. The two- and three-dimensional crystallographic point groups and their crystal systems are summarized in Tables 3.2.1.1 and 3.2.1.2. They are described in detail in Section 3.2.1.2. The two one-dimensional point groups are discussed in Section 2.1.3.16. The noncrystallographic point groups are treated in Section 3.2.1.4.

Crystallographic point groups occur:

- (i) in vector space as symmetries of the external shapes of crystals, *i.e.* of the set of vectors normal to the crystal faces (morphological symmetry); this also includes the symmetries of sets of symmetry-equivalent net planes in crystals (reciprocal-lattice points), fundamental for the theory of X-ray diffraction.
- (ii) in point space as site symmetries of points in lattices or in crystal structures and as symmetries of molecules, atomic groups and coordination polyhedra.

General point groups, *i.e.* crystallographic and noncrystallographic point groups, occur as:

- (iii) symmetries of (rigid) molecules (molecular symmetry);
- (iv) symmetries of physical properties of crystals (*e.g.* tensor symmetries); here noncrystallographic point groups with axes of order infinity are of particular importance, as in the symmetries of circles, spheres or rotation ellipsoids;
- (v) approximate symmetries of the local environment of a point in a crystal structure, *i.e.* as *local* site symmetries. Examples are sphere-like atoms or ions in crystals, as well as icosahedral atomic groups. These noncrystallographic symmetries, however, are only approximate, even for the close neighbourhood of a site.

A (geometric) *crystal class* (point-group type) is the set of all crystals having the same point-group symmetry. The word ‘class’, therefore, denotes a classificatory pigeonhole and should not be

used as synonymous with the point group of a particular crystal. The symbol of a crystal class is that of the common point group. (For geometric and arithmetic crystal classes of space groups, see Sections 1.3.4.2 and 1.3.4.4.)

Of particular importance for the structure determination of crystals are the 11 *centrosymmetric crystallographic point groups*, because they describe the possible symmetries of the diffraction record of a crystal: $\bar{1}$; $2/m$; mmm ; $4/m$; $4/mmm$; $\bar{3}$; $\bar{3}m$; $6/m$; $6/mmm$; $m\bar{3}$; $m\bar{3}m$. This is due to Friedel’s rule, which states that, provided anomalous dispersion is neglected, the diffraction record of any crystal is centrosymmetric, even if the crystal is noncentrosymmetric. The symmetry of the diffraction record determines the *Laue class* of the crystal; this is further explained in Chapter 1.6. For a given crystal, its Laue class is obtained if a symmetry centre is added to its point group, as shown in Table 3.2.2.1.

In two dimensions, six ‘centrosymmetric’ crystallographic point groups and hence six two-dimensional Laue classes exist: 2 ; $2mm$; 4 ; $4mm$; 6 ; $6mm$. These point groups are, for instance, the only possible symmetries of zero-layer X-ray photographs.

Among the centrosymmetric crystallographic point groups in three dimensions, the seven *lattice point groups* (holohedral point groups, *holohedries*) are of special importance because they constitute the possible point symmetries of lattices, *i.e.* the site symmetries of their nodes. In three dimensions, the seven holohedries are: $\bar{1}$; $2/m$; mmm ; $4/mmm$; $\bar{3}m$; $6/mmm$; $m\bar{3}m$. Note that $\bar{3}m$ is the point symmetry of the rhombohedral lattice and $6/mmm$ the point symmetry of the hexagonal lattice; both occur in the hexagonal crystal family (cf. Chapter 2.1). Point groups that are, within a crystal family, subgroups of a holohedry are called merohedries; they are called specifically hemihedries for subgroups of index 2, tetartohedries for index 4 and ogdohedries for index 8.

In two dimensions, four holohedries exist: 2 ; $2mm$; $4mm$; $6mm$. Note that the hexagonal crystal family in two dimensions contains only one lattice type, with point symmetry $6mm$.

Another classification of the crystallographic point groups is that into isomorphism classes. Here all those point groups that have the same kind of group table appear in one class. These isomorphism classes are also known under the name of *abstract point groups*.

There are 18 abstract crystallographic point groups in three dimensions: the point groups in each of the following lines are isomorphous and belong to the same abstract group:

Table 3.2.1.1

The ten two-dimensional crystallographic point groups, arranged according to crystal system

The dashed line separates point groups with different Laue classes within one crystal system.

General symbol	Crystal system			
	Oblique (top) Rectangular (bottom)	Square	Hexagonal	
n	1	2	4	3
nmm	m	$2mm$	$4mm$	$3m$
				$6mm$

¹ For reasons of simplicity, in this chapter the same term ‘point group’ is used for a ‘particular point group’ and a ‘type of point group’. For space groups, this distinction is explained in Section 1.3.4.1. For a different use of the term ‘point group’ see Section 1.3.3.1.

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Table 3.2.1.2

The 32 three-dimensional crystallographic point groups, arranged according to crystal system (*cf.* Chapter 2.1)

Full Hermann–Mauguin (left) and Schoenflies symbols (right). The dashed line separates point groups with different Laue classes within one crystal system. A brief introduction to point-group symbols is provided in Hahn & Klapper (2005).

	Crystal system											
General symbol	Triclinic		Monoclinic (top) Orthorhombic (bottom)		Tetragonal		Trigonal		Hexagonal		Cubic	
n	1	C_1	2	C_2	4	C_4	3	C_3	6	C_6	23	T
$\bar{n}$	$\bar{1}$	C_i	$m \equiv \bar{2}$	C_s	$\bar{4}$	S_4	$\bar{3}$	C_{3i}	$\bar{6} \equiv 3/m$	C_{3h}	–	–
n/m			$2/m$	C_{2h}	$4/m$	C_{4h}	–	–	$6/m$	C_{6h}	$2/m\bar{3}$	T_h
$n22$			222	D_2	422	D_4	32	D_3	622	D_6	432	O
$nm\bar{m}$			$mm2$	C_{2v}	$4mm$	C_{4v}	$3m$	C_{3v}	$6mm$	C_{6v}	–	–
$\bar{n}2m$			–	–	$\bar{4}2m$	D_{2d}	$\bar{3}2/m$	D_{3d}	$\bar{6}2m$	D_{3h}	$\bar{4}3m$	T_d
$n/m\ 2/m\ 2/m$			$2/m\ 2/m\ 2/m$	D_{2h}	$4/m\ 2/m\ 2/m$	D_{4h}	–	–	$6/m\ 2/m\ 2/m$	D_{6h}	$4/m\ \bar{3}\ 2/m$	O_h

Order 1: 1

$$2: \overline{1}, 2, m$$

3: 3

4: $2/m, 222, m\bar{m}2$

$$4: 4.\overline{4}$$
$$6:\overline{3}, 6:\overline{6}$$

6: 32, 3m

8: *mmm*8: $4/m$ Order 8: $422, 4mm, \bar{4}2m$

12: $6/m$

12: $\bar{3}m$, 622, $6mm$, $\bar{6}2m$

12: 23

16: $4/mmm$

24: 6/*mmm*

24: $m\bar{3}$

24: 432, $\bar{4}3m$

48: $m\bar{3}m$.

(b) The two point groups $\bar{4}2m$ and $\bar{6}m2$ are described for two orientations with respect to the crystal axes, as $\bar{4}2m$ and $\bar{4}m2$ and as $\bar{6}m2$ and $\bar{6}2m$.

(c) The five trigonal point groups 3 , $\bar{3}$, 32 , $3m$ and $\bar{3}m$ are treated with two axial systems, 'hexagonal axes' and 'rhombohedral axes'.

(d) The hexagonal-axes description of the three trigonal point groups 32 , $3m$ and $\bar{3}m$ is given for two orientations, as 321 and 312 , as $3m1$ and $31m$, and as $\bar{3}m1$ and $\bar{3}1m$; this applies also to the two-dimensional point group $3m$.

The presentation of the point groups is similar to that of the space groups in Part 2. The *headline* contains the short Hermann–Mauguin and the Schoenflies symbols. The full Hermann–Mauguin symbol, if different, is given below the short symbol. No Schoenflies symbols exist for two-dimensional groups. For an explanation of the symbols see Sections 1.4.1 and 2.1.3.4, and Chapter 3.3.

Next to the headline, a pair of *stereographic projections* is given. The diagram on the left displays a general crystal or point form, that on the right shows the ‘framework of symmetry elements’. Except as noted below, the c axis is always normal to the plane of the figure, the a axis points down the page and the b axis runs horizontally from left to right. For the five trigonal point groups, the c axis is normal to the page only for the description with ‘hexagonal axes’; if described with ‘rhombohedral axes’, the direction [111] is normal and the positive a axis slopes towards the observer. The conventional coordinate systems used for the various crystal systems are listed in Table 2.1.1.1 and illustrated in Figs. 2.1.3.1 to 2.1.3.10.

In the *right-hand projection*, the graphical symbols of the symmetry elements are the same as those used in the space-group diagrams; they are listed in Chapter 2.1. Note that the symbol of a symmetry centre, a small circle, is also used for a face pole in the left-hand diagram. Mirror planes are indicated by heavy solid lines or circles; thin lines are used for the projection circle, for symmetry axes in the plane and for some special zones in the cubic system.

In the *left-hand projection*, the projection circle and the coordinate axes are indicated by thin solid lines, as are again some special zones in the cubic system. The dots and circles in this projection can be interpreted in two ways.

(i) As *general face poles*, where they represent general crystal faces which form a polyhedron, the ‘general crystal form’ (face form) $\{hkl\}$ of the point group (see below). In two dimensions, edges, edge poles, edge forms and polygons take the place of faces, face poles, crystal forms (face forms) and polyhedra in three dimensions.

In two dimensions, the ten crystallographic point groups form nine abstract groups; the groups 2 and m are isomorphous and belong to the same abstract group, the remaining eight point groups correspond to one abstract group each.

3.2.1.2. Crystallographic point groups

3.2.1.2.1. Description of point groups

In crystallography, point groups usually are described

- (i) by means of their Hermann–Mauguin or Schoenflies symbols;
- (ii) by means of their stereographic projections;
- (iii) by means of the matrix representations of their symmetry operations, frequently listed in the form of Miller indices (hkl) of the equivalent general crystal faces;
- (iv) by means of drawings of actual crystals, natural or synthetic.

Descriptions (i) through (iii) are given in this section, whereas for crystal drawings and actual photographs reference is made to textbooks of crystallography and mineralogy [Buerger (1956, ch. 10) and Phillips (1971, chs. 3, 4 and 6) are particularly rich in pictures of crystal morphologies]; this also applies to the construction and the properties of the stereographic projection.

In Tables 3.2.3.1 and 3.2.3.2, the two- and three-dimensional crystallographic point groups are listed and described. The tables are arranged according to crystal systems and Laue classes. Within each crystal system and Laue class, the sequence of the point groups corresponds to that in the space-group tables of this volume: pure rotation groups are followed by groups containing reflections, rotoinversions and inversions. The holohedral point group is always given last.

In Tables 3.2.3.1 and 3.2.3.2, some point groups are described in *two or three versions*, in order to bring out the relations to the corresponding space groups (*cf.* Section 2.1.3.2):

(a) The three monoclinic point groups 2, m and $2/m$ are given with two settings, one with ‘unique axis b ’ and one with ‘unique axis c ’.

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Face poles marked as dots lie above the projection plane and represent faces which intersect the positive c axis² (positive Miller index l), those marked as circles lie below the projection plane (negative Miller index l). In two dimensions, edge poles always lie on the pole circle.

- (ii) As *general points* (centres of atoms) that span a polyhedron or polygon, the ‘general crystallographic point form’ x, y, z . This interpretation is of interest in the study of coordination polyhedra, atomic groups and molecular shapes. The polyhedron or polygon of a point form is dual to the polyhedron of the corresponding crystal form.³

The general, special and limiting *crystal forms* and *point forms* constitute the main part of the table for each point group. The theoretical background is given below in Section 3.2.1.2.2, *Crystal and point forms* and an explanation of the listed data is to be found in Section 3.2.1.2.3, *Description of crystal and point forms*.

The last entry for each point group contains the *Symmetry of special projections*, i.e. the plane point group that is obtained if the three-dimensional point group is projected along a symmetry direction. The special projection directions are the same as for the space groups; they are listed in Section 2.1.3.14. The relations between the axes of the three-dimensional point group and those of its two-dimensional projections can easily be derived with the help of the stereographic projection. No projection symmetries are listed for the two-dimensional point groups.

Note that the symmetry of a projection along a certain direction may be higher than the symmetry of the crystal face normal to that direction. For example, in point group $\bar{1}$ all faces have face symmetry 1, whereas projections along any direction have symmetry 2; in point group 422, the faces (001) and (00 $\bar{1}$) have face symmetry 4, whereas the projection along [001] has symmetry 4mm.

3.2.1.2.2. Crystal and point forms

For a point group $\mathcal{P}$ a *crystal form* is a set of all symmetry-equivalent faces; a *point form* is a set of all symmetry-equivalent points. Crystal and point forms in point groups correspond to ‘crystallographic orbits’ in space groups; cf. Sections 1.1.7 and 1.4.4.1.

Two kinds of crystal and point forms with respect to $\mathcal{P}$ can be distinguished. They are defined as follows:

- (i) *General form*: A face is called ‘general’ if only the identity operation transforms the face onto itself. Each complete set of symmetry-equivalent ‘general faces’ is a *general crystal form*. The *multiplicity* of a general form, i.e. the number of its faces, is the order of $\mathcal{P}$. In the stereographic projection, the poles of general faces do *not* lie on symmetry elements of $\mathcal{P}$.
A *point* is called ‘general’ if its *site symmetry*, i.e. the group of symmetry operations that transforms this point onto itself, is 1. A *general point form* is a complete set of symmetry-equivalent ‘general points’.
- (ii) *Special form*: A face is called ‘special’ if it is transformed onto itself by at least one symmetry operation of $\mathcal{P}$, in addition to the identity. Each complete set of symmetry-equivalent ‘special faces’ is called a *special crystal form*. The *face symmetry* of a special face is the group of symmetry operations that transforms this face onto itself; it is a subgroup of $\mathcal{P}$. The multiplicity of a special crystal form is the multiplicity of the general form divided by the order of the face-

symmetry group. In the stereographic projection, the poles of special faces lie on symmetry elements of $\mathcal{P}$. The Miller indices of a special crystal form obey restrictions like $\{hk0\}$, $\{hhl\}$, $\{100\}$.

A *point* is called ‘special’ if its *site symmetry* is higher than 1. A special point form is a complete set of symmetry-equivalent ‘special points’. The multiplicity of a special point form is the multiplicity of the general form divided by the order of the site-symmetry group. It is thus the same as that of the corresponding special crystal form. The coordinates of the points of a special point form obey restrictions, like $x, y, 0$; x, x, z ; $x, 0, 0$.

In two dimensions, point groups 1, 2, 3, 4 and 6 and, in three dimensions, point groups 1 and $\bar{1}$ have no special crystal forms.

General and special crystal and point forms can be represented by their sets of equivalent Miller indices $\{hkl\}$ and point coordinates x, y, z . Each set of these ‘triplets’ stands for infinitely many crystal forms or point forms which are obtained by independent variation of the values and signs of the Miller indices h, k, l or the point coordinates x, y, z .

It should be noted that for crystal forms, owing to the well known ‘law of rational indices’, the indices h, k, l must be integers; no such restrictions apply to the coordinates x, y, z , which can be rational or irrational numbers.

Example

In point group 4, the general crystal form $\{hkl\}$ stands for the set of all possible tetragonal pyramids, pointing either upwards or downwards, depending on the sign of l ; similarly, the general point form x, y, z includes all possible squares, lying either above or below the origin, depending on the sign of z . For the limiting cases $l = 0$ or $z = 0$, see below.

In order to survey the infinite number of possible forms of a point group, they are classified into *Wyckoff positions of crystal and point forms*, for short *Wyckoff positions*. This name has been chosen in analogy to the Wyckoff positions of space groups; cf. Sections 1.4.4.2 and 2.1.3.11. In point groups, the term ‘position’ can be visualized as the position of the face poles and points in the stereographic projection. Each ‘Wyckoff position’ is labelled by a *Wyckoff letter*.

Definition

A ‘Wyckoff position of crystal and point forms’ consists of all those crystal forms (point forms) of a point group $\mathcal{P}$ for which the face poles (points) are positioned on the same set of conjugate symmetry elements of $\mathcal{P}$; i.e. for each face (point) of one form there is one face (point) of every other form of the same ‘Wyckoff position’ that has exactly the same face (site) symmetry.

Each point group contains one ‘general Wyckoff position’ comprising all *general* crystal and point forms. In addition, up to two ‘special Wyckoff positions’ may occur in two dimensions and up to six in three dimensions. They are characterized by the different sets of conjugate face and site symmetries and correspond to the seven positions of a pole in the interior, on the three edges, and at the three vertices of the so-called ‘characteristic triangle’ of the stereographic projection.

Examples

- (1) All tetragonal pyramids $\{hkl\}$ and tetragonal prisms $\{hk0\}$ in point group 4 have face symmetry 1 and belong to the same general ‘Wyckoff position’ 4b, with Wyckoff letter b.

² This does not apply to ‘rhombohedral axes’: here the positive directions of all three axes slope upwards from the plane of the paper: cf. Fig. 2.1.3.9.

³ Dual polyhedra have the same number of edges, but the numbers of faces and vertices are interchanged; cf. textbooks of geometry.

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- (2) All tetragonal pyramids *and* tetragonal prisms in point group $4mm$ belong to two special 'Wyckoff positions', depending on the orientation of their face-symmetry groups m with respect to the crystal axes: For the 'oriented face symmetry' $.m.$, the forms $\{h0l\}$ and $\{100\}$ belong to Wyckoff position $4c$; for the oriented face symmetry $.m$, the forms $\{hhl\}$ and $\{110\}$ belong to Wyckoff position $4b$. The face symmetries $.m.$ and $.m$ are not conjugate in point group $4mm$. For the analogous 'oriented site symmetries' in space groups, see Section 2.1.3.12.

It is instructive to subdivide the crystal forms (point forms) of one Wyckoff position further, into *characteristic* and *noncharacteristic* forms. For this, one has to consider two symmetries that are connected with each crystal (point) form:

- (i) the point group $\mathcal{P}$ by which a form is generated (*generating point group*), i.e. the point group in which it occurs;
- (ii) the full symmetry (inherent symmetry) of a form (considered as a polyhedron by itself), here called eigensymmetry $\mathcal{E}$. The *eigensymmetry point group* $\mathcal{E}$ is either the generating point group itself or a supergroup of it.

Examples

- (1) Each tetragonal pyramid $\{hkl\}$ ($l \neq 0$) of Wyckoff position $4b$ in point group 4 has generating symmetry 4 and eigensymmetry $4mm$; each tetragonal prism $\{hk0\}$ of the same Wyckoff position has generating symmetry 4 again, but eigensymmetry $4/mmm$.
- (2) A cube $\{100\}$ may have generating symmetry $23, m\bar{3}, 432, \bar{4}3m$ or $m\bar{3}m$, but its eigensymmetry is always $m\bar{3}m$.

The eigensymmetries and the generating symmetries of the 47 crystal forms (point forms) are listed in Table 3.2.1.3. With the help of this table, one can find the various point groups in which a given crystal form (point form) occurs, as well as the face (site) symmetries that it exhibits in these point groups; for experimental methods see Sections 3.2.2.2 and 3.2.2.3. Diagrams of the 47 crystal forms are presented in Fig. 3.2.1.1.

With the help of the two groups $\mathcal{P}$ and $\mathcal{E}$, each crystal or point form occurring in a particular point group can be assigned to one of the following two categories:

- (i) *characteristic* form, if eigensymmetry $\mathcal{E}$ and generating symmetry $\mathcal{P}$ are the same;
- (ii) *noncharacteristic* form, if $\mathcal{E}$ is a proper supergroup of $\mathcal{P}$.

The importance of this classification will be apparent from the following examples.

Examples

- (1) A pedion and a pinacoid are noncharacteristic forms in all crystallographic point groups in which they occur;
- (2) all other crystal or point forms occur as characteristic forms in their eigensymmetry group $\mathcal{E}$;
- (3) a tetragonal pyramid is noncharacteristic in point group 4 and characteristic in $4mm$;
- (4) a hexagonal prism can occur in nine point groups (12 Wyckoff positions) as a noncharacteristic form; in $6/mmm$, it occurs in two Wyckoff positions as a characteristic form.

The general forms of the 13 point groups with no, or only one, symmetry direction ('monoaxial groups') $1, 2, 3, 4, 6, \bar{1}, m, \bar{3}, \bar{4}, \bar{6} = 3/m, 2/m, 4/m, 6/m$ are always noncharacteristic, i.e. their eigensymmetries are enhanced in comparison with the generating point groups. The general positions of the other 19 point groups always contain characteristic crystal forms that may be used to

determine the point group of a crystal uniquely (cf. Section 3.2.2).⁴

So far, we have considered the occurrence of one crystal or point form in different point groups and different Wyckoff positions. We now turn to the occurrence of different kinds of crystal or point forms in one and the same Wyckoff position of a particular point group.

In a Wyckoff position, crystal forms (point forms) of different eigensymmetries may occur; the crystal forms (point forms) with the lowest eigensymmetry (which is always well defined) are called *basic forms* (German: *Grundformen*) of that Wyckoff position. The crystal and point forms of higher eigensymmetry are called *limiting forms* (German: *Grenzformen*) (cf. Table 3.2.1.3). These forms are always noncharacteristic.

Limiting forms⁵ occur for certain restricted values of the Miller indices or point coordinates. They always have the same multiplicity and oriented face (site) symmetry as the corresponding basic forms because they belong to the same Wyckoff position. The enhanced eigensymmetry of a limiting form may or may not be accompanied by a change in the topology⁶ of its polyhedra, compared with that of a basic form. In every case, however, the name of a limiting form is different from that of a basic form.

The face poles (or points) of a limiting form lie on symmetry elements of a supergroup of the point group that are not symmetry elements of the point group itself. There may be several such supergroups.

Examples

- (1) In point group 4 , the (noncharacteristic) crystal forms $\{hkl\}$ ($l \neq 0$) (tetragonal pyramids) of eigensymmetry $4mm$ are basic forms of the general Wyckoff position $4b$, whereas the forms $\{hk0\}$ (tetragonal prisms) of higher eigensymmetry $4/mmm$ are 'limiting general forms'. The face poles of forms $\{hk0\}$ lie on the horizontal mirror plane of the supergroup $4/m$.
- (2) In point group $4mm$, the (characteristic) special crystal forms $\{h0l\}$ with eigensymmetry $4mm$ are 'basic forms' of the special Wyckoff position $4c$, whereas $\{100\}$ with eigensymmetry $4/mmm$ is a 'limiting special form'. The face poles of $\{100\}$ are located on the intersections of the vertical mirror planes of the point group $4mm$ with the horizontal mirror plane of the supergroup $4/mmm$, i.e. on twofold axes of $4/mmm$.

Whereas basic and limiting forms belonging to one 'Wyckoff position' are always clearly distinguished, closer inspection shows that a Wyckoff position may contain different 'types' of limiting forms. We need, therefore, a further criterion to classify the limiting forms of one Wyckoff position into types: A *type of limiting form of a Wyckoff position* consists of all those limiting forms for which the face poles (points) are located on the same set of additional conjugate symmetry elements of the holohedral point group (for the trigonal point groups, the hexagonal holo-

⁴ For a survey of these relations, as well as of the 'limiting forms', it is helpful to consider the (seven) *normalizers* of the crystallographic point groups in the group of all rotations and reflections (orthogonal group, sphere group); normalizers of the crystallographic and noncrystallographic point groups are listed in Tables 3.5.4.1 and 3.5.4.2.

⁵ The treatment of 'limiting forms' in the literature is quite ambiguous. In some textbooks, limiting forms are omitted or treated as special forms in their own right; other authors define only limiting *general* forms and consider limiting *special* forms as if they were new special forms. For additional reading, see P. Niggli (1941, pp. 80–98).

⁶ The topology of a polyhedron is determined by the numbers of its vertices, edges and faces, by the number of vertices of each face and by the number of faces meeting in each vertex.

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Table 3.2.1.3

The 47 crystallographic face and point forms, their names, eigensymmetries, and their occurrence in the crystallographic point groups (generating point groups)

The oriented face (site) symmetries of the forms are given in parentheses after the Hermann–Mauguin symbol (column 6); a symbol such as $mm2(.m., .m.)$ indicates that the form occurs in point group $mm2$ twice, with face (site) symmetries $.m.$ and $.m..$ Basic (general and special) forms are printed in bold face, limiting (general and special) forms in normal type. The various settings of point groups 32 , $3m$, $\bar{3}m$, $\bar{4}2m$ and $\bar{6}m2$ are connected by braces. The 47 crystal forms are shown in Fig. 3.2.1.1. (Note that the numbering of the forms in column 1 does not correspond to the numbering used in Fig. 3.2.1.1.)

No.	Crystal form	Point form	Number of faces or points	Eigensymmetry	Generating point groups with oriented face (site) symmetries between parentheses
1	Pedion or monohedron	Single point	1	∞m	1(1); 2(2); $m(m)$; 3(3); 4(4); 6(6); $mm2(mm2)$; $4mm(4mm)$; $3m(3m)$; $6mm(6mm)$
2	Pinacoid or parallelohedron	Line segment through origin	2	$\frac{\infty}{m}m$	$\bar{1}(1)$; 2(1); $m(1)$; $\frac{2}{m}(2, m)$; $222(2.., .2., .2)$; $mm2(.m., .m.)$; $mmm(2mm, m2m, mm2)$; $\bar{4}(2..)$; $\frac{4}{m}(4..)$; $422(4..)$; $\left\{ \begin{array}{l} \bar{4}2m(2.mm) \\ \bar{4}m2(2mm.) \end{array} \right\}$; $\frac{4}{m}mm(4mm)$; $\bar{3}(3)$; $\left\{ \begin{array}{l} 321(3..) \\ 312(3..) \\ 32(3.) \end{array} \right\}$; $\left\{ \begin{array}{l} \bar{3}m1(3m.) \\ \bar{3}1m(3.m) \\ \bar{3}m(3m) \end{array} \right\}$; $\bar{6}(3..)$; $\frac{6}{m}(6..)$; $622(6..)$; $\left\{ \begin{array}{l} \bar{6}m2(3m.) \\ \bar{6}2m(3.m) \end{array} \right\}$; $\frac{6}{m}mm(6mm)$
3	Sphenoid, dome, or dihedron	Line segment	2	$mm2$	2(1); $m(1)$; $mm2(.m., .m.)$
4	Rhombic disphenoid or rhombic tetrahedron	Rhombic tetrahedron	4	222	222(1)
5	Rhombic pyramid	Rectangle	4	$mm2$	$mm2(1)$
6	Rhombic prism	Rectangle through origin	4	mmm	$2/m(1)$; $222(1)\dagger$; $mm2(1)$; $mmm(m.., .m., .m)$
7	Rhombic dipyrmaid	Rectangular prism	8	mmm	$mmm(1)$
8	Tetragonal pyramid	Square	4	$4mm$	4(1); $4mm(.m., .m.)$
9	Tetragonal disphenoid or tetragonal tetrahedron	Tetragonal tetrahedron	4	$\bar{4}2m$	$\bar{4}(1)$; $\left\{ \begin{array}{l} \bar{4}2m(.m) \\ \bar{4}m2(.m.) \end{array} \right\}$
10	Tetragonal prism	Square through origin	4	$\frac{4}{m}mm$	4(1); $\bar{4}(1)$; $\frac{4}{m}(m..)$; $422(.2, .2)$; $4mm(.m., .m.)$; $\dagger \left\{ \begin{array}{l} \bar{4}2m(.2) \text{ and } \bar{4}2m(.m) \\ \bar{4}m2(.2) \text{ and } \bar{4}m2(.m.) \end{array} \right\}$; $\frac{4}{m}mm(m.m2, m2m.)$
11	Tetragonal trapezohedron	Twisted tetragonal antiprism	8	422	422(1)
12	Ditetragonal pyramid	Truncated square	8	$4mm$	4mm(1)
13	Tetragonal scalenohedron	Tetragonal tetrahedron cut off by pinacoid	8	$\bar{4}2m$	$\left\{ \begin{array}{l} \bar{4}2m(1) \\ \bar{4}m2(1) \end{array} \right\}$
14	Tetragonal dipyrmaid	Tetragonal prism	8	$\frac{4}{m}mm$	$\frac{4}{m}(1)$; $422(1)\dagger$; $\left\{ \begin{array}{l} \bar{4}2m(1) \\ \bar{4}m2(1) \end{array} \right\}$; $\frac{4}{m}mm(.m., .m.)$
15	Ditetragonal prism	Truncated square through origin	8	$\frac{4}{m}mm$	$422(1)$; $4mm(1)$; $\left\{ \begin{array}{l} \bar{4}2m(1) \\ \bar{4}m2(1) \end{array} \right\}$; $\frac{4}{m}mm(m..)$
16	Ditetragonal dipyrmaid	Edge-truncated tetragonal prism	16	$\frac{4}{m}mm$	$\frac{4}{m}mm(1)$
17	Trigonal pyramid	Trigon	3	$3m$	3(1); $\left\{ \begin{array}{l} 3m1(.m.) \\ 31m(.m) \\ 3m(.m) \end{array} \right\}$

3.2. POINT GROUPS AND CRYSTAL CLASSES

Table 3.2.1.3 (continued)

No.	Crystal form	Point form	Number of faces or points	Eigensymmetry	Generating point groups with oriented face (site) symmetries between parentheses
18	Trigonal prism	Trigon through origin	3	$\bar{6}2m$	$3(1); \left\{ \begin{matrix} 321(.2.) \\ 312(..2) \\ 32 (.2) \end{matrix} \right\}; \left\{ \begin{matrix} 3m1(.m.) \\ 31m(..m) \\ 3m (.m) \end{matrix} \right\};$ $\bar{6}(m..); \left\{ \begin{matrix} \bar{6}m2(mm2) \\ \bar{6}2m(m2m) \end{matrix} \right\}$
19	Trigonal trapezohedron	Twisted trigonal antiprism	6	32	$\left\{ \begin{matrix} 321(1) \\ 312(1) \\ 32 (1) \end{matrix} \right\}$
20	Ditrigonal pyramid	Truncated trigon	6	3m	$\left\{ \begin{matrix} 3m1(1) \\ 31m(1) \\ 3m (1) \end{matrix} \right\}$
21	Rhombohedral	Trigonal antiprism	6	$\bar{3}m$	$\bar{3}(1); \left\{ \begin{matrix} 321(1) \\ 312(1); \\ 32 (1) \end{matrix} \right\}; \left\{ \begin{matrix} \bar{3}m1(.m.) \\ \bar{3}1m(..m) \\ \bar{3}m (.m) \end{matrix} \right\}$
22	Ditrigonal prism	Truncated trigon through origin	6	$\bar{6}2m$	$\left\{ \begin{matrix} 321(1) \\ 312(1); \\ 32 (1) \end{matrix} \right\}; \left\{ \begin{matrix} 3m1(1) \\ 31m(1); \\ 3m (1) \end{matrix} \right\};$ $\left\{ \begin{matrix} \bar{6}m2(m..) \\ \bar{6}2m(m..) \end{matrix} \right\}$
23	Hexagonal pyramid	Hexagon	6	6mm	$\left\{ \begin{matrix} 3m1(1) \\ 31m(1); \\ 3m (1) \end{matrix} \right\}; 6(1); 6mm(..m., .m.)$
24	Trigonal dipyrmaid	Trigonal prism	6	$\bar{6}2m$	$\left\{ \begin{matrix} 321(1) \\ 312(1); \\ 32 (1) \end{matrix} \right\}; \bar{6}(1); \left\{ \begin{matrix} \bar{6}m2(.m.) \\ \bar{6}2m(..m) \end{matrix} \right\}$
25	Hexagonal prism	Hexagon through origin	6	$\frac{6}{m}mm$	$\bar{3}(1); \left\{ \begin{matrix} 321(1) \\ 312(1); \\ 32 (1) \end{matrix} \right\}; \left\{ \begin{matrix} 3m1(1) \\ 31m(1); \\ 3m (1) \end{matrix} \right\};$ $\dagger \left\{ \begin{matrix} \bar{3}m1(.2.) \text{ and } \bar{3}m1(.m.) \\ \bar{3}1m(..2) \text{ and } \bar{3}1m(..m); \\ \bar{3}m(.2) \text{ and } \bar{3}m(.m) \end{matrix} \right\};$ $6(1); \frac{6}{m}(m..); 622(.2., .2.);$ $6mm(..m., .m.); \left\{ \begin{matrix} \bar{6}m2(m..) \\ \bar{6}2m(m..) \end{matrix} \right\};$ $\frac{6}{m}mm(m2m, mm2)$
26	Ditrigonal scalenohedron or hexagonal scalenohedron	Trigonal antiprism sliced off by pinacoid	12	$\bar{3}m$	$\left\{ \begin{matrix} \bar{3}m1(1) \\ \bar{3}1m(1) \\ \bar{3}m (1) \end{matrix} \right\}$
27	Hexagonal trapezohedron	Twisted hexagonal antiprism	12	622	622(1)
28	Dihexagonal pyramid	Truncated hexagon	12	6mm	6mm(1)
29	Ditrigonal dipyrmaid	Edge-truncated trigonal prism	12	$\bar{6}2m$	$\left\{ \begin{matrix} \bar{6}m2(1) \\ \bar{6}2m(1) \end{matrix} \right\}$
30	Dihexagonal prism	Truncated hexagon	12	$\frac{6}{m}mm$	$\left\{ \begin{matrix} \bar{3}m1(1) \\ \bar{3}1m(1); \\ \bar{3}m (1) \end{matrix} \right\}; 622(1); 6mm(1);$ $\frac{6}{m}mm(m..)$
31	Hexagonal dipyrmaid	Hexagonal prism	12	$\frac{6}{m}mm$	$\left\{ \begin{matrix} \bar{3}m1(1) \\ \bar{3}1m(1); \\ \bar{3}m (1) \end{matrix} \right\}; \frac{6}{m}(1); 622(1)\dagger;$ $\left\{ \begin{matrix} \bar{6}m2(1); \\ \bar{6}2m(1) \end{matrix} \right\}; \frac{6}{m}mm(..m., .m.)$

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Table 3.2.1.3 (continued)

No.	Crystal form	Point form	Number of faces or points	Eigensymmetry	Generating point groups with oriented face (site) symmetries between parentheses
32	Dihexagonal dipyramid	Edge-truncated hexagonal prism	24	$\frac{6}{m}mm$	$\frac{6}{m}mm(1)$
33	Tetrahedron	Tetrahedron	4	$\bar{4}3m$	$23(.3.); \bar{4}3m(.3m)$
34	Cube or hexahedron	Octahedron	6	$m\bar{3}m$	$23(2..); m\bar{3}(2mm..); 432(4..); \bar{4}3m(2.mm); m\bar{3}m(4m.m)$
35	Octahedron	Cube	8	$m\bar{3}m$	$m\bar{3}(.3.); 432(.3.); m\bar{3}m(.3m)$
36	Pentagon-tritetrahedron or tetartoid or tetrahedral pentagon-dodecahedron	Snub tetrahedron (= pentagon-tritetrahedron + two tetrahedra)	12	23	$23(1)$
37	Pentagon-dodecahedron or dihexahedron or pyritohedron	Irregular icosahedron (= pentagon-dodecahedron + octahedron)	12	$m\bar{3}$	$23(1); m\bar{3}(m..)$
38	Tetragon-tritetrahedron or deltohedron or deltoid-dodecahedron	Cube and two tetrahedra	12	$\bar{4}3m$	$23(1); \bar{4}3m(..m)$
39	Trigon-tritetrahedron or tristetrahedron	Tetrahedron truncated by tetrahedron	12	$\bar{4}3m$	$23(1); \bar{4}3m(..m)$
40	Rhomb-dodecahedron	Cuboctahedron	12	$m\bar{3}m$	$23(1); m\bar{3}(m..); 432(..2); \bar{4}3m(..m); m\bar{3}m(m.m2)$
41	Didodecahedron or diploid or dyakisdodecahedron	Cube & octahedron & pentagon-dodecahedron	24	$m\bar{3}$	$m\bar{3}(1)$
42	Trigon-trioctahedron or trisoctahedron	Cube truncated by octahedron	24	$m\bar{3}m$	$m\bar{3}(1); 432(1); m\bar{3}m(..m)$
43	Tetragon-trioctahedron or trapezohedron or deltoid-icositetrahedron	Cube & octahedron & rhomb-dodecahedron	24	$m\bar{3}m$	$m\bar{3}(1); 432(1); m\bar{3}m(..m)$
44	Pentagon-trioctahedron or gyroid	Cube + octahedron + pentagon-trioctahedron	24	432	$432(1)$
45	Hexatetrahedron or hexakistetrahedron	Cube truncated by two tetrahedra	24	$\bar{4}3m$	$\bar{4}3m(1)$
46	Tetrahexahedron or tetrakisohedron	Octahedron truncated by cube	24	$m\bar{3}m$	$432(1); \bar{4}3m(1); m\bar{3}m(m..)$
47	Hexaoctahedron or hexakisohedron	Cube truncated by octahedron and by rhomb-dodecahedron	48	$m\bar{3}m$	$m\bar{3}m(1)$

† These limiting forms occur in three or two non-equivalent orientations (different types of limiting forms); cf. Table 3.2.3.2. ‡ In point groups $\bar{4}2m$ and $\bar{3}m$, the tetragonal prism and the hexagonal prism occur twice, as a 'basic special form' and as a 'limiting special form'. In these cases, the point groups are listed twice, as ' $\bar{4}2m(.2.)$ ' and ' $\bar{4}2m(..m)$ ' and as ' $\bar{3}m(1,2.)$ ' and ' $\bar{3}m(1,m..)$ '.

hedry $6/mmm$ has to be taken). Different types of limiting forms may have the same eigensymmetry and the same topology, as shown by the examples below. The occurrence of two topologically different polyhedra as two 'realizations' of one type of limiting form in point groups 23, $m\bar{3}$ and 432 is explained below in Section 3.2.1.2.4, *Notes on crystal and point forms*, item (viii).

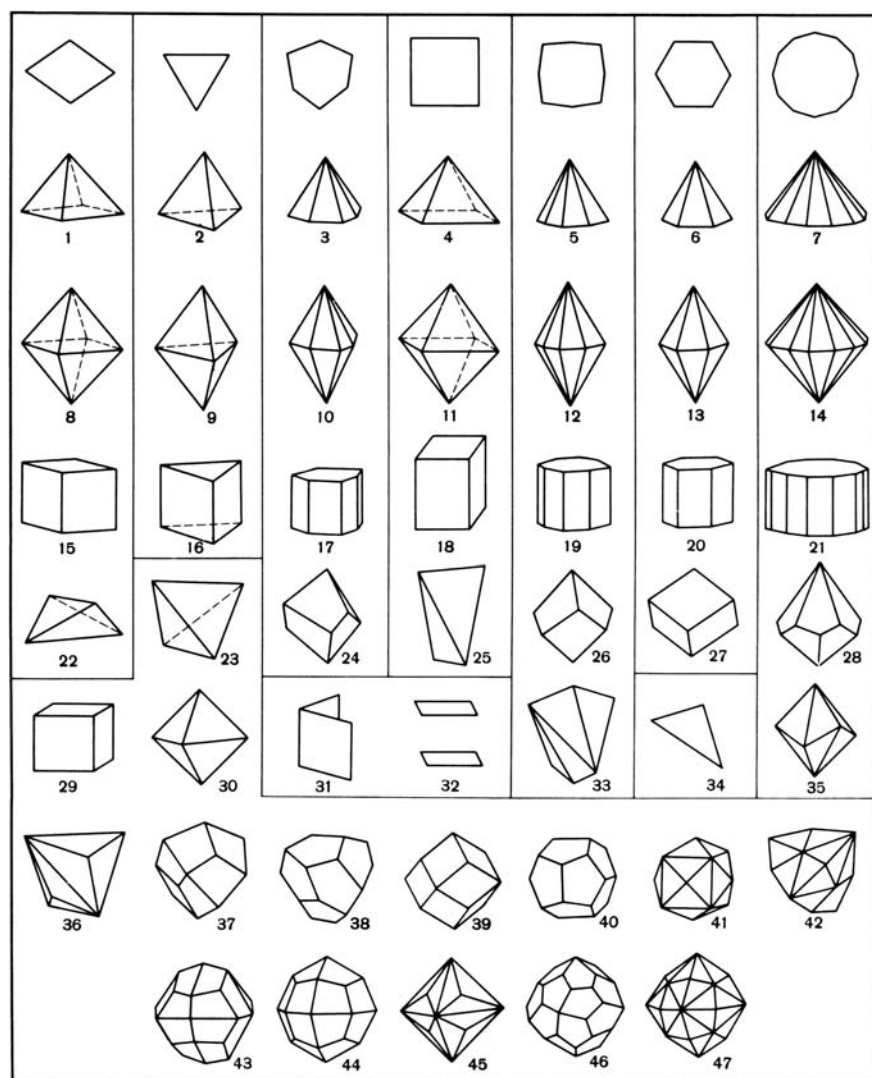
Examples

- (1) In point group 32, the limiting general crystal forms are of four types:
 - (i) ditrigonal prisms, eigensymmetry $\bar{6}2m$ (face poles on horizontal mirror plane of holohedry $6/mmm$);
 - (ii) trigonal dipyramids, eigensymmetry $\bar{6}2m$ (face poles on one kind of vertical mirror plane);

- (iii) rhombohedra, eigensymmetry $\bar{3}m$ (face poles on second kind of vertical mirror plane);
- (iv) hexagonal prisms, eigensymmetry $6/mmm$ (face poles on horizontal twofold axes).

Types (i) and (ii) have the same eigensymmetry but different topologies; types (i) and (iv) have the same topology but different eigensymmetries; type (iii) differs from the other three types in both eigensymmetry and topology.

- (2) In point group 222, the face poles of the three types of general limiting forms, rhombic prisms, are located on the three (non-equivalent) symmetry planes of the holohedry mmm . Geometrically, the axes of the prisms are directed along the three non-equivalent orthorhombic symmetry directions. The three types of limiting forms have the same

**Figure 3.2.1.1**

The 47 crystal forms that crystals may take (from Shubnikov & Koptsik, 1974, p. 74): (1)–(7) Pyramids: orthorhombic, trigonal, ditrigonal, tetragonal, ditetragonal, hexagonal, dihexagonal; (8)–(14) bipyramids of the same types; (15)–(21) prisms of the same types; (22), (23), (25) tetrahedra: orthorhombic, regular and tetragonal; (24), (26), (28) trapezohedra: trigonal, tetragonal, hexagonal; (27) rhombohedron; (34) scalene triangle; (33), (35) scalenohedra: tetragonal and ditrigonal; (31) dihedron (axial or non-axial); (32) pinacoid; (23), (29), (30), (36)–(47) simple forms of the cubic system: (23) tetrahedron; (29) hexahedron (cube); (30) octahedron; (36) trigonal tristetrahedron; (37) tetragonal tristetrahedron; (38) pentagonal tristetrahedron; (39) rhombic dodecahedron; (40) pentagonal dodecahedron; (41) tetrahexahedron; (42) hexatetrahedron; (43) didodecahedron; (44) tetragonal trisoctahedron; (45) trigonal trisoctahedron; (46) pentagonal trisoctahedron; (47) hexoctahedron. The central cross sections of all the figures above the stepped line dividing the table are the regular polygons indicated in the top row. Note that the numbers in this figure do not correspond to the numbers used in column 1 of Table 3.2.1.3.

eigensymmetry and the same topology but different orientations.

Similar cases occur in point groups 422 and 622 (*cf.* the first footnote to Table 3.2.1.3).

Not considered in this volume are limiting forms of another kind, namely those that require either special metrical conditions for the axial ratios or irrational indices or coordinates (which always can be closely approximated by rational values). For instance, a rhombic disphenoid can, for special axial ratios, appear as a tetragonal or even as a cubic tetrahedron; similarly, a rhombohedron can degenerate to a cube. For special irrational

indices, a ditetragonal prism changes to a (noncrystallographic) octagonal prism, a dihexagonal pyramid to a dodecagonal pyramid or a crystallographic pentagon-dodecahedron to a regular pentagon-dodecahedron. These kinds of limiting forms are listed by A. Niggli (1963).

In conclusion, each general or special Wyckoff position always contains one set of basic crystal (point) forms. In addition, it may contain one or more sets of limiting forms of different types. As a rule,⁷ each type comprises polyhedra of the same eigensymmetry and topology and, hence, of the same name, for instance ‘ditetragonal pyramid’. The name of the *basic general* forms is often used to designate the corresponding crystal class, for instance ‘ditetragonal-pyramidal class’; some of these names are listed in Table 3.2.1.4.

3.2.1.2.3. Description of crystal and point forms

The main part of each point-group table in Section 3.2.3 describes the general and special *crystal and point forms* of that point group, in a manner analogous to the *positions* in a space group. The general Wyckoff position is given at the top, followed downwards by the special Wyckoff positions with decreasing multiplicity. Within each Wyckoff position, the first block of column 6 refers to the basic forms, the subsequent blocks list the various types of limiting form, if any.

The columns, from left to right, contain the following data (further details are to be found below in Section 3.2.1.2.4, *Notes on crystal and point forms*):

Column 1: *Multiplicity* of the ‘Wyckoff position’, *i.e.* the number of equivalent faces and points of a crystal or point form.

Column 2: *Wyckoff letter*. Each general or special ‘Wyckoff position’ is designated by a ‘Wyckoff letter’, analogous to the Wyckoff letter of a position in a space group (*cf.* Sections 1.4.4.2 and 2.1.3.11).

Column 3: *Face symmetry* or *site symmetry*, given in the form of an ‘oriented point-group symbol’, analogous to the oriented site-symmetry symbols of space groups (*cf.* Sections 1.4.4.2 and 2.1.3.12). The face symmetry is also the symmetry of etch pits, striations and other

face markings. For the two-dimensional point groups, this column contains the *edge symmetry*, which can be either 1 or *m*.

Column 4: *Coordinates* x, y, z of the symmetry-equivalent points of a point form.

Column 5: *Name of crystal form*. If more than one name is in common use, several are listed. The names of the limiting forms are also given. The crystal forms, their names, eigensymmetries and occurrence in the point groups are summarized in Table 3.2.1.3, which may be useful for determinative purposes, as explained in Sections 3.2.2.2 and 3.2.2.3. There are 47 different

⁷ For the exceptions in the cubic crystal system *cf.* Section 3.2.1.2.4, *Notes on crystal and point forms*, item (viii).

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Table 3.2.1.4

Names and symbols of the 32 crystal classes

System used in this volume	Point group		Schoenflies symbol	Class names		
	International symbol			Class names		
	Short	Full		Groth (1921)	Friedel (1926)	
Triclinic	1 $\bar{1}$	1 $\bar{1}$	C_1 $C_i(S_2)$	Pedial (asymmetric) Pinacoidal	Hemihedry Holohedry	
Monoclinic	2 m $2/m$	2 m $\frac{2}{m}$	C_2 $C_s(C_{1h})$ C_{2h}	Sphenoidal Domatic Prismatic	Holoaxial hemihedry Antihemihedry Holohedry	
Orthorhombic	222 $mm2$ mmm	222 $mm2$ $\frac{2}{m} \frac{2}{m} \frac{2}{m}$	$D_2(V)$ C_{2v} $D_{2h}(V_h)$	Disphenoidal Pyramidal Dipyramidal	Holoaxial hemihedry Antihemihedry Holohedry	
Tetragonal	4 $\bar{4}$ $4/m$ 422 $4mm$ $\bar{4}2m$ $4/mmm$	4 $\bar{4}$ $\frac{4}{m}$ 422 $4mm$ $\bar{4}2m$ $\frac{4}{m} \frac{2}{m} \frac{2}{m}$	C_4 S_4 C_{4h} D_4 C_{4v} $D_{2d}(V_d)$ D_{4h}	Pyramidal Disphenoidal Dipyramidal Trapezohedral Ditetragonal-pyramidal Scalenohedral Ditetragonal-dipyramidal	Tetartohedry with 4-axis Sphenohedral tetartohedry Parahemihedry Holoaxial hemihedry Antihemihedry with 4-axis Sphenohedral antihemihedry Holohedry	
Trigonal	3 $\bar{3}$ 32 $3m$ $\bar{3}m$	3 $\bar{3}$ 32 $3m$ $\bar{3} \frac{2}{m}$	C_3 $C_{3i}(S_6)$ D_3 C_{3v} D_{3d}	Pyramidal Rhombohedral Trapezohedral Ditrigonal-pyramidal Ditrigonal-scalenohedral	<i>Hexagonal</i> Ogdohedry Paratetartohedry Holoaxial tetartohedry with 3-axis Hemimorphic antitetartohedry Parahemihedry with 3-axis	<i>Rhombohedral</i> Tetartohedry Parahemihedry Holoaxial hemihedry Antihemihedry Holohedry
Hexagonal	6 $\bar{6}$ $6/m$ 622 $6mm$ $\bar{6}2m$ $6/mmm$	6 $\bar{6}$ $\frac{6}{m}$ 622 $6mm$ $\bar{6}2m$ $\frac{6}{m} \frac{2}{m} \frac{2}{m}$	C_6 C_{3h} C_{6h} D_6 C_{6v} D_{3h} D_{6h}	Pyramidal Trigonal-dipyramidal Dipyramidal Trapezohedral Dihexagonal-pyramidal Ditrigonal-dipyramidal Dihexagonal-dipyramidal	Tetartohedry with 6-axis Trigonal antitetartohedry Parahemihedry with 6-axis Holoaxial hemihedry Antihemihedry with 6-axis Trigonal antihemihedry Holohedry	
Cubic	23 $m\bar{3}$ 432 $\bar{4}3m$ $m\bar{3}m$	23 $\frac{2}{m} \bar{3}$ 432 $\bar{4}3m$ $\frac{4}{m} \frac{3}{m} \frac{2}{m}$	T T_h O T_d O_h	Tetrahedral-pentagondodecahedral (= tetartoidal) Disdodecahedral (= diploidal) Pentagon-icositetrahedral (= gyroidal) Hexakistetrahedral (= hextetrahedral) Hexakisoctahedral (= hexoctahedral)	Tetartohedry Parahemihedry Holoaxial hemihedry Antihemihedry Holohedry	

types of crystal form. Frequently, 48 are quoted because ‘sphenoid’ and ‘dome’ are considered as two different forms. It is customary, however, to regard them as the same form, with the name ‘dihedron’.

Name of point form (printed in italics). There exists no general convention on the names of the point forms. Here, only one name is given, which does not always agree with that of other authors. The names of the point forms are also contained in Table 3.2.1.3.

Note that the same point form, ‘line segment’, corresponds to both sphenoid and dome.

Column 6: *Miller indices* (hkl) for the symmetry-equivalent faces (edges) of a crystal form. In the trigonal and hexagonal crystal systems, when referring to hexagonal axes, Bravais–Miller indices ($hkil$) are used, with $h + k + i = 0$.

With a few exceptions, the triplets of Miller indices (hkl) and point coordinates x, y, z are arranged in such a way as to show

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analogous sequences; they are both based on the same set of generators, as described in Sections 1.4.3 and 2.1.3.10. For all point groups, except those referred to a hexagonal coordinate system, the correspondence between the (hkl) and the x, y, z triplets is immediately obvious.⁸

The sets of symmetry-equivalent crystal faces also represent the sets of equivalent reciprocal-lattice points, as well as the sets of equivalent X-ray (neutron) reflections. This important aspect is treated in Klapper & Hahn (2010).

Examples

- (1) In point group $\bar{4}$, the general crystal form $4b$ is listed as $(hkl) (\bar{h}\bar{k}l) (k\bar{h}\bar{l}) (\bar{k}h\bar{l})$; the corresponding general position $4h$ of the symmorphic space group $P\bar{4}$ reads $x, y, z; \bar{x}, \bar{y}, z; y, \bar{x}, \bar{z}; \bar{y}, x, \bar{z}$.
- (2) In point group 3 (hexagonal axes), the general crystal form $3b$ is listed as $(hkl) (ihkl) (kihl)$ with $i = -(h + k)$; the corresponding general point form $3b$ is $x, y, z; \bar{y}, x - y, z; \bar{x} + y, \bar{x}, z$.
- (3) The Miller indices of the *cubic point groups* are arranged in one, two or four blocks of (3×4) entries. The first block belongs to point group 23. The second block belongs to the diagonal twofold axes in 432 and $m\bar{3}m$ or to the diagonal mirror plane in $\bar{4}3m$. In point groups $m\bar{3}$ and $m\bar{3}m$, the lower one or two blocks are derived from the upper blocks by application of the inversion.

Further discussion of the data in Tables 3.2.3.1 and 3.2.3.2 as far as molecular symmetry is concerned can be found in Section 3.2.4.3.

3.2.1.2.4. Notes on crystal and point forms

- (i) As mentioned in Section 3.2.1.1, each set of Miller indices of a given point group represents infinitely many face forms with the same name. Exceptions occur for the following cases.

Some special crystal forms occur with only *one* representative. Examples are the pinacoid {001}, the hexagonal prism {10 $\bar{1}$ 0} and the cube {100}. The Miller indices of these forms consist of fixed numbers and signs and contain no variables.

In a few noncentrosymmetric point groups, a special crystal form is realized by *two* representatives: they are related by a centre of symmetry that is not part of the point-group symmetry. These cases are

- (a) the two opposite, polar pedions {001} and {00 $\bar{1}$ };
 - (b) the two trigonal prisms {10 $\bar{1}$ 0} and { $\bar{1}$ 010}; similarly for two dimensions;
 - (c) the two trigonal prisms {11 $\bar{2}$ 0} and { $\bar{1}\bar{1}$ 20}; similarly for two dimensions;
 - (d) the positive and negative tetrahedra {111} and { $\bar{1}\bar{1}\bar{1}$ }.
- In the point-group tables, both representatives of these forms are listed, separated by 'or', for instance '(001) or (00 $\bar{1}$)'.

⁸ The matrices of corresponding triplets $(\bar{h}\bar{k}\bar{l})$ and $\bar{x}, \bar{y}, \bar{z}$, i.e. of triplets generated by the same symmetry operation from (hkl) and x, y, z , are inverse to each other, provided the x, y, z and $\bar{x}, \bar{y}, \bar{z}$ are regarded as columns and the (hkl) and $(\bar{h}\bar{k}\bar{l})$ as rows: this is due to the contravariant and covariant nature of the point coordinates and Miller indices, respectively. Note that for orthogonal matrices the inverse matrix equals the transposed matrix; in crystallography, this applies to all coordinate systems (including the rhombohedral one), except for the hexagonal system. The matrices for the symmetry operations occurring in the crystallographic point groups are listed in Tables 1.2.2.1 and 1.2.2.2.

- (ii) In crystallography, the terms tetragonal, trigonal, hexagonal, as well as tetragon, trigon and hexagon, imply that the cross sections of the corresponding polyhedra, or the polygons, are *regular* tetragons (squares), trigons or hexagons. Similarly, ditetragonal, ditrigonal, dihexagonal, as well as ditetragon, ditrigon and dihexagon, refer to *semi-regular* cross sections or polygons.

- (iii) Crystal forms can be 'open' or 'closed'. A crystal form is 'closed' if its faces form a closed polyhedron; the minimum number of faces for a closed form is 4. Closed forms are disphenoids, dipyramids, rhombohedra, trapezohedra, scalenohedra and all cubic forms; open forms are pedions, pinacoids, sphenoids (domes), pyramids and prisms.

A point form is always closed. It should be noted, however, that a point form dual to a *closed* face form is a *three-dimensional* polyhedron, whereas the dual of an *open* face form is a *two- or one-dimensional* polygon, which, in general, is located 'off the origin' but may be centred at the origin (here called 'through the origin').

- (iv) Crystal forms are well known; they are described and illustrated in many textbooks. Crystal forms are 'isohedral' polyhedra that have all faces equivalent but may have more than one kind of vertex; they include regular polyhedra. The in-sphere of the polyhedron touches all the faces.

Crystallographic point forms are less known; they are described in a few places only, notably by A. Niggli (1963), by Fischer *et al.* (1973), and by Burzlaff & Zimmermann (1977). The latter publication contains drawings of the polyhedra of all point forms. Point forms are 'isogonal' polyhedra (polygons) that have all vertices equivalent but may have more than one kind of face;⁹ again, they include regular polyhedra. The circumsphere of the polyhedron passes through all the vertices.

In most cases, the names of the point-form polyhedra can easily be derived from the corresponding crystal forms: the duals of n -gonal pyramids are regular n -gons off the origin, those of n -gonal prisms are regular n -gons through the origin. The duals of di- n -gonal pyramids and prisms are truncated (regular) n -gons, whereas the duals of n -gonal dipyramids are n -gonal prisms.

In a few cases, however, the relations are not so evident. This applies mainly to some cubic point forms [see item (v) below]. A further example is the rhombohedron, whose dual is a trigonal antiprism (in general, the duals of n -gonal streptohedra are n -gonal antiprisms).¹⁰ The duals of n -gonal trapezohedra are polyhedra intermediate between n -gonal prisms and n -gonal antiprisms; they are called here 'twisted n -gonal antiprisms' (example: point group 622). Finally, the duals of di- n -gonal scalenohedra are n -gonal antiprisms 'sliced off' perpendicular to the prism axis by the pinacoid {001}.¹¹

- (v) Some cubic point forms have to be described by 'combinations' of 'isohedral' polyhedra because no common

⁹ Thus, the name 'prism' for a *point form* implies combination of the prism with a pinacoid.

¹⁰ A tetragonal tetrahedron is a digonal streptohedron; hence, its dual is a 'digonal antiprism', which is again a tetragonal tetrahedron.

¹¹ The dual of a tetragonal (= di-digonal) scalenohedron is a 'digonal antiprism', which is 'cut off' by the pinacoid {001}.

3. ADVANCED TOPICS ON SPACE-GROUP SYMMETRY

names exist for 'isogonal' polyhedra. The maximal number of polyhedra required is three. The *shape* of the combination that describes the point form depends on the relative sizes of the polyhedra involved, *i.e.* on the relative values of their central distances. Moreover, in some cases even the *topology* of the point form may change.

Example

'Cube truncated by octahedron' and 'octahedron truncated by cube'. Both forms have 24 vertices, 14 faces and 36 edges, but the faces of the first combination are octagons and trigons, those of the second are hexagons and tetragons. These combinations represent different special point forms x, y, z and $0, y, z$. One form can change into the other only *via* the (semi-regular) cuboctahedron $0, y, y$, which has 12 vertices, 14 faces and 24 edges.

The unambiguous description of the cubic point forms by combinations of 'isohedral' polyhedra requires restrictions on the relative sizes of the polyhedra of a combination. The permissible range of the size ratios is limited on the one hand by vanishing, on the other hand by splitting of vertices of the combination. Three cases have to be distinguished:

- (a) The relative sizes of the polyhedra of the combination can vary *independently*. This occurs whenever three edges meet in one vertex. In Table 3.2.3.2, the names of these point forms contain the term 'truncated'.

Examples

- (1) 'Octahedron truncated by cube' (24 vertices, dual to tetrahexahedron).
 (2) 'Cube truncated by two tetrahedra' (24 vertices, dual to hexatetrahedron), implying independent variation of the relative sizes of the two truncating tetrahedra.
- (b) The relative sizes of the polyhedra are *interdependent*. This occurs for combinations of three polyhedra whenever four edges meet in one vertex. The names of these point forms contain the symbol '&'.

Example

'Cube & two tetrahedra' (12 vertices, dual to tetragon-tritetrahedron); here the interdependence results from the requirement that in the combination a cube edge is reduced to a vertex in which faces of the two tetrahedra meet. The location of this vertex on the cube edge is free. A higher-symmetry 'limiting' case of this combination is the 'cuboctahedron', where the two tetrahedra have the same sizes and thus form an octahedron.

- (c) The relative sizes of the polyhedra are *fixed*. This occurs for combinations of three polyhedra if five edges meet in one vertex. These point forms are designated by special names (snub tetrahedron, snub cube, irregular icosahedron), or their names contain the symbol '+'.
 The cuboctahedron appears here too, as a limiting form of the snub tetrahedron (dual to pentagon-tritetrahedron) and of the irregular icosahedron (dual to pentagon-dodecahedron) for the special coordinates $0, y, y$.

- (vi) Limiting crystal forms result from general or special crystal forms for special values of certain geometrical parameters of the form.

Examples

- (1) A pyramid degenerates into a prism if its apex angle becomes 0, *i.e.* if the apex moves towards infinity.
 (2) In point group 32, the general form is a trigonal trapezohedron $\{hkl\}$; this form can be considered as two opposite trigonal pyramids, rotated with respect to each other by an angle χ . The trapezohedron changes into the limiting forms 'trigonal dipyrmaid' $\{hhl\}$ for $\chi = 0^\circ$ and 'rhombohedral' $\{h0l\}$ for $\chi = 60^\circ$.

- (vii) One and the same type of polyhedron can occur as a general, special or limiting form.

Examples

- (1) A tetragonal dipyrmaid is a general form in point group $4/m$, a special form in point group $4/mmm$ and a limiting general form in point groups 422 and $\bar{4}2m$.
 (2) A tetragonal prism appears in point group $\bar{4}2m$ both as a basic special form (4b) and as a limiting special form (4c).

- (viii) A peculiarity occurs for the cubic point groups. Here the crystal forms $\{hhl\}$ are realized as two topologically different kinds of polyhedra with the same face symmetry, multiplicity and, in addition, the same eigensymmetry. The realization of one or other of these forms depends upon whether the Miller indices obey the conditions $|h| > |l|$ or $|h| < |l|$, *i.e.* whether, in the stereographic projection, a face pole is located between the directions $[110]$ and $[111]$ or between the directions $[111]$ and $[001]$. These two kinds of polyhedra have to be considered as two *realizations of one type* of crystal form because their face poles are located on the same set of conjugate symmetry elements. Similar considerations apply to the point forms x, x, z .

In the point groups $m\bar{3}m$ and $\bar{4}3m$, the two kinds of polyhedra represent two realizations of one *special* 'Wyckoff position'; hence, they have the same Wyckoff letter. In the groups 23, $m\bar{3}$ and 432, they represent two realizations of the same type of limiting *general* forms. In the tables of the cubic point groups, the two entries are always connected by braces.

The same kind of peculiarity occurs for the two icosahedral point groups, as mentioned in Section 3.2.1.4.2 and listed in Table 3.2.2.1.

3.2.1.2.5. Names and symbols of the crystal classes

Several different sets of names have been devised for the 32 crystal classes. Their use, however, has greatly declined since the introduction of the international point-group symbols. As examples, two sets (both translated into English) that are frequently found in the literature are given in Table 3.2.1.4. To the name of the class the name of the system has to be added: *e.g.* 'tetragonal pyramidal' or 'tetragonal tetartohedry'.

Note that Friedel (1926) based his nomenclature on the point symmetry of the lattice. Hence, two names are given for the five trigonal point groups, depending whether the lattice is hexagonal

3.2. POINT GROUPS AND CRYSTAL CLASSES

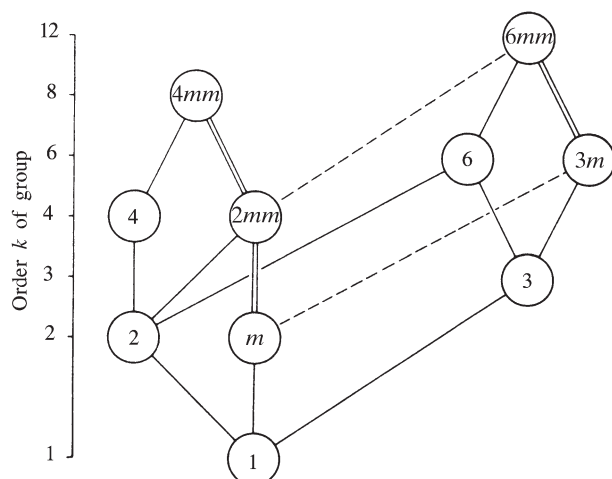


Figure 3.2.1.2

Maximal subgroups and minimal supergroups of the two-dimensional crystallographic point groups. Solid lines indicate maximal normal subgroups; double solid lines mean that there are two maximal normal subgroups with the same symbol. Dashed lines refer to sets of maximal conjugate subgroups. The group orders are given on the left.

or rhombohedral: e.g. ‘hexagonal ogdohedry’ and ‘rhombohedral tetartohedry’.

3.2.1.3. Subgroups and supergroups of the crystallographic point groups

In this section, the sub- and supergroup relations between the crystallographic point groups are presented in the form of a ‘family tree’.¹² Figs. 3.2.1.2 and 3.2.1.3 apply to two and three dimensions. The sub- and supergroup relations between two groups are represented by solid or dashed lines. For a given point group $\mathcal{P}$ of order k_P the lines to groups of lower order connect $\mathcal{P}$ with all its *maximal subgroups* $\mathcal{H}$ with orders k_H ; the index $[i]$ of each subgroup is given by the ratio of the orders k_P/k_H . The lines to groups of higher order connect $\mathcal{P}$ with all its *minimal supergroups* $\mathcal{S}$ with orders k_S ; the index $[i]$ of each supergroup is given by the ratio k_S/k_P . In other words: if the diagram is read downwards, subgroup relations are displayed; if it is read upwards, supergroup relations are revealed. The index is always an integer (theorem of Lagrange) and can be easily obtained from the group orders given on the left of the diagrams. The highest index of a maximal subgroup is [3] for two dimensions and [4] for three dimensions.

Two important kinds of subgroups, namely sets of conjugate subgroups and normal subgroups, are distinguished by dashed and solid lines. They are characterized as follows:

The subgroups $\mathcal{H}_1, \mathcal{H}_2, \dots, \mathcal{H}_n$ of a group $\mathcal{P}$ are *conjugate subgroups* if $\mathcal{H}_1, \mathcal{H}_2, \dots, \mathcal{H}_n$ are symmetry-equivalent in $\mathcal{P}$, i.e. if for every pair $\mathcal{H}_i, \mathcal{H}_j$ at least one symmetry operation W of $\mathcal{P}$ exists which maps $\mathcal{H}_i$ onto $\mathcal{H}_j$: $W^{-1}\mathcal{H}_iW = \mathcal{H}_j$; cf. Sections 1.1.5 and 1.1.8.

Examples

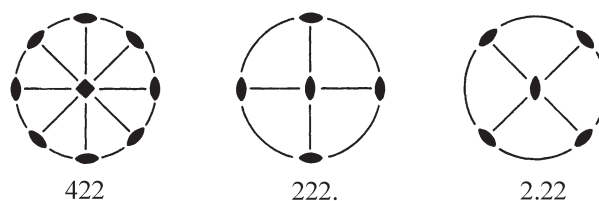
- (1) Point group $3m$ has three different mirror planes which are equivalent due to the threefold axis. In each of the three maximal subgroups of type m , one of these mirror planes is retained. Hence, the three subgroups m are conjugate in $3m$. This set of conjugate subgroups is represented by one dashed line in Figs. 3.2.1.2 and 3.2.1.3.

- (2) Similarly, group 432 has three maximal conjugate subgroups of type 422 and four maximal conjugate subgroups of type 32 .

The subgroup $\mathcal{H}$ of a group $\mathcal{P}$ is a *normal* (or invariant) subgroup if no subgroup $\mathcal{H}'$ of $\mathcal{P}$ exists that is conjugate to $\mathcal{H}$ in $\mathcal{P}$. Note that this does not imply that $\mathcal{H}$ is also a normal subgroup of any supergroup of $\mathcal{P}$. Subgroups of index [2] are always normal and maximal (cf. Section 1.1.5). (The role of normal subgroups for the structure of space groups is discussed in Sections 1.3.3 and 1.4.2.3.)

Examples

- (1) Fig. 3.2.1.3 shows two solid lines between point groups 422 and 222 , indicating that 422 has two maximal normal subgroups 222 of index [2]. The symmetry elements of one subgroup are rotated by 45° around the c axis with respect to those of the other subgroup. Thus, in one subgroup the symmetry elements of the two secondary, in the other those of the two tertiary tetragonal symmetry directions (cf. Table 2.1.3.1) are retained, whereas the primary twofold axis is the same for both subgroups. There exists no symmetry operation of 422 that maps one subgroup onto the other. This is illustrated by the stereograms below. The two normal subgroups can be indicated by the ‘oriented symbols’ 222_1 and 222_2 .



- (2) Similarly, group 432 has one maximal normal subgroup, 23 .

Figs. 3.2.1.2 and 3.2.1.3 show that there exist two ‘summits’ in both two and three dimensions from which all other point groups can be derived by ‘chains’ of maximal subgroups. These summits are formed by the square and the hexagonal holohedry in two dimensions and by the cubic and the hexagonal holohedry in three dimensions.

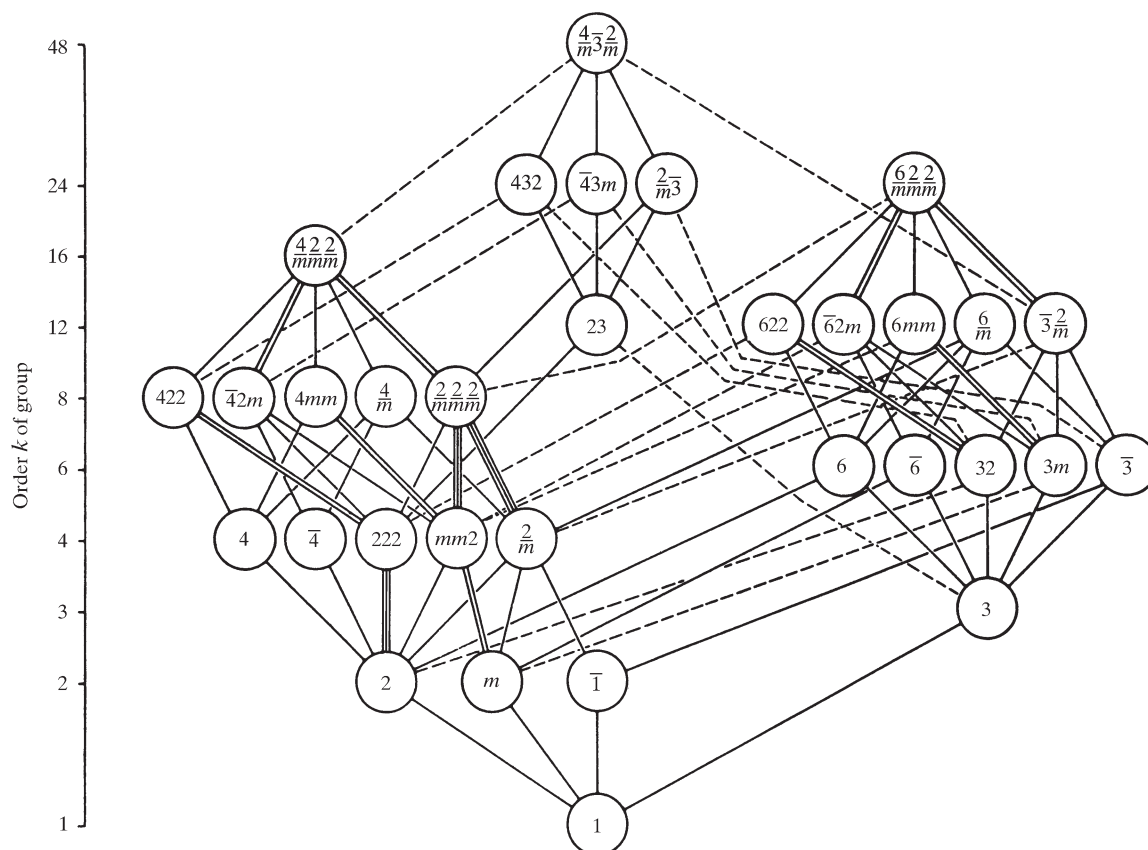
The sub- and supergroups of the point groups are useful both in their own right and as a basis of the *translationengleiche* or *t* subgroups and supergroups of space groups (cf. Section 1.7.1). Tables of the sub- and supergroups of the plane groups and space groups are contained in Volume A1 of *International Tables for Crystallography* (2010). A general discussion of sub- and supergroups of crystallographic groups, together with further explanations and examples, is given in Section 1.7.1.

3.2.1.4. Noncrystallographic point groups

3.2.1.4.1. Description of general point groups

In Sections 3.2.1.2 and 3.2.1.3, only the 32 *crystallographic* point groups (crystal classes) are considered. In addition, infinitely many *noncrystallographic* point groups exist that are of interest as possible symmetries of molecules and of quasicrystals and as approximate local site symmetries in crystals. Crystallographic and noncrystallographic point groups are collected here under the name *general point groups*. They are reviewed in this section and listed in Tables 3.2.1.5 and 3.2.1.6.

¹² This type of diagram was first used in *International Tables for the Determination of Crystal Structures* (1935); in *International Tables for X-ray Crystallography* (1952) a somewhat different approach was employed.

**Figure 3.2.1.3**

Maximal subgroups and minimal supergroups of the three-dimensional crystallographic point groups. Solid lines indicate maximal normal subgroups; double or triple solid lines mean that there are two or three maximal normal subgroups with the same symbol. Dashed lines refer to sets of maximal conjugate subgroups. The group orders are given on the left. Full Hermann-Mauguin symbols are used.

Because of the infinite number of these groups only *classes of general point groups (general classes)*¹³ can be listed. They are grouped into *general systems*, which are similar to the crystal systems. The ‘general classes’ are of two kinds: in the cubic, icosahedral, circular, cylindrical and spherical system, each general class contains *one* point group only, whereas in the $4N$ -gonal, $(2N+1)$ -gonal and $(4N+2)$ -gonal system, each general class contains *infinitely* many point groups, which differ in their principal n -fold symmetry axis, with $n = 4, 8, 12, \dots$ for the $4N$ -gonal system, $n = 1, 3, 5, \dots$ for the $(2N+1)$ -gonal system and $n = 2, 6, 10, \dots$ for the $(4N+2)$ -gonal system.

Furthermore, some general point groups are of order infinity because they contain symmetry axes (rotation or rotoinversion axes) of order infinity¹⁴ (∞ -fold axes). These point groups occur in the circular system (two dimensions) and in the cylindrical and spherical systems (three dimensions).

The Hermann-Mauguin and Schoenflies symbols for the general point groups follow the rules of the crystallographic point

groups (cf. Sections 1.4.1, 2.1.3.4 and 3.3.1). This extends also to the infinite groups where symbols like ∞m or $C_{\infty v}$ are immediately obvious.

In *two dimensions* (Table 3.2.1.5), the eight general classes are collected into three systems. Two of these, the $4N$ -gonal and the $(4N+2)$ -gonal systems, contain only point groups of finite order with one n -fold rotation point each. These systems are generalizations of the square and hexagonal crystal systems. The circular system consists of two infinite point groups, with one ∞ -fold rotation point each.

In *three dimensions* (Table 3.2.1.6), the 33 general classes are collected into seven systems. Three of these, the $4N$ -gonal, the $(2N+1)$ -gonal and the $(4N+2)$ -gonal systems,¹⁵ contain only point groups of finite order with one principal n -fold symmetry axis each. These systems are generalizations of the tetragonal, trigonal and hexagonal crystal systems (cf. Table 3.2.3.2). The five cubic groups are well known as crystallographic groups. The two icosahedral groups of orders 60 and 120, characterized by special combinations of twofold, threefold and fivefold symmetry axes, are discussed in more detail below. The groups of the cylindrical and the spherical systems are all of order infinity; they describe the symmetries of cylinders, cones, rotation ellipsoids, spheres etc.¹⁶

¹³ The ‘classes of general point groups’ are not the same as the commonly used ‘crystal classes’ because some of them contain point groups of *different* orders. All these orders, however, follow a common scheme. In this sense, the ‘general classes’ are an extension of the concept of (geometric) crystal classes. For example, the general class nmm of the $4N$ -gonal system contains the point groups $4mm$ (tetragonal), $8mm$ (octagonal), $12mm$ (dodecagonal), $16mm$ etc.

¹⁴ The axes of order infinity, as considered here, do not correspond to cyclic groups (as do the axes of finite order) because there is no smallest rotation from which all other rotations can be derived as higher powers, i.e. by successive application. Instead, rotations of all possible angles exist. Nevertheless, it is customary to symbolize these axes as ∞ or C_{∞} ; note that the Hermann-Mauguin symbols ∞/m and ∞ are equivalent, and so are the Schoenflies symbols $C_{\infty h}$, S_{∞} and $C_{\infty v}$. (There exist also axes of order infinity that do correspond to cyclic groups, namely axes based upon smallest rotations with irrational values of the rotation angle.)

¹⁵ Here, the $(2N+1)$ -gonal and the $(4N+2)$ -gonal systems are distinguished in order to bring out the analogy with the trigonal and the hexagonal crystal systems. They could equally well be combined into one, in correspondence with the hexagonal ‘crystal family’ (cf. Sections 1.3.4.4 and 2.1.1).

¹⁶ The terms ‘rotating’ and ‘stationary’ in the circular, cylindrical and spherical systems do not imply any relation to dynamical properties (motions) of crystals or molecules. They only serve to illustrate the absence (group ∞) or presence (∞m , $\infty \bar{m}$) of ‘vertical’ mirror planes in these groups or order ∞ .

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Table 3.2.1.5

Classes of general point groups in two dimensions ($N = \text{integer} \geq 0$)

General Hermann–Mauguin symbol	Order of group	General edge form	General point form	Crystallographic groups
$4N$ -gonal system (n -fold rotation point with $n = 4N$)				
n	n	Regular n -gon	Regular n -gon	4
nmm	$2n$	Semiregular di- n -gon	Truncated n -gon	$4mm$
$(4N + 2)$ -gonal system (n -fold or $\frac{1}{2}n$ -fold rotation point with $n = 4N + 2$)				
$\frac{1}{2}n$	$\frac{1}{2}n$	Regular $\frac{1}{2}n$ -gon	Regular $\frac{1}{2}n$ -gon	1, 3
$\frac{1}{2}nm$	n	Semiregular di- $\frac{1}{2}n$ -gon	Truncated $\frac{1}{2}n$ -gon	$m, 3m$
n	n	Regular n -gon	Regular n -gon	2, 6
nmm	$2n$	Semiregular di- n -gon	Truncated n -gon	$2mm, 6mm$
Circular system†				
∞	∞	Rotating circle	Rotating circle	–
∞m	∞	Stationary circle	Stationary circle	–

† A rotating circle has no mirror lines; there exist two enantiomorphic circles with opposite senses of rotation. A stationary circle has infinitely many mirror lines through its centre.

It is possible to define the three-dimensional point groups on the basis of either rotoinversion axes $\bar{n}$ or rotoreflection axes $\tilde{n}$. The equivalence between these two descriptions is apparent from the following examples:

$$\begin{aligned}
 n = 4N: & \quad \bar{4} = \tilde{4} & \bar{8} = \tilde{8} & \dots & \bar{n} = \tilde{n} \\
 n = 2N + 1: & \quad \bar{1} = \tilde{2} & \bar{3} = \tilde{6} = 3 \times \bar{1} & \dots & \bar{n} = \tilde{2n} = n \times \bar{1} \\
 n = 4N + 2: & \quad \bar{2} = \tilde{1} = m & \bar{6} = \tilde{3} = 3/m & \dots & \bar{n} = \frac{1}{2}n = \frac{1}{2}n/m.
 \end{aligned}$$

In the present tables, the standard convention of using roto-inversion axes is followed.

Tables 3.2.1.5 and 3.2.1.6 contain for each class its general Hermann–Mauguin and Schoenflies symbols, the group order and the names of the general face form and its dual, the general point form.¹⁷ Special and limiting forms are not given, nor are ‘Miller indices’ (hkl) and point coordinates x, y, z . They can be derived easily from Tables 3.2.3.1 and 3.2.3.2 for the crystallographic groups.¹⁸

3.2.1.4.2. The two icosahedral groups

The two point groups 235 and $m\bar{3}5$ of the icosahedral system (orders 60 and 120) are of particular interest among the noncrystallographic groups because of the occurrence of fivefold axes and their increasing importance as symmetries of molecules (viruses), of quasicrystals, and as approximate local site symmetries in crystals (alloys, B_{12} icosahedron). Furthermore, they contain as special forms the two noncrystallographic *platonic solids*, the regular icosahedron (20 faces, 12 vertices) and its dual, the regular pentagon-dodecahedron (12 faces, 20 vertices).

The icosahedral groups (cf. diagrams in Table 3.2.3.3) are characterized by six fivefold axes that include angles of 63.43° . Each fivefold axis is surrounded by five threefold and five twofold axes, with angular distances of 37.38° between a fivefold and a threefold axis and of 31.72° between a fivefold and a twofold axis. The angles between neighbouring threefold axes are 41.81° ,

between neighbouring twofold axes 36° . The smallest angle between a threefold and a twofold axis is 20.90° .

Each of the six fivefold axes is perpendicular to five twofold axes; there are thus six maximal conjugate pentagonal subgroups of types 52 (for 235) and $\bar{5}m$ (for $m\bar{3}5$) with index [6]. Each of the ten threefold axes is perpendicular to three twofold axes, leading to ten maximal conjugate trigonal subgroups of types 32 (for 235) and $\bar{3}m$ (for $m\bar{3}5$) with index [10]. There occur, furthermore, five maximal conjugate cubic subgroups of types 23 (for 235) and $m\bar{3}$ (for $m\bar{3}5$) with index [5].

The two icosahedral groups are listed in Table 3.2.3.3, in a form similar to the cubic point groups in Table 3.2.3.2. Each group is illustrated by stereographic projections of the symmetry elements and the general face poles (general points); the complete sets of symmetry elements are listed below the stereograms. Both groups are referred to a cubic coordinate system, with the coordinate axes along three twofold rotation axes and with four threefold axes along the body diagonals. This relation is well brought out by symbolizing these groups as 235 and $m\bar{3}5$ instead of the customary symbols 532 and $\bar{5}3m$.

The table contains also the multiplicities, the Wyckoff letters and the names of the general and special face forms and their duals, the point forms, as well as the oriented face- and site-symmetry symbols. In the icosahedral ‘holohedry’ $m\bar{3}5$, the *special* ‘Wyckoff position’ $60d$ occurs in three realizations, i.e. with three types of polyhedra. In 235, however, these three types of polyhedra are different realizations of the limiting *general* forms, which depend on the location of the poles with respect to the axes 2, 3 and 5. For this reason, the three entries are connected by braces; cf. Section 3.2.1.2.4, *Notes on crystal and point forms*, item (viii).

Not included are the sets of equivalent Miller indices and point coordinates. Instead, only the ‘initial’ triplets (hkl) and x, y, z for each type of form are listed. The complete sets of indices and coordinates can be obtained in two steps¹⁹ as follows:

¹⁷ The noncrystallographic face and point forms are extensions of the corresponding crystallographic forms: cf. Section 3.2.1.2.4, *Notes on crystal and point forms*. The name *streptohedron* applies to the general face forms of point groups $\bar{n}$ with $n = 4N$ and $n = 2N + 1$; it is thus a generalization of the tetragonal disphenoid or tetragonal tetrahedron ($\bar{4}$) and the rhombohedron ($\bar{3}$).

¹⁸ The term ‘Miller indices’ is used here also for the noncrystallographic point groups. Note that these indices do not have to be integers or rational numbers, as for the crystallographic point groups. Irrational indices, however, can always be closely approximated by integers, quite often even by small integers.

¹⁹ A one-step procedure applies to the icosahedral ‘Wyckoff position’ $12a$, the face poles and points of which are located on the fivefold axes. Here, step (ii) is redundant and can be omitted. The forms $\{01\tau\}$ and $0, y, \tau y$ are contained in the cubic point groups 23 and $m\bar{3}$ and in the cubic space groups $P23$ and $Pm\bar{3}$ as limiting cases of Wyckoff positions $\{0kl\}$ and $0, y, z$ with specialized (irrational) values of the indices and coordinates. In geometric terms, the regular pentagon-dodecahedron is a noncrystallographic ‘limiting polyhedron’ of the ‘crystallographic’ pentagon-dodecahedron and the regular icosahedron is a ‘limiting polyhedron’ of the ‘irregular’ icosahedron (cf. Section 3.2.1.2.2, *Crystal and point forms*).

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Table 3.2.1.6

Classes of general point groups in three dimensions ($N = \text{integer} \geq 0$)

Short general Hermann–Mauguin symbol, followed by full symbol where different	Schoenflies symbol	Order of group	General face form	General point form	Crystallographic groups
$4N$-gonal system (single n-fold symmetry axis with $n = 4N$)					
n	C_n	n	n -gonal pyramid	Regular n -gon	4
$\bar{n}$	S_n	n	$\frac{1}{2}n$ -gonal streptohedron	$\frac{1}{2}n$ -gonal antiprism	$\bar{4}$
n/m	C_{nh}	$2n$	n -gonal dipyrmaid	n -gonal prism	$4/m$
$n22$	D_n	$2n$	n -gonal trapezohedron	Twisted n -gonal antiprism	422
$nm\bar{m}$	C_{nv}	$2n$	Di- n -gonal pyramid	Truncated n -gon	$4mm$
$\bar{n}2m$	$D_{\frac{n}{2}nd}$	$2n$	n -gonal scalenohedron	$\frac{1}{2}n$ -gonal antiprism sliced off by pinacoid	$42m$
$n/mmm, \frac{n}{m} \frac{2}{m} \frac{2}{m}$	D_{nh}	$4n$	Di- n -gonal dipyrmaid	Edge-truncated n -gonal prism	$4/mmm$
$(2N + 1)$-gonal system (single n-fold symmetry axis with $n = 2N + 1$)					
n	C_n	n	n -gonal pyramid	Regular n -gon	1, 3
$\bar{n} = n \times \bar{1}$	C_{ni}	$2n$	n -gonal streptohedron	n -gonal antiprism	$\bar{1}, \bar{3} = 3 \times \bar{1}$
$n2$	D_n	$2n$	n -gonal trapezohedron	Twisted n -gonal antiprism	32
nm	C_{nv}	$2n$	Di- n -gonal pyramid	Truncated n -gon	$3m$
$\bar{n}m, \bar{n} \frac{2}{m}$	D_{nd}	$4n$	Di- n -gonal scalenohedron	n -gonal antiprism sliced off by pinacoid	$\bar{3}m$
$(4N + 2)$-gonal system (single n-fold symmetry axis with $n = 4N + 2$)					
n	C_n	n	n -gonal pyramid	Regular n -gon	2, 6
$\bar{n} = \frac{1}{2}n/m$	$C_{\frac{n}{2}nh}$	n	$\frac{1}{2}n$ -gonal dipyrmaid	$\frac{1}{2}n$ -gonal prism	$\bar{2} \equiv m, \bar{6} \equiv 3/m$
n/m	C_{nh}	$2n$	n -gonal dipyrmaid	n -gonal prism	$2/m, 6/m$
$n22$	D_n	$2n$	n -gonal trapezohedron	Twisted n -gonal antiprism	222, 622
$nm\bar{m}$	C_{nv}	$2n$	Di- n -gonal pyramid	Truncated n -gon	$mm2, 6mm$
$\bar{n}2m = \frac{1}{2}n/m2m$	$D_{\frac{n}{2}nh}$	$2n$	Di- $\frac{1}{2}n$ -gonal dipyrmaid	Truncated $\frac{1}{2}n$ -gonal prism	$\bar{6}2m$
$n/mmm, \frac{n}{m} \frac{2}{m} \frac{2}{m}$	D_{nh}	$4n$	Di- n -gonal dipyrmaid	Edge-truncated n -gonal prism	$mmm, 6/mmm$
Cubic system (for details see Table 3.2.3.2)					
23	T	12	Pentagon-tritetrahedron	Snub tetrahedron	23
$m\bar{3}, \frac{2}{m} \bar{3}$	T_h	24	Didodecahedron	Cube & octahedron & pentagon-dodecahedron	$m\bar{3}$
432	O	24	Pentagon-trioctahedron	Snub cube	432
$\bar{4}3m$	T_d	24	Hexatetrahedron	Cube truncated by two tetrahedra	$\bar{4}3m$
$m\bar{3}m, \frac{4}{m} \frac{2}{m} \frac{2}{m}$	O_h	48	Hexaoctahedron	Cube truncated by octahedron and by rhomb-dodecahedron	$m\bar{3}m$
Icosahedral system† (for details see Table 3.2.3.3)					
235	I	60	Pentagon-hexecontahedron	Snub pentagon-dodecahedron	–
$m\bar{3}\bar{5}, \frac{2}{m} \bar{3}\bar{5}$	I_h	120	Hecatonicosahedron	Pentagon-dodecahedron truncated by icosahedron and by rhomb-triacontahedron	–
Cylindrical system‡					
∞	C_∞	∞	Rotating cone	Rotating circle	–
$\infty/m \equiv \infty$	$C_{\infty h} \equiv S_\infty \equiv C_{\infty i}$	∞	Rotating double cone	Rotating finite cylinder	–
$\infty 2$	D_∞	∞	‘Anti-rotating’ double cone	‘Anti-rotating’ finite cylinder	–
∞m	$C_{\infty v}$	∞	Stationary cone	Stationary circle	–
$\infty/m\bar{m} \equiv \infty m, \frac{\infty}{m} \frac{2}{m} \equiv \frac{2}{m}$	$D_{\infty h} \equiv D_{\infty d}$	∞	Stationary double cone	Stationary finite cylinder	–
Spherical system§					
$2\infty, \infty\infty$	K	∞	Rotating sphere	Rotating sphere	–
$m\infty, \frac{2}{m} \infty, \infty\infty m$	K_h	∞	Stationary sphere	Stationary sphere	–

† The Hermann–Mauguin symbols of the two icosahedral point groups are often written as 532 and $\bar{5}3m$ (see text). ‡ Rotating and ‘anti-rotating’ forms in the cylindrical system have no ‘vertical’ mirror planes, whereas stationary forms have infinitely many vertical mirror planes. In classes ∞ and $\infty 2$, enantiomorphism occurs, *i.e.* forms with opposite senses of rotation. Class $\infty/m \equiv \infty$ exhibits no enantiomorphism due to the centre of symmetry, even though the double cone is rotating in one direction. This can be understood as follows: The handedness of a rotating cone depends on the sense of rotation with respect to the axial direction from the base to the tip of the cone. Thus, the rotating double cone consists of two cones with opposite handedness and opposite orientations related by the (single) horizontal mirror plane. In contrast, the ‘anti-rotating’ double cone in class $\infty 2$ consists of two cones of equal handedness and opposite orientations, which are related by the (infinitely many) twofold axes. The term ‘anti-rotating’ means that upper and lower halves of the forms rotate in opposite directions. § The spheres in class 2∞ of the spherical system must rotate around an axis with at least two different orientations, in order to suppress all mirror planes. This class exhibits enantiomorphism, *i.e.* it contains spheres with either right-handed or left-handed senses of rotation around the axes (*cf.* Section 3.2.2.4, *Optical properties*). The stationary spheres in class $m\infty$ contain infinitely many mirror planes through the centres of the spheres. Group 2∞ is sometimes symbolized by $\infty\infty$; group $m\infty$ by $\infty\infty$ or $\infty\infty m$. The symbols used here indicate the minimal symmetry necessary to generate the groups; they show, furthermore, the relation to the cubic groups. The Schoenflies symbol K is derived from the German name *Kugelgruppe*.

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- (i) For the face forms the cubic point groups 23 and $m\bar{3}$ (Table 3.2.3.2), and for the point forms the cubic space groups $P23$ (195) and $Pm\bar{3}$ (200) have to be considered. For each 'initial' triplet (hkl) , the set of Miller indices of the (general or special) crystal form with the same face symmetry in 23 (for group 235) or $m\bar{3}$ (for $m\bar{3}5$) is taken. For each 'initial' triplet x, y, z , the coordinate triplets of the (general or special) position with the same site symmetry in $P23$ or $Pm\bar{3}$ are taken.
- (ii) To obtain the complete set of icosahedral Miller indices and point coordinates, the 'cubic' (hkl) triplets (as rows) and x, y, z triplets (as columns) have to be multiplied with the identity matrix and with
- the matrices $\mathbf{Y}, \mathbf{Y}^2, \mathbf{Y}^3$ and $\mathbf{Y}^4$ for the Miller indices;
 - the matrices $\mathbf{Y}^{-1}, \mathbf{Y}^{-2}, \mathbf{Y}^{-3}$ and $\mathbf{Y}^{-4}$ for the point coordinates.

This sequence of matrices ensures the same correspondence between the Miller indices and the point coordinates as for the crystallographic point groups in Table 3.2.3.2.

The matrices²⁰ are

$$\mathbf{Y} = \mathbf{Y}^{-4} = \begin{pmatrix} \frac{1}{2} & g & G \\ g & G & -\frac{1}{2} \\ -G & \frac{1}{2} & g \end{pmatrix}, \quad \mathbf{Y}^2 = \mathbf{Y}^{-3} = \begin{pmatrix} -g & G & \frac{1}{2} \\ G & \frac{1}{2} & -g \\ -\frac{1}{2} & g & -G \end{pmatrix},$$

$$\mathbf{Y}^3 = \mathbf{Y}^{-2} = \begin{pmatrix} -g & G & -\frac{1}{2} \\ G & \frac{1}{2} & g \\ \frac{1}{2} & -g & -G \end{pmatrix}, \quad \mathbf{Y}^4 = \mathbf{Y}^{-1} = \begin{pmatrix} \frac{1}{2} & g & -G \\ g & G & \frac{1}{2} \\ G & -\frac{1}{2} & g \end{pmatrix},$$

with²¹

$$G = \frac{\sqrt{5} + 1}{4} = \frac{\tau}{2} = \cos 36^\circ = 0.80902 \simeq \frac{72}{89}$$

$$g = \frac{\sqrt{5} - 1}{4} = \frac{\tau - 1}{2} = \cos 72^\circ = 0.30902 \simeq \frac{17}{55}.$$

These matrices correspond to counter-clockwise rotations of 72° , 144° , 216° and 288° around a fivefold axis parallel to $[1\tau 0]$.

The resulting indices h, k, l and coordinates x, y, z are irrational but can be approximated closely by rational (or integral) numbers. This explains the occurrence of almost regular icosahedra or pentagon-dodecahedra as crystal forms (for instance pyrite) or atomic groups (for instance B_{12} icosahedron).

Further descriptions (including diagrams) of noncrystallographic groups are contained in papers by Nowacki (1933) and A. Niggli (1963) and in the textbooks by P. Niggli (1941, pp. 78–80, 96), Shubnikov & Koptsik (1974) and Vainshtein (1994). For the geometry of polyhedra, the well known books by H. S. M. Coxeter (especially Coxeter, 1973) are recommended.

3.2.1.4.3. Sub- and supergroups of the general point groups

In Figs. 3.2.1.4 to 3.2.1.6, the subgroup and supergroup relations between the two-dimensional and three-dimensional general point groups are illustrated. It should be remembered that the index of a group–subgroup relation between two groups of order infinity may be finite or infinite. For the two spherical

²⁰ Note that for orthogonal matrices $\mathbf{Y}^{-1} = \mathbf{Y}^t$ (t = transposed).

²¹ The number $\tau = 2G = 2g + 1 = (\sqrt{5} + 1)/2 = 1.618034$ (Fibonacci number) is the characteristic value of the *golden section* $(\tau + 1) : \tau = \tau : 1$, i.e. $\tau(\tau - 1) = 1$. Furthermore, τ is the distance between alternating vertices of a regular pentagon of unit edge length and the distance from centre to vertex of a regular decagon of unit edge length.

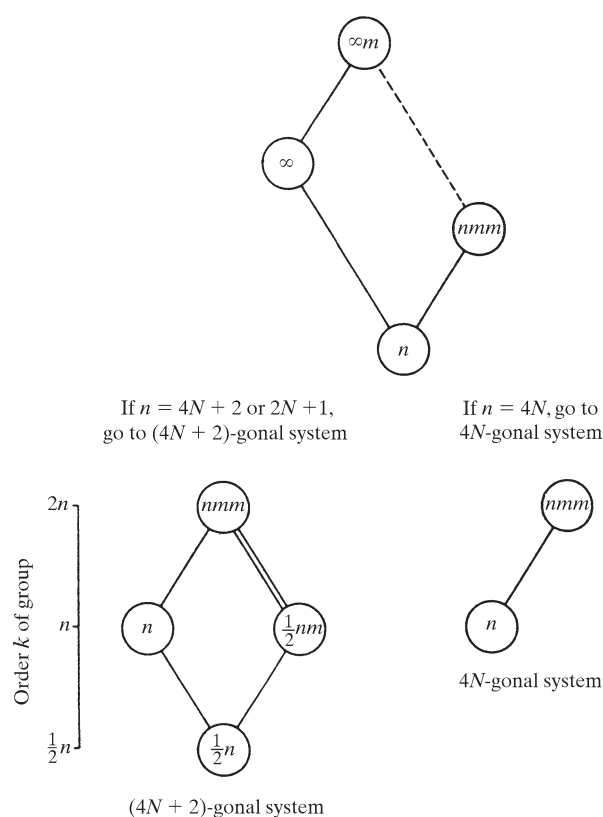


Figure 3.2.1.4

Subgroups and supergroups of the two-dimensional general point groups. Solid lines indicate maximal normal subgroups, double solid lines mean that there are two maximal normal subgroups with the same symbol. Dashed lines refer to sets of maximal conjugate subgroups. For the finite groups, the orders are given on the left. Note that the subgroups of the two circular groups are not maximal and the diagram applies only to a specified value of N (see text). For complete examples see Fig. 3.2.1.5.

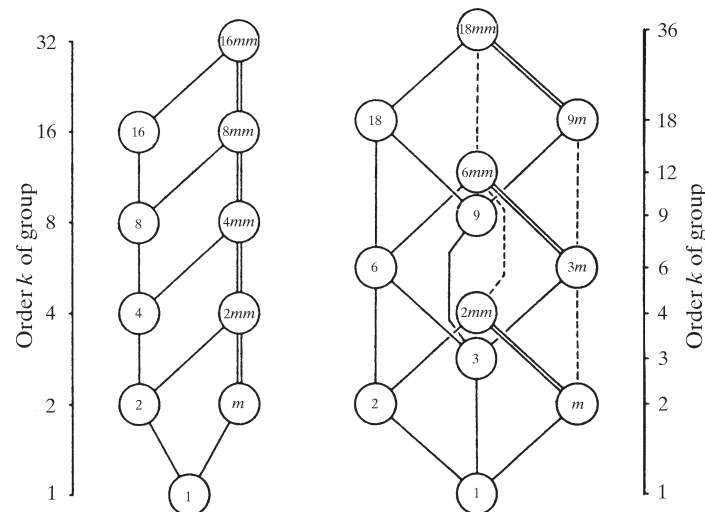
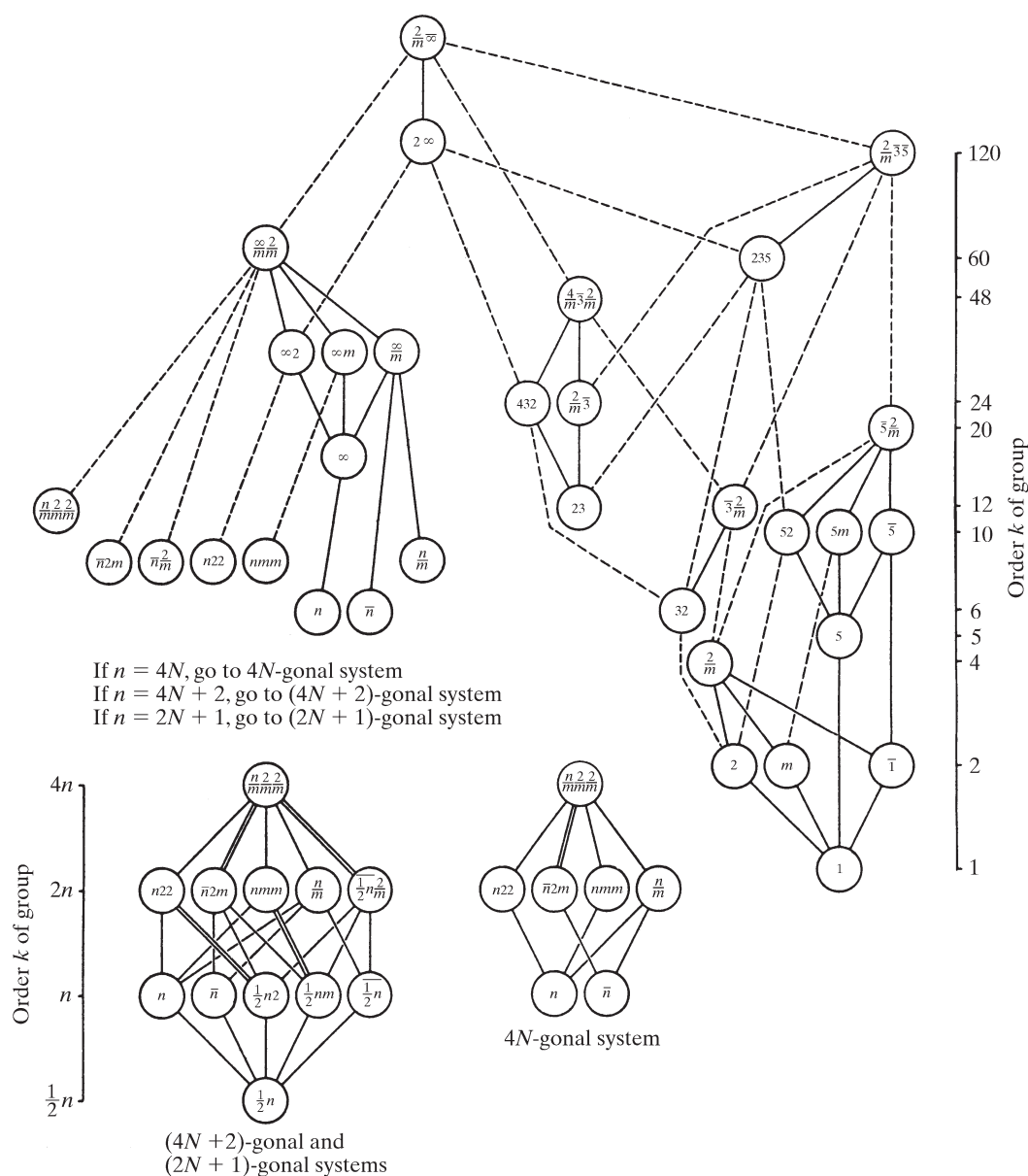


Figure 3.2.1.5

The subgroups of the two-dimensional general point groups $16mm$ ($4N$ -gonal system) and $18mm$ [$(4N + 2)$ -gonal system], including the $(2N + 1)$ -gonal groups]. Compare with Fig. 3.2.1.4 which applies only to one value of N .

groups, for instance, the index is $[2]$; the cylindrical groups, on the other hand, are subgroups of index $[\infty]$ of the spherical groups.

Fig. 3.2.1.4 for *two dimensions* shows that the two circular groups ∞m and ∞ have subgroups of types nmm and n ,

**Figure 3.2.1.6**

Subgroups and supergroups of the three-dimensional general point groups. Solid lines indicate maximal normal subgroups, double solid lines mean that there are two maximal normal subgroups with the same symbol. Dashed lines refer to sets of maximal conjugate subgroups. For the finite groups, the orders are given on the left and on the right. Note that the subgroups of the five cylindrical groups are not maximal and that the diagram applies only to a specified value of N (see text). Only those crystallographic point groups are included that are maximal subgroups of noncrystallographic point groups, cf. Fig. 3.2.1.3. Full Hermann-Mauguin symbols are used.

respectively, each of index $[\infty]$. The order of the rotation point may be $n = 4N$, $n = 4N + 2$ or $n = 2N + 1$. In the first case, the subgroups belong to the $4N$ -gonal system, in the latter two cases, they belong to the $(4N + 2)$ -gonal system. [In the diagram of the $(4N + 2)$ -gonal system, the $(2N + 1)$ -gonal groups appear with the symbols $\frac{1}{2}nm$ and $\frac{1}{2}n$.] The subgroups of the circular groups are not maximal because for any given value of N there exists a group with $N' = 2N$ which is both a subgroup of the circular group and a supergroup of the initial group.

The subgroup relations, for a specified value of N , within the $4N$ -gonal and the $(4N + 2)$ -gonal system, are shown in the lower part of the figure. They correspond to those of the crystallographic groups. A finite number of further maximal subgroups is obtained for lower values of N , until the crystallographic groups (Fig. 3.2.1.2) are reached. This is illustrated for both systems in Fig. 3.2.1.5.

Fig. 3.2.1.6 for *three dimensions* illustrates that the two spherical groups $2/m\infty$ and 2∞ each have one infinite set of

cylindrical maximal conjugate subgroups, as well as one infinite set of cubic and one infinite set of icosahedral maximal finite conjugate subgroups, all of index $[\infty]$.

Each of the two icosahedral groups 235 and $2/m\bar{3}5$ has one set of five cubic, one set of six pentagonal and one set of ten trigonal maximal conjugate subgroups of indices $[5]$, $[6]$ and $[10]$, respectively (cf. Section 3.2.1.4.2, *The two icosahedral groups*); they are listed on the right of Fig. 3.2.1.6. For the crystallographic groups, Fig. 3.2.1.3 applies. The subgroup types of the five cylindrical point groups are shown on the upper left part of Fig. 3.2.1.6. As explained above for two dimensions, these subgroups are *not maximal* and of index $[\infty]$. Depending upon whether the main symmetry axis has the multiplicity $4N$, $4N + 2$ or $2N + 1$, the subgroups belong to the $4N$ -gonal, $(4N + 2)$ -gonal or $(2N + 1)$ -gonal system.

The subgroup and supergroup relations within these three systems are displayed in the lower left part of Fig. 3.2.1.6. They are analogous to the crystallographic groups. To facilitate the

3.2. POINT GROUPS AND CRYSTAL CLASSES

use of the diagrams, the $(4N + 2)$ -gonal and the $(2N + 1)$ -gonal systems are combined, with the consequence that the five classes of the $(2N + 1)$ -gonal system now appear with the symbols $\frac{1}{2}n\frac{2}{m}$, $\frac{1}{2}n2$, $\frac{1}{2}nm$, $\frac{1}{2}n$ and $\frac{1}{2}n$. Again, the diagrams apply to a specified value of N . A finite number of further maximal subgroups is obtained for lower values of N , until the crystallographic groups (Fig. 3.2.1.3) are reached (*cf.* the two-dimensional examples in Fig. 3.2.1.5).

3.2.2. Point-group symmetry and physical properties of crystals

BY H. KLAPPER AND TH. HAHN

In the previous section (Section 3.2.1), the crystallographic and noncrystallographic point groups are treated under geometrical aspects only. In the present section the point-group symmetries of the physical properties are considered. Among the physical properties, those represented by tensors are accessible to a mathematical treatment of their symmetries, which are apparent by the invariance of tensor components under symmetry operations. For more details the reader is referred to Nye (1957, 1985), Paufler (1986), Schwarzenbach & Chapuis (2006) and Shuvalov (1988). A similar comprehensive treatment of non-tensorial properties, such as cleavage, plasticity, hardness or crystal growth does not exist, but these properties are, of course, also governed by the crystallographic point-group symmetry.

3.2.2.1. General restrictions on physical properties imposed by symmetry

3.2.2.1.1. Neumann's principle

Neumann's principle (Neumann, 1885) describes the relation between the symmetry Q of a physical property and the crystallographic point group P of a crystal. It states that the symmetry of any physical property of a crystal is higher than, or at least equal to, its crystallographic point-group symmetry, or in the language of groups, the symmetry Q of any physical property of a crystal is a proper or improper supergroup of its crystallographic symmetry P : $Q \subseteq P$. This is easily illustrated for polar second-rank tensors, which are represented by ellipsoids and hyperboloids. The representation surfaces of triclinic, monoclinic and orthorhombic crystals are general ellipsoids or hyperboloids of symmetry $Q = 2/m2/m2/m$, which is a proper supergroup of all triclinic, monoclinic and orthorhombic point groups with the only exception of the orthorhombic holohedry $P = 2/m2/m2/m$, for which the two symmetries are the same. For the uniaxial crystals of the trigonal, tetragonal and hexagonal systems the representation ellipsoids and hyperboloids have rotation symmetries with the cylindrical point group $Q = \infty/m2/m$ (see Table 3.2.1.6), which is a proper supergroup of all uniaxial crystallographic point groups. For cubic crystals, second-rank tensors are isotropic and represented by a sphere with point-group symmetry $Q = 2/m\bar{3} (\infty\infty m)$, a proper supergroup of all cubic crystallographic point groups. For more details the reader is referred to Paufler (1986) and Authier (2014). A short resume is given by Klapper & Hahn (2005).

As a consequence of the invariance of tensor components under a symmetry operation (or alternatively: under a transformation of the coordinate system to a symmetry-equivalent one),

some of the tensor components are equal or even zero. The number of independent components decreases when the symmetry of the crystal increases. Thus, an increase of the point-group symmetry from 1 (triclinic) to $4/m\bar{3}2/m$ (cubic) or to the sphere group $\infty\infty m$ (*i.e.* isotropy) reduces the number of tensor components for symmetrical second-rank tensors from 6 to 1. Even more drastic is this reduction for all tensors of odd rank (such as pyroelectricity and piezoelectricity) or axial tensors of second rank (*e.g.* optical activity): all components are zero if an inversion centre is present, *i.e.* properties described by these tensors do not exist in centrosymmetric crystals (see the textbooks of tensor physics mentioned above). These properties, which exist only in noncentrosymmetric crystals are, as a rule, the most important ones, not only for physical applications but also for structure determination, because they allow a proof of the absence of a symmetry centre.

For the description of noncentrosymmetric crystals and their specific properties, certain notions are of importance and these are explained in the following two sections.

3.2.2.1.2. Curie's principle

Curie's principle (Curie, 1894) describes the crystallographic symmetry $\mathcal{P}_F$ of a macroscopic crystal which is subject to an external influence F , for example to an electric field $\mathbf{E}$, to uniaxial stress σ_{ii} , to a temperature change ΔT *etc.* For this treatment, the point-group symmetries $\mathcal{R}$ of the external influences (Curie groups) are defined as follows (see Authier, 2014, p. 11; Paufler, 1986, p. 29):

Homogeneous electric field $\mathbf{E}$: $\mathcal{R} = \infty m$ (polar continuous rotation axis with 'parallel' mirror planes, *i.e.* symmetry of a stationary cone) [Note that a rotating cone (left- or right-handed) represents geometrically the polar enantiomorphous group ∞ ; a stationary cone represents the polar group ∞m with 'vertical' mirror planes, *cf.* Section 3.2.1.4.1.];

Homogeneous magnetic field $\mathbf{H}$: $\mathcal{R} = \infty/m$ (symmetry of a rotating cylinder);

Uniaxial stress σ_{ii} : $\mathcal{R} = \infty/m2/m$ (symmetry of a stationary centrosymmetric cylinder);

Temperature change ΔT or hydrostatic pressure p (scalars): $\mathcal{R} = 2/m\bar{3} (\infty\infty m)$ (symmetry of a stationary centrosymmetric sphere);

Shear stress σ_{ij} : $\mathcal{R} = 2/m2/m2/m$ (orthorhombic).

According to Curie's principle, the point-group symmetry $\mathcal{P}_F$ of the crystal under the external field F is the intersection symmetry of the two point groups: $\mathcal{P}$ of the crystal without field and $\mathcal{R}$ of the field without crystal: $\mathcal{P}_F = \mathcal{R} \cap \mathcal{P}$; *i.e.* $\mathcal{P}_F$ is a (proper or improper) subgroup of both groups $\mathcal{R}$ and $\mathcal{P}$.

As examples we consider the effect of an electric field ($\mathcal{R} = \infty m$) and of a uniaxial stress ($\mathcal{R} = \infty/m2/m$) along one of the (four) threefold rotoinversion axes $\bar{3}$ of cubic crystals with point groups $\mathcal{P} = 2/m\bar{3}$ and $\mathcal{P} = 4/m\bar{3}2/m$.

Electric field parallel to $[111]$:

$2/m\bar{3} \cap \infty m$:	$\mathcal{P}_F = 3$	along $[111]$ (polar, pyroelectric, optically active; <i>cf.</i> Sections 3.2.2.5 and 3.2.2.4.2)
$4/m\bar{3}2/m \cap \infty m$:	$\mathcal{P}_F = 3m$	along $[111]$ (polar, pyroelectric, not optically active)

3. ADVANCED TOPICS ON SPACE-GROUP SYMMETRY

Table 3.2.2.1

The 11 Laue classes, the 21 noncentrosymmetric crystallographic point groups (crystal classes) and the occurrence (+) of specific crystal properties

The number of non-zero independent tensor components is given in parentheses. For comparison, the icosahedral and the sphere groups (isotropy) are added. For first-rank tensors the direction $[uvw]$ of the property vector is given. There are 10 pyroelectric, 20 piezoelectric, 11 enantiomorphic and 15 optically active crystal classes.

Crystal system	Laue class	Noncentrosymmetric point group	First-rank tensor (e.g. pyroelectricity, piezoelectricity under hydrostatic pressure)	Third-rank tensor (e.g. piezoelectricity, second-harmonic generation)	Enantiomorphism	Axial second-rank tensor (optical activity = optical gyration)
Triclinic	$\bar{1}$	1	+ (3) $[uvw]$	+ (18)	+	+ (6)
Monoclinic	$2/m$	2 m	+ (1) $[010]^\dagger$ + (2) $[u0w]^\dagger$	+ (8) + (10)	+ —	+ (4) + (2)
Orthorhombic	$2/m 2/m 2/m$	222 $mm2$	— + (1) $[001]$	+ (3) + (5)	+ —	+ (3) + (1)
Tetragonal	$4/m$	4 $\bar{4}$	+ (1) $[001]$ —	+ (4) + (4)	+ —	+ (2) + (2)
	$4/m 2/m 2/m$	422 $4mm$ $\bar{4}2m$	— + (1) $[001]$ —	+ (1) + (3) + (2)	+ — —	+ (2) — + (1)
Trigonal	$\bar{3}$	3	+ (1) $[001]^\ddagger$	+ (6)	+	+ (2)
	$\bar{3} 2/m$	32 $3m$	— + (1) $[001]^\ddagger$	+ (2) + (4)	+ —	+ (2) —
Hexagonal	$6/m$	6 $\bar{6}$	+ (1) $[001]$ —	+ (4) + (2)	+ —	+ (2) —
	$6/m 2/m 2/m$	622 $6mm$ $\bar{6}2m$	— + (1) $[001]$ —	+ (1) + (3) + (1)	+ — —	+ (2) — —
Cubic	$2/m \bar{3}$	23	—	+ (1)	+	+ (1)
	$4/m \bar{3} 2/m$	432 $\bar{4}3m$	— —	— + (1)	+ —	+ (1) —
Icosahedral	$m \bar{3} \bar{5} (\bar{5} \bar{3} m)$	235 (532)	—	—	+	+ (1)
Spherical	$m \bar{\infty} (\infty \infty m)$	2 ∞ ($\infty \infty$)	—	—	+	+ (1)

† Unique axis b . ‡ Hexagonal axes.

Uniaxial stress parallel to $[111]$:

$$\begin{aligned} 2/m \bar{3} \cap \infty/m 2/m: & \quad \mathcal{P}_F = \bar{3} \quad \text{along } [111] \\ 4/m \bar{3} 2/m \cap \infty/m 2/m: & \quad \mathcal{P}_F = \bar{3} 2/m \quad \text{along } [111]. \end{aligned}$$

Note that, in contrast to uniaxial stress, the electric field destroys centrosymmetry, leading to pyroelectricity and even optical gyration. The polarity and the gyration sense are reversed upon reversal of the electric field. If the electric field and the uniaxial stress were applied to an arbitrary (non-symmetry) direction of the above cubic crystals, point groups 1 and $\bar{1}$ would result in the two cases.

If a scalar influence $\mathcal{R} = 2/m \bar{\infty} (\infty \infty m)$ (centrosymmetric sphere group) is applied to a crystal, its symmetry $\mathcal{P}_F = \mathcal{P}$ is not changed, provided that no phase transition is induced.

For further reading on tensor properties of crystals the textbooks in the references and *International Tables for Crystallography*, Volume D (2014) are recommended.

3.2.2.1.3. Enantiomorphism, enantiomerism, chirality, dissymmetry

All these terms refer to the same symmetry restriction, the absence of improper rotations (rotoinversions, roto reflections) in a crystal or in a molecule. This implies in particular the absence of a centre of symmetry, $\bar{1}$, and of a mirror plane, $m = \bar{2}$, but also of a $\bar{4}$ axis. As a consequence, such *chiral* crystals or molecules can

occur in two different forms, which are related as a right and a left hand; hence, they are called right-handed and left-handed forms. These two forms of a molecule or a crystal are mirror-related and not superimposable (not congruent). Thus, the only symmetry operations that are allowed for *chiral* objects are proper rotations. Such objects are also called *dissymmetric*, in contrast to *asymmetric* objects, which have no symmetry.

The terms *enantiomerism* and *chirality* are mainly used in chemistry and applied to molecules, whereas the term *enantiomorphism* is preferred in crystallography if reference is made to crystals.

Enantiomorphic crystals, as well as solutions or melts containing chiral molecules of one handedness, exhibit optical activity (*cf.* Section 3.2.2.4.2). Crystals and molecules of the other handedness show optical activity with the opposite sense of rotation. For this reason, two molecules of opposite chirality are also called *optical isomers*.

Chiral molecules form enantiomorphic crystals of the corresponding handedness. These crystals belong, therefore, to one of the 11 crystal classes allowing enantiomorphism (Table 3.2.2.1). Racemic mixtures (containing equal amounts of molecules of each chirality), however, may crystallize in non-enantiomorphic or even centrosymmetric crystal classes. Racemization (*i.e.* the switching of molecules from one chirality to the other) of an optically active melt or solution can occur in some cases during crystallization, leading to non-enantiomorphic crystals.

3.2. POINT GROUPS AND CRYSTAL CLASSES

Table 3.2.2.2

Polar axes and nonpolar directions in the 21 noncentrosymmetric crystal classes

All directions normal to an evenfold rotation axis and along rotoinversion axes are nonpolar. All directions other than those in the column 'Nonpolar directions' are polar. A symbol like $[u0w]$ refers to the set of directions obtained for all possible values of u and w , in this case to all directions normal to the b axis, *i.e.* parallel to the plane (010). Symmetry-equivalent sets of nonpolar directions are placed between semicolons; the sequence of these sets follows the sequence of the symmetry directions in Table 2.1.3.1.

System	Class	Polar (symmetry) axes	Nonpolar directions
Triclinic	1	None†	None
Monoclinic Unique axis b	2 m	[010] None†	$[u0w]$ [010]
Monoclinic Unique axis c	2 m	[001] None†	$[uv0]$ [001]
Orthorhombic	222 $mm2$	None [001]	$[0vw]$; $[u0w]$; $[uv0]$ $[uv0]$
Tetragonal	4	[001]	$[uv0]$
	$\bar{4}$	None	[001]; $[uv0]$
	422	None	$[uv0]$; $[0vw]$ $[u0w]$; $[uuw]$ $[u\bar{u}w]$
	4mm	[001]	$[uv0]$
	42m	None	$[uv0]$; $[0vw]$ $[u0w]$
	4m2	None	$[uv0]$; $[uuw]$ $[u\bar{u}w]$
Trigonal (Hexagonal axes)	3	[001]	None
	321	[100], [010], $[\bar{1}\bar{1}0]$	$[u2uw]$ $[\bar{2}u\bar{u}w]$ $[u\bar{u}w]$
	312	$[\bar{1}\bar{1}0]$, [120], $[\bar{2}\bar{1}0]$	$[uuw]$ $[\bar{u}0w]$ $[0\bar{v}w]$
	3m1	[001]	[100] [010] $[\bar{1}\bar{1}0]$
	31m	[001]	$[\bar{1}\bar{1}0]$ [120] $[\bar{2}\bar{1}0]$
Trigonal (Rhombohedral axes)	3	[111]	None
	32	$[\bar{1}\bar{1}0]$, $[01\bar{1}]$, $[\bar{1}01]$	$[uuw]$ $[uvv]$ $[uvu]$
	3m	[111]	$[\bar{1}\bar{1}0]$ $[01\bar{1}]$ $[\bar{1}01]$
Hexagonal	6	[001]	$[uv0]$
	$\bar{6}$	None	[001]
	622	None	$[u2uw]$ $[\bar{2}u\bar{u}w]$ $[u\bar{u}w]$ $[uuw]$ $[\bar{u}0w]$ $[0\bar{v}w]$
	6mm	[001]	$[uv0]$
	$\bar{6}m2$ $\bar{6}2m$	$[\bar{1}\bar{1}0]$, [120], $[\bar{2}\bar{1}0]$ [100], [010], $[\bar{1}\bar{1}0]$	$[uuw]$ $[\bar{u}0w]$ $[0\bar{v}w]$ $[u2uw]$ $[\bar{2}u\bar{u}w]$ $[u\bar{u}w]$
Cubic	23 $\bar{4}3m$ }	Four threefold axes along $\langle 111 \rangle$	$[0vw]$ $[u0w]$ $[uv0]$ $[0vw]$ $[u0w]$ $[uv0]$
	432	None	$[0vw]$ $[u0w]$ $[uv0]$; $[uuw]$ $[uvv]$ $[uvu]$; $[u\bar{u}w]$ $[uv\bar{v}]$ $[\bar{u}vu]$

† In class 1 any direction is polar; in class m all directions except [010] (or [001]) are polar.

Enantiomorphic crystals can also be built up from achiral molecules or atom groups. In these cases, the achiral molecules or atom groups form chiral configurations in the structure. The best known examples are quartz and NaClO_3 . For details, reference should be made to Rogers (1975).

3.2.2.1.4. Polar directions, polar axes, polar point groups

A *direction* is called *polar* if its two directional senses are geometrically or physically different. A *polar symmetry direction* of a crystal is called a *polar axis*. Only proper rotation or screw axes can be polar. The polar and nonpolar directions in the 21 noncentrosymmetric point groups are listed in Table 3.2.2.2.

The terms *polar point group* or *polar crystal class* are used in two different meanings. In crystal physics, a crystal class is

considered as polar if it allows the existence of a permanent dipole moment, *i.e.* if it is capable of pyroelectricity (*cf.* Section 3.2.2.5). In crystallography, however, the term *polar crystal class* is frequently used synonymously with *noncentrosymmetric crystal class*. The synonymous use of polar and acentric, however, can be misleading, as is shown by the following example. Consider an optically active liquid. Its symmetry can be represented as a right-handed or a left-handed sphere (*cf.* Sections 3.2.1.4 and 3.2.2.4). The optical activity is isotropic, *i.e.* magnitude and rotation sense are the same in any direction and its counterdirection. Thus, no polar direction exists, although the liquid is noncentrosymmetric with respect to optical activity.

According to Neumann's principle, information about the point group of a crystal may be obtained by the observation of physical effects. Here, the term 'physical properties' includes crystal morphology and etch figures. The use of any of the techniques described below does not necessarily result in the complete determination of symmetry but, when used in conjunction with other methods, much information may be obtained. It is important to realize that the evidence from these methods is often negative, *i.e.* that symmetry conclusions drawn from such evidence must be considered as only provisional.

In the following sections, the physical properties suitable for the determination of symmetry are outlined briefly. For more details, reference should be made to the monographs by Bhagavantam (1966), Nye (1957) and Wooster (1973).

3.2.2.2. Morphology

If a crystal shows well developed faces, information on its symmetry may be derived from the external form of the crystal. By means of the optical goniometer, faces related by symmetry can be determined even for crystals far below 1 mm in diameter. By this procedure, however, only the eigensymmetry (*cf.* Section 3.2.1.2.2) of the crystal morphology (which may consist of a single form or a combination of forms) can be established. The determination of the point group is unique in all cases where the observed eigensymmetry group is compatible with only one generating group.

Column 6 in Table 3.2.1.3 lists all point groups for which a given crystal form (characterized by its name and eigensymmetry) can occur. In 19 cases, the point group can be uniquely determined because only one entry appears in column 6. These crystal forms are always characteristic general forms, for which eigensymmetry and generating point-group symmetry are identical. They belong to the 19 point groups with more than one symmetry direction.

If a crystal exhibits a combination of forms which by themselves do not permit unambiguous determination of the point group, those generating point groups are possible that are common to all crystal forms of the combination. The mutual orientation of the forms, if variable, has to be taken into account, too.

Example

Two tetragonal pyramids, each of eigensymmetry $4mm$, rotated with respect to each other by an angle $\neq 45^\circ$, determine the point group 4 uniquely because the eigensymmetry of the *combination* is only 4.

In practice, however, unequal or incomplete development of the faces of a form often simulates a symmetry that is lower than the actual crystal symmetry. In such cases, or when the

3. ADVANCED TOPICS ON SPACE-GROUP SYMMETRY

morphological analysis is ambiguous, the crystallization of a small amount of the compound on a seed crystal, ground to a sphere, is useful. By this procedure, faces of additional forms (and often of the characteristic general form) appear as small facets on the sphere and their interfacial angles can be measured.

In favourable cases, even the space group can be derived from the morphology of a crystal: this is based on the so-called *Bravais–Donnay–Harker principle*. A textbook description is given by Phillips (1971, ch. 13).

Furthermore, measurements of the interfacial angles by means of the optical goniometer permit the determination of the relative dimensions of a ‘morphological unit cell’ with good accuracy. Thus, for instance, the interaxial angles α , β , γ and the axial ratio $a:b:c$ of a triclinic crystal may be derived. The ratio $a:b:c$, however, may contain an uncertainty by an integral factor with respect to the actual cell edges of the crystal. This means that any one unit length may have to be multiplied by an integer in order to obtain correspondence to the ‘structural’ unit cell.

3.2.2.3. Etch figures

Additional information on the point group of a crystal can be gained from the face symmetry, which is usually determined by observation of etch figures, striations and other face markings. Etch pits are produced by heating the crystal in vacuum (evaporation from the surface) or by attacking it with an appropriate reagent, which should not be optically active. The etch pits generally appear at the end points of dislocation lines on the face. They exhibit the symmetry of one of the ten two-dimensional point groups which, in general,²² corresponds to the symmetry of the crystal face under investigation.

The observation of etch figures is important when the morphological analysis is ambiguous (*cf.* Section 3.2.2.2). For instance, a tetragonal pyramid, which is compatible with point groups 4 and $4mm$, can be uniquely attributed to point group 4 if its face symmetry is found to be 1. For face symmetry m , group $4mm$ would result. The (oriented) face symmetries of the 47 crystal forms in the various point groups are listed in column 6 of Table 3.2.1.3 and in column 3 of Table 3.2.3.2.

In noncentrosymmetric crystals, the etch pits on parallel but opposite faces, even though they have the same symmetry, may be of different size or shape, thus proving the absence of a symmetry centre. Note that the orientation of etch pits with respect to the edges of the face is significant (*cf.* Buerger, 1956), as well as the mutual arrangement of etch pits on opposite faces. Thus, for a pinacoid with face symmetry 1, the possible point groups $\bar{1}$, 2 and m of the crystal can be distinguished by the mutual orientation of etch pits on the two faces. Moreover, twinning by merohedry and the true symmetry of the two (or more) twin partners (‘twin domains’) may be detected.

The method of etching can be applied not only to growth faces but also to cleavage faces or arbitrarily cut faces.

3.2.2.4. Optical properties

Optical studies provide good facilities with which to determine the symmetry of transparent crystals. The following optical properties may be used.

²² It should be noted, however, that asymmetric etch figures may occur that are due, for example, to an inclination of dislocation lines against the surface.

Table 3.2.2.3

Categories of crystal systems distinguished according to the different forms of the indicatrix

Crystal system	Shape of indicatrix	Optical character
Cubic	Sphere	Isotropic (not doubly refracting)
Tetragonal } Trigonal } Hexagonal }	Rotation ellipsoid	Uniaxial } Anisotropic (doubly refracting)
Orthorhombic } Monoclinic } Triclinic }	General ellipsoid	Biaxial }

3.2.2.4.1. Refraction

The dependence of the *refractive index* on the vibration direction of a plane-polarized light wave travelling through the crystal can be obtained from the optical indicatrix. This surface is an ellipsoid, which can degenerate into a rotation ellipsoid or even into a sphere. Thus, the lowest symmetry of the property ‘refraction’ is $2/m\ 2/m\ 2/m$, the point group of the general ellipsoid. According to the three different forms of the indicatrix, three categories of crystal systems have to be distinguished (Table 3.2.2.3).

The orientation of the indicatrix is related to the symmetry directions of the crystal. In tetragonal, trigonal and hexagonal crystals, the rotation axis of the indicatrix (which is the unique optic axis) is parallel to the main symmetry axis. For orthorhombic crystals, the three principal axes of the indicatrix are oriented parallel to the three symmetry directions of the crystal. In the monoclinic system, one of the axes of the indicatrix coincides with the monoclinic symmetry direction, whereas in the triclinic case, the indicatrix can, in principle, have any orientation relative to a chosen reference system. Thus, in triclinic and, with restrictions, in monoclinic crystals, the *orientation* of the indicatrix can change with wavelength λ and temperature T (orientation dispersion). In any system, the *size* of the indicatrix and, in all but the cubic system, its *shape* can also vary with λ and T .

When studying the symmetry of a crystal by optical means, note that strain can lower the apparent symmetry owing to the high sensitivity of optical properties to strain.

3.2.2.4.2. Optical activity

The symmetry information obtained from *optical activity* is quite different from that given by optical refraction. Optical activity is in principle confined to the 21 noncentrosymmetric classes but it can occur in only 15 of them (Table 3.2.2.1). In the 11 enantiomorphism classes, a single crystal is either right- or left-handed. In the four non-enantiomorphous classes m , $mm2$, $\bar{4}$ and $\bar{4}2m$, optical activity may also occur; here directions of both right- and left-handed rotations of the plane of polarization exist in the same crystal. In the other six noncentrosymmetric classes, $3m$, $4mm$, $\bar{6}$, $6mm$, $\bar{6}2m$, $\bar{4}3m$, optical activity is not possible.

In the two cubic enantiomorphous classes 23 and $\bar{4}32$, the optical activity is isotropic and can be observed along any direction.²³ For the other optically active crystals, the rotation of the plane of polarization can, in practice, be detected only in directions parallel (or approximately parallel) to the optic axes. This is because of the dominating effect of double refraction. No optical activity, however, is present along an inversion axis or along a direction parallel or perpendicular to a

²³ This property can be represented by enantiomorphic spheres of point group 2∞ , *cf.* Table 3.2.1.6.

3.2. POINT GROUPS AND CRYSTAL CLASSES

mirror plane. Thus, no activity occurs along the optic axis in crystal classes $\bar{4}$ and $\bar{4}2m$. In classes m and $mm2$, no activity can be present along the two optic axes if these axes lie in m . If they are not parallel to m , they show optical rotation(s) of opposite sense.

3.2.2.4.3. Second-harmonic generation (SHG)

Light waves passing through a noncentrosymmetric crystal induce new waves of twice the incident frequency. This second-harmonic generation is due to the nonlinear optical susceptibility. The *second-harmonic coefficients* form a third-rank tensor, which is subject to the same symmetry constraints as the piezoelectric tensor (see Section 3.2.2.6). Thus, only 20 noncentrosymmetric crystals, except those of class 432, can show the second-harmonic effect; cf. Table 3.2.2.1.

Second-harmonic generation is a powerful method of testing crystalline materials for the absence of a symmetry centre. With an appropriate experimental device, very small amounts (less than 10 mg) of powder are sufficient to detect the second-harmonic signals, even for crystals with small deviations from centrosymmetry (Dougherty & Kurtz, 1976).

3.2.2.5. Pyroelectricity and ferroelectricity

In principle, *pyroelectricity* can only exist in crystals with a permanent electric dipole moment. This moment is changed by heating and cooling, thus giving rise to electric charges on certain crystal faces, which can be detected by simple experimental procedures.

An electric dipole moment can be present only along a polar direction that has no symmetry-equivalent directions.²⁴ Such polar directions occur in the following ten classes: $6mm$, $4mm$, and their subgroups 6 , 4 , $3m$, 3 , $mm2$, 2 , m , 1 (cf. Table 3.2.2.1). In point groups with a rotation axis, the electric moment is along this axis. In class m , the electric moment is parallel to any direction in the mirror plane (direction $[u0w]$). In class 1 , any direction $[uvw]$ is possible. In point groups 1 and m , besides a change in magnitude, a directional variation of the electric moment can also occur during heating or cooling.

In practice, it is difficult to prevent strains from developing throughout the crystal as a result of temperature gradients in the sample. This gives rise to piezoelectrically induced charges superposed on the true pyroelectric effect. Consequently, when the development of electric charges by a change in temperature is observed, the only safe deduction is that the specimen must lack a centre of symmetry. Failure to detect pyroelectricity may be due to extreme weakness of the effect, although modern methods are very sensitive.

A crystal is *ferroelectric* if the direction of the permanent electric dipole moment can be changed by an electric field. Thus, ferroelectricity can only occur in the ten pyroelectric crystal classes, mentioned above.

3.2.2.6. Piezoelectricity

In piezoelectric crystals, an electric dipole moment can be induced by compressional and torsional stress. For a uniaxial compression, the induced moment may be parallel, normal or inclined to the compression axis. These cases are called longitudinal, transverse or mixed compressional piezoeffect, respectively. Correspondingly, for torsional stress, the electric moment may be parallel, normal or inclined to the torsion axis.

The *piezoelectricity* is described by a third-rank tensor, the moduli of which vanish for all centrosymmetric point groups. Additionally, in class 432, all piezoelectric moduli are zero owing to the high symmetry. Thus, piezoelectricity can only occur in 20 noncentrosymmetric crystal classes (Table 3.2.2.1).

The piezoelectric point groups 422 and 622 show the following peculiarity: there is no direction for which a longitudinal component of the electric moment is induced under uniaxial compression. Thus, no longitudinal or mixed compressional effects occur. The moment is always normal to the compression axis (pure transverse compressional effect). This means that, with the compression pistons as electrodes, no electric charges can be found, since only transverse compressional or torsional piezoeffects occur. In all other piezoelectric classes, there exist directions in which both longitudinal and transverse components of the electric dipole moment are induced under uniaxial compression.

An electric moment can also develop under hydrostatic pressure. This kind of piezoelectricity, like pyroelectricity, can be represented by a first-rank tensor (vector), whereby the hydrostatic pressure is regarded as a scalar. Thus, piezoelectricity under hydrostatic pressure is subject to the same symmetry constraints as pyroelectricity.

Like 'second-harmonic generation' (Section 3.2.2.4.3), the piezoelectric effect is very useful for testing crystals for the absence of a symmetry centre. There exist powerful methods for testing powder samples or even small single crystals. In the old technique of Giebe & Scheibe (cf. Wooster & Brenton, 1970), the absorption and emission of radio-frequency energy by electromechanical oscillations of piezoelectric particles are detected. In the more modern method of observing 'polarization echoes', radio-frequency pulses are applied to powder samples. By this procedure, electromechanical vibration pulses are induced in piezoelectric particles, the echoes of which can be detected (cf. Melcher & Shiren, 1976).

²⁴ In the literature, the requirement for pyroelectricity is frequently expressed as 'a unique (or singular) polar axis'. This term, however, is misleading for point groups 1 and m .

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3.2.3. Tables of the crystallographic point-group types

BY H. KLAPPER, TH. HAHN AND M. I. AROYO

The crystallographic point-group types are listed in Tables 3.2.3.1 and 3.2.3.2 for two-dimensional and for three-dimensional space, respectively. No listings are presented for the noncrystallographic point-group types (*i.e.* having axes of orders other than 1, 2, 3, 4 and 6), but their symbols can be found in Tables 3.2.1.5, 3.2.1.6 and 3.2.3.3 (*cf.* Section 3.2.1.4 for a review of noncrystallographic point groups). The two icosahedral point groups 235 and $m\bar{3}5$ are treated in detail in Section 3.2.1.4.2, while their crystallographic data are shown in Table 3.2.3.3.

There is a physical difference between the point groups of crystals and those of molecules. For a molecule, the point group is the set of all symmetry operations that map its atoms onto one another. Macroscopic crystals, however, hardly ever exhibit their ideal symmetry because of the defects that occur during crystal

growth. For a crystal, its point group is the set of all symmetry operations that map the set of the vectors normal to the crystal faces onto one another; it does not operate in point space, but in vector space.

In Tables 3.2.3.1 and 3.2.3.2 the point-group types are presented for crystals as well as for molecules. However, parts of the tables concern either crystals only or molecules only. The names of crystal forms in the fifth column (in roman type) and the hkl face indices in the last column are only relevant for crystals. The names of the point forms in the second line of each pair of entries in the fifth column (given in italics) and the other data (multiplicities, Wyckoff letters, site symmetries and sets of symmetry-equivalent coordinates) concern both crystals and molecules. However, for point groups with an origin fixed by symmetry, the Wyckoff position with the Wyckoff letter o is only of interest for molecules.

Because the meanings of the entries are not identical for crystals and for molecules, they are not explained here, but in Sections 3.2.1 (for crystals) and 3.2.4 (for molecules).

(continued on page 772)

Table 3.2.3.1

The ten two-dimensional crystallographic point groups

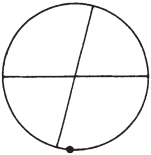
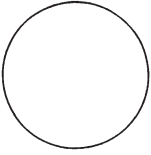
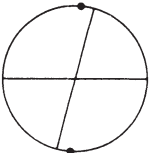
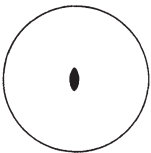
The point groups are listed in blocks according to crystal system and are specified by their Hermann–Mauguin symbols. For each point group, the stereographic projections show (on the left) the general position and (on the right) the symmetry elements.

The list of Wyckoff positions includes:

Columns 1 to 4: multiplicity, Wyckoff letter, oriented site-symmetry symbol, coordinate doublets;

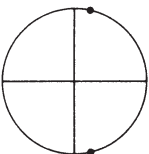
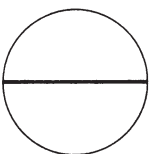
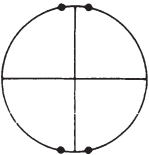
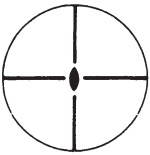
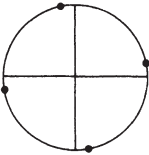
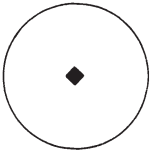
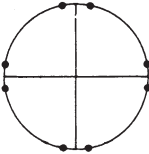
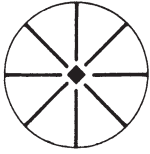
Under the stereographic projections: edge forms (in roman type) and point forms (in italics); if there are two entries, the second entry refers to a limiting (noncharacteristic) form;

Last column: Miller indices of equivalent edges [for hexagonal groups, Bravais–Miller indices (hki) are used].

OBLIQUE SYSTEM						
1						
1	a	1	x, y		Single edge <i>Single point</i>	(hk)
2						
2	a	1	$x, y \quad \bar{x}, \bar{y}$		Pair of parallel edges <i>Line segment through origin</i>	$(hk) \quad (\bar{h}\bar{k})$
1	o	2	$0, 0$		Point in origin	

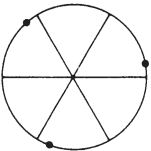
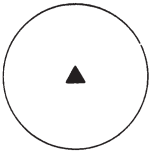
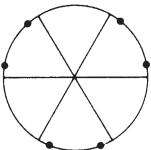
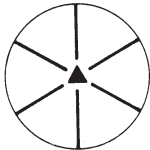
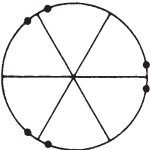
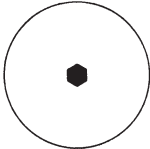
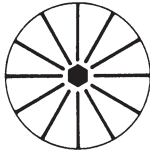
3.2. POINT GROUPS AND CRYSTAL CLASSES

Table 3.2.3.1 (continued)

RECTANGULAR SYSTEM				
<i>m</i>				
				 
2	<i>b</i>	1	$x, y \quad \bar{x}, y$	Pair of edges <i>Line segment</i> $(hk) \quad (\bar{h}k)$
				Pair of parallel edges <i>Line segment through origin</i> $(10) \quad (\bar{1}0)$
1	<i>a</i>	<i>.m.</i>	$0, y$	Single edge <i>Single point</i> $(01) \text{ or } (0\bar{1})$
<i>2mm</i>				
				 
4	<i>c</i>	1	$x, y \quad \bar{x}, \bar{y} \quad \bar{x}, y \quad x, \bar{y}$	Rhomb <i>Rectangle</i> $(hk) \quad (\bar{h}\bar{k}) \quad (\bar{h}k) \quad (h\bar{k})$
2	<i>b</i>	<i>.m.</i>	$0, y \quad 0, \bar{y}$	Pair of parallel edges <i>Line segment through origin</i> $(01) \quad (0\bar{1})$
2	<i>a</i>	<i>..m</i>	$x, 0 \quad \bar{x}, 0$	Pair of parallel edges <i>Line segment through origin</i> $(10) \quad (\bar{1}0)$
1	<i>o</i>	<i>2mm</i>	$0, 0$	<i>Point in origin</i>
SQUARE SYSTEM				
4				
				 
4	<i>a</i>	1	$x, y \quad \bar{x}, \bar{y} \quad \bar{y}, x \quad y, \bar{x}$	Square <i>Square</i> $(hk) \quad (\bar{h}\bar{k}) \quad (\bar{k}h) \quad (k\bar{h})$
1	<i>o</i>	<i>4..</i>	$0, 0$	<i>Point in origin</i>
<i>4mm</i>				
				 
8	<i>c</i>	1	$x, y \quad \bar{x}, \bar{y} \quad \bar{y}, x \quad y, \bar{x}$ $\bar{x}, y \quad x, \bar{y} \quad y, x \quad \bar{y}, \bar{x}$	Ditetragon <i>Truncated square</i> $(hk) \quad (\bar{h}\bar{k}) \quad (\bar{k}h) \quad (k\bar{h})$ $(\bar{h}k) \quad (h\bar{k}) \quad (kh) \quad (k\bar{h})$
4	<i>b</i>	<i>..m</i>	$x, x \quad \bar{x}, \bar{x} \quad \bar{x}, x \quad x, \bar{x}$	Square <i>Square</i> $(11) \quad (\bar{1}\bar{1}) \quad (\bar{1}1) \quad (1\bar{1})$
4	<i>a</i>	<i>.m.</i>	$x, 0 \quad \bar{x}, 0 \quad 0, x \quad 0, \bar{x}$	Square <i>Square</i> $(10) \quad (\bar{1}0) \quad (01) \quad (0\bar{1})$
1	<i>o</i>	<i>4mm</i>	$0, 0$	<i>Point in origin</i>

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Table 3.2.3.1 (continued)

HEXAGONAL SYSTEM						
3						
3	<i>a</i>	1	x, y $\bar{y}, x - y$ $\bar{x} + y, \bar{x}$		Trigon Trigon	(hki) (ihk) (kih)
1	<i>o</i>	3..	0, 0		Point in origin	
3m1						
6	<i>b</i>	1	x, y $\bar{y}, x - y$ $\bar{x} + y, \bar{x}$ $\bar{y}, \bar{x}$ $\bar{x} + y, y$ $x, x - y$		Ditrigon Truncated trigon	(hki) (ihk) (kih) $(\bar{h}\bar{k}\bar{i})$ $(\bar{i}\bar{h}\bar{k})$ $(\bar{k}\bar{i}\bar{h})$
					Hexagon Hexagon	$(11\bar{2})$ $(\bar{2}11)$ $(\bar{1}\bar{2}1)$ $(\bar{1}\bar{1}2)$ $(2\bar{1}\bar{1})$ $(\bar{1}\bar{2}\bar{1})$
3	<i>a</i>	. <i>m</i> .	$x, \bar{x}$ $x, 2x$ $2\bar{x}, \bar{x}$		Trigon Trigon	$(10\bar{1})$ $(\bar{1}10)$ $(0\bar{1}1)$ or $(\bar{1}01)$ $(1\bar{1}0)$ $(01\bar{1})$
1	<i>o</i>	3 <i>m</i> .	0, 0		Point in origin	
31m						
6	<i>b</i>	1	x, y $\bar{y}, x - y$ $\bar{x} + y, \bar{x}$ y, x $x - y, \bar{y}$ $\bar{x}, \bar{x} + y$		Ditrigon Truncated trigon	(hki) (ihk) (kih) (khi) (ikh) (hik)
					Hexagon Hexagon	$(10\bar{1})$ $(\bar{1}10)$ $(0\bar{1}1)$ $(01\bar{1})$ $(\bar{1}01)$ $(1\bar{1}0)$
3	<i>a</i>	.. <i>m</i>	$x, 0$ $0, x$ $\bar{x}, \bar{x}$		Trigon Trigon	$(11\bar{2})$ $(\bar{2}11)$ $(\bar{1}\bar{2}1)$ or $(\bar{1}\bar{1}2)$ $(2\bar{1}\bar{1})$ $(\bar{1}\bar{2}\bar{1})$
1	<i>o</i>	3. <i>m</i>	0, 0		Point in origin	
6						
6	<i>a</i>	1	x, y $\bar{y}, x - y$ $\bar{x} + y, \bar{x}$ $\bar{x}, \bar{y}$ $y, \bar{x} + y$ $x - y, x$		Hexagon Hexagon	(hki) (ihk) (kih) $(\bar{h}\bar{k}\bar{i})$ $(\bar{i}\bar{h}\bar{k})$ $(\bar{k}\bar{i}\bar{h})$
1	<i>o</i>	6..	0, 0		Point in origin	
6mm						
12	<i>c</i>	1	x, y $\bar{y}, x - y$ $\bar{x} + y, \bar{x}$ $\bar{x}, \bar{y}$ $y, \bar{x} + y$ $x - y, x$ $\bar{y}, \bar{x}$ $\bar{x} + y, y$ $x, x - y$ y, x $x - y, \bar{y}$ $\bar{x}, \bar{x} + y$		Dihexagon Truncated hexagon	(hki) (ihk) (kih) $(\bar{h}\bar{k}\bar{i})$ $(\bar{i}\bar{h}\bar{k})$ $(\bar{k}\bar{i}\bar{h})$ $(\bar{k}\bar{h}\bar{i})$ $(\bar{i}\bar{k}\bar{h})$ $(\bar{h}\bar{i}\bar{k})$ (khi) (ikh) (hik)
6	<i>b</i>	. <i>m</i> .	$x, \bar{x}$ $x, 2x$ $2\bar{x}, \bar{x}$ $\bar{x}, x$ $\bar{x}, 2\bar{x}$ $2x, x$		Hexagon Hexagon	$(10\bar{1})$ $(\bar{1}10)$ $(0\bar{1}1)$ $(\bar{1}01)$ $(1\bar{1}0)$ $(01\bar{1})$
6	<i>a</i>	.. <i>m</i>	$x, 0$ $0, x$ $\bar{x}, \bar{x}$ $\bar{x}, 0$ $0, \bar{x}$ x, x		Hexagon Hexagon	$(11\bar{2})$ $(\bar{2}11)$ $(\bar{1}\bar{2}1)$ $(\bar{1}\bar{1}2)$ $(2\bar{1}\bar{1})$ $(\bar{1}\bar{2}\bar{1})$
1	<i>o</i>	6 <i>mm</i>	0, 0		Point in origin	

3.2. POINT GROUPS AND CRYSTAL CLASSES

Table 3.2.3.2

The 32 three-dimensional crystallographic point groups

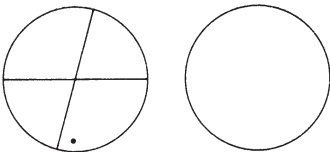
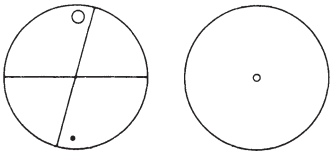
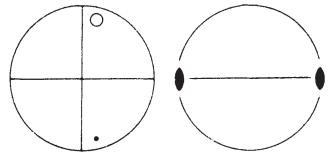
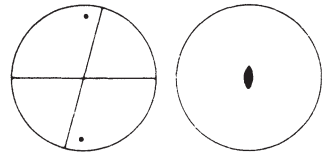
The point groups are listed in blocks according to crystal system and are specified by their short and (if different) full Hermann–Mauguin symbols and their Schoenflies symbols. For each point group, the stereographic projections show (on the left) the general position and (on the right) the symmetry elements.

The list of Wyckoff positions includes:

Columns 1 to 4: multiplicity, Wyckoff letter, oriented site-symmetry symbol, coordinate triplets;

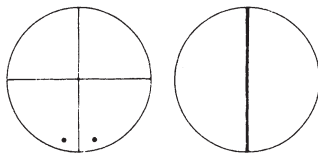
Under the stereographic projections: face forms (in roman type) and point forms (in italics); if there is more than one entry, subsequent entries refer to limiting (noncharacteristic) forms;

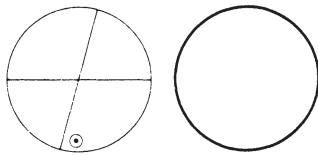
Last column: Miller indices of equivalent faces [for trigonal and hexagonal groups, Bravais–Miller indices (*hkl*) are used if referred to hexagonal axes].

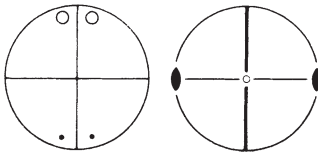
TRICLINIC SYSTEM			
1	C_1		
1	<i>a</i>	1	x, y, z
			Pedion or monohedron <i>Single point</i>
			Symmetry of special projections Along any direction 1
$\bar{1}$	C_i		
2	<i>a</i>	1	$x, y, z \quad \bar{x}, \bar{y}, \bar{z}$
			Pinacoid or parallelohedron <i>Line segment through origin</i>
1	<i>o</i>	$\bar{1}$	0, 0, 0
			Point in origin
			Symmetry of special projections Along any direction 2
MONOCLINIC SYSTEM			
2	C_2		
UNIQUE AXIS <i>b</i>			
2	<i>b</i>	1	$x, y, z \quad \bar{x}, \bar{y}, \bar{z}$
			Sphenoid or dihedron <i>Line segment</i>
			Pinacoid or parallelohedron <i>Line segment through origin</i>
1	<i>a</i>	2	0, <i>y</i> , 0
			Pedion or monohedron <i>Single point</i>
			Symmetry of special projections Along [100] Along [010] Along [001] <i>m</i> 2 <i>m</i>
2	C_2		
UNIQUE AXIS <i>c</i>			
2	<i>b</i>	1	$x, y, z \quad \bar{x}, \bar{y}, \bar{z}$
			Sphenoid or dihedron <i>Line segment</i>
			Pinacoid or parallelohedron <i>Line segment through origin</i>
1	<i>a</i>	2	0, 0, <i>z</i>
			Pedion or monohedron <i>Single point</i>
			Symmetry of special projections Along [100] Along [010] Along [001] <i>m</i> <i>m</i> 2

3. ADVANCED TOPICS ON SPACE-GROUP SYMMETRY

Table 3.2.3.2 (continued)

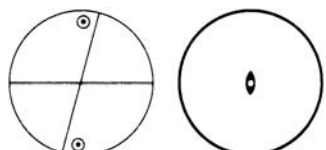
MONOCLINIC SYSTEM (<i>cont.</i>)												
m		C_s										
UNIQUE AXIS b												
2	b	1	x, y, z	$x, \bar{y}, z$	Dome or dihedron		(hkl)	$(h\bar{k}l)$				
					Line segment							
					Pinacoid or parallelohedron		(010)	$(0\bar{1}0)$				
					Line segment through origin							
1	a	m	$x, 0, z$	Pedion or monohedron		$(h0l)$						
					Single point							
Symmetry of special projections												
Along $[100]$		Along $[010]$		Along $[001]$								
m		1		m								

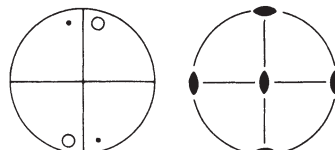
m		C_s										
UNIQUE AXIS c												
2	b	1	x, y, z	$x, y, \bar{z}$	Dome or dihedron		(hkl)	$(hk\bar{l})$				
					Line segment							
					Pinacoid or parallelohedron		(001)	$(00\bar{1})$				
					Line segment through origin							
1	a	m	$x, y, 0$	Pedion or monohedron		$(hk0)$						
					Single point							
Symmetry of special projections												
Along $[100]$		Along $[010]$		Along $[001]$								
m		m		1								

$2/m$		C_{2h}										
UNIQUE AXIS b												
4	c	1	x, y, z	$\bar{x}, y, \bar{z}$	$\bar{x}, \bar{y}, \bar{z}$	$x, \bar{y}, z$	Rhombic prism		(hkl) $(\bar{h}k\bar{l})$ $(\bar{h}\bar{k}\bar{l})$ $(h\bar{k}l)$			
					Rectangle through origin							
2	b	m	$x, 0, z$	$\bar{x}, 0, \bar{z}$	Pinacoid or parallelohedron		$(h0l)$	$(\bar{h}0\bar{l})$				
					Line segment through origin							
2	a	2	$0, y, 0$	$0, \bar{y}, 0$	Pinacoid or parallelohedron		(010)	$(0\bar{1}0)$				
					Line segment through origin							
1	o	$2/m$	$0, 0, 0$	Point in origin								
Symmetry of special projections												
Along $[100]$		Along $[010]$		Along $[001]$								
$2mm$		2		$2mm$								

3.2. POINT GROUPS AND CRYSTAL CLASSES

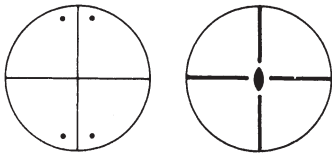
Table 3.2.3.2 (continued)

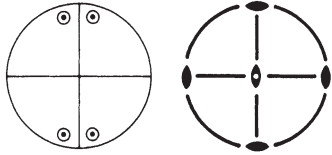
MONOCLINIC SYSTEM (cont.)									
$2/m$		C_{2h}							
UNIQUE AXIS c									
4	c	1	x, y, z $\bar{x}, \bar{y}, z$ $\bar{x}, \bar{y}, \bar{z}$ $x, y, \bar{z}$	Rhombic prism <i>Rectangle through origin</i>		(hkl) $(\bar{h}\bar{k}l)$ $(\bar{h}\bar{k}\bar{l})$ $(h\bar{k}\bar{l})$			
2	b	m	$x, y, 0$ $\bar{x}, \bar{y}, 0$	Pinacoid or parallelohedron <i>Line segment through origin</i>		$(hk0)$ $(\bar{h}\bar{k}0)$			
2	a	2	$0, 0, z$ $0, 0, \bar{z}$	Pinacoid or parallelohedron <i>Line segment through origin</i>		(001) $(00\bar{1})$			
1	o	$2/m$	$0, 0, 0$	Point in origin					
Symmetry of special projections									
Along $[100]$		Along $[010]$		Along $[001]$					
$2mm$		$2mm$		2					

ORTHORHOMBIC SYSTEM									
222		D_2							
4	d	1	x, y, z $\bar{x}, \bar{y}, z$ $\bar{x}, y, \bar{z}$ $x, \bar{y}, \bar{z}$	Rhombic disphenoid or rhombic tetrahedron <i>Rhombic tetrahedron</i>		(hkl) $(\bar{h}\bar{k}l)$ $(\bar{h}k\bar{l})$ $(h\bar{k}\bar{l})$			
				Rhombic prism <i>Rectangle through origin</i>		$(hk0)$ $(\bar{h}\bar{k}0)$ $(\bar{h}k0)$ $(h\bar{k}0)$			
				Rhombic prism <i>Rectangle through origin</i>		$(h0l)$ $(\bar{h}0l)$ $(\bar{h}0\bar{l})$ $(h0\bar{l})$			
				Rhombic prism <i>Rectangle through origin</i>		$(0kl)$ $(0\bar{k}l)$ $(0k\bar{l})$ $(0\bar{k}\bar{l})$			
2	c	$\dots 2$	$0, 0, z$ $0, 0, \bar{z}$	Pinacoid or parallelohedron <i>Line segment through origin</i>		(001) $(00\bar{1})$			
2	b	$\dots 2$	$0, y, 0$ $0, \bar{y}, 0$	Pinacoid or parallelohedron <i>Line segment through origin</i>		(010) $(0\bar{1}0)$			
2	a	$2\dots$	$x, 0, 0$ $\bar{x}, 0, 0$	Pinacoid or parallelohedron <i>Line segment through origin</i>		(100) $(\bar{1}00)$			
1	o	222	$0, 0, 0$	Point in origin					
Symmetry of special projections									
Along $[100]$		Along $[010]$		Along $[001]$					
$2mm$		$2mm$		$2mm$					

3. ADVANCED TOPICS ON SPACE-GROUP SYMMETRY

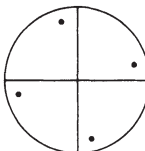
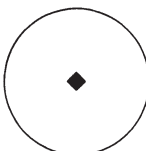
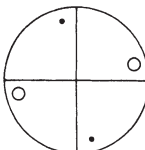
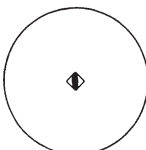
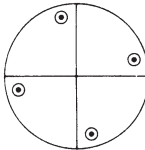
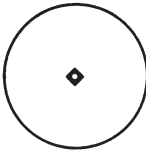
Table 3.2.3.2 (continued)

ORTHORHOMBIC SYSTEM (<i>cont.</i>)									
$mm2$		C_{2v}							
4	d	1	x, y, z	$\bar{x}, \bar{y}, z$	$x, \bar{y}, z$	$\bar{x}, y, z$	Rhombic pyramid <i>Rectangle through origin</i>	(hkl)	$(\bar{h}\bar{k}l)$ $(h\bar{k}l)$ $(\bar{h}kl)$
							Rhombic prism <i>Rectangle through origin</i>	$(hk0)$	$(\bar{h}\bar{k}0)$ $(h\bar{k}0)$ $(\bar{h}k0)$
2	c	$m..$	$0, y, z$	$0, \bar{y}, z$			Dome or dihedron <i>Line segment</i>	$(0kl)$	$(0\bar{k}l)$
							Pinacoid or parallelohedron <i>Line segment through origin</i>	(010)	$(0\bar{1}0)$
2	b	$.m.$	$x, 0, z$	$\bar{x}, 0, z$			Dome or dihedron <i>Line segment</i>	$(h0l)$	$(\bar{h}0l)$
							Pinacoid or parallelohedron <i>Line segment through origin</i>	(100)	$(\bar{1}00)$
1	a	$mm2$	$0, 0, z$				Pedion or monohedron <i>Single point</i>	(001) or $(00\bar{1})$	
Symmetry of special projections									
			Along $[100]$	Along $[010]$	Along $[001]$				
			m	m	$2mm$				

$m m m$		D_{2h}							
8	g	1	x, y, z	$\bar{x}, \bar{y}, z$	$\bar{x}, y, \bar{z}$	$x, \bar{y}, \bar{z}$	Rhombic dipyrarnid <i>Rectangular prism</i>	(hkl)	$(\bar{h}\bar{k}l)$ $(\bar{h}k\bar{l})$ $(h\bar{k}\bar{l})$
			$\bar{x}, \bar{y}, \bar{z}$	$x, y, \bar{z}$	$x, \bar{y}, z$	$\bar{x}, y, z$		$(\bar{h}\bar{k}\bar{l})$	$(hk\bar{l})$ $(h\bar{k}l)$ $(\bar{h}kl)$
4	f	$..m$	$x, y, 0$	$\bar{x}, \bar{y}, 0$	$\bar{x}, y, 0$	$x, \bar{y}, 0$	Rhombic prism <i>Rectangle through origin</i>	$(hk0)$	$(\bar{h}\bar{k}0)$ $(\bar{h}k0)$ $(h\bar{k}0)$
4	e	$.m.$	$x, 0, z$	$\bar{x}, 0, z$	$\bar{x}, 0, \bar{z}$	$x, 0, \bar{z}$	Rhombic prism <i>Rectangle through origin</i>	$(h0l)$	$(\bar{h}0l)$ $(\bar{h}0\bar{l})$ $(h0\bar{l})$
4	d	$m..$	$0, y, z$	$0, \bar{y}, z$	$0, y, \bar{z}$	$0, \bar{y}, \bar{z}$	Rhombic prism <i>Rectangle through origin</i>	$(0kl)$	$(0\bar{k}l)$ $(0k\bar{l})$ $(0\bar{k}\bar{l})$
2	c	$mm2$	$0, 0, z$	$0, 0, \bar{z}$			Pinacoid or parallelohedron <i>Line segment through origin</i>	(001)	$(00\bar{1})$
2	b	$m2m$	$0, y, 0$	$0, \bar{y}, 0$			Pinacoid or parallelohedron <i>Line segment through origin</i>	(010)	$(0\bar{1}0)$
2	a	$2mm$	$x, 0, 0$	$\bar{x}, 0, 0$			Pinacoid or parallelohedron <i>Line segment through origin</i>	(100)	$(\bar{1}00)$
1	o	mmm	$0, 0, 0$				Point in origin		
Symmetry of special projections									
			Along $[100]$	Along $[010]$	Along $[001]$				
			$2mm$	$2mm$	$2mm$				

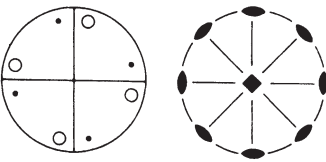
3.2. POINT GROUPS AND CRYSTAL CLASSES

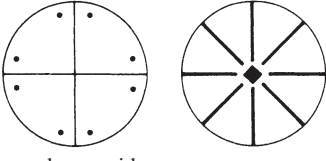
Table 3.2.3.2 (continued)

TETRAGONAL SYSTEM									
4			C_4						
4	b	1	x, y, z	$\bar{x}, \bar{y}, z$	$\bar{y}, x, z$	$y, \bar{x}, z$	Tetragonal pyramid <i>Square</i>	(hkl)	$(\bar{h}\bar{k}l)$ $(\bar{k}hl)$ $(k\bar{h}l)$
							Tetragonal prism <i>Square through origin</i>	$(hk0)$	$(\bar{h}\bar{k}0)$ $(\bar{k}h0)$ $(k\bar{h}0)$
1	a	4..	0, 0, z				Pedion or monohedron <i>Single point</i>	(001) or $(00\bar{1})$	
Symmetry of special projections									
			Along [001]	Along [100]	Along [110]				
			4	m	m				
$\bar{4}$			S_4						
4	b	1	x, y, z	$\bar{x}, \bar{y}, z$	$y, \bar{x}, \bar{z}$	$\bar{y}, x, \bar{z}$	Tetragonal disphenoid or tetragonal tetrahedron <i>Tetragonal tetrahedron</i>	(hkl)	$(\bar{h}\bar{k}l)$ $(\bar{k}hl)$ $(k\bar{h}l)$
							Tetragonal prism <i>Square through origin</i>	$(hk0)$	$(\bar{h}\bar{k}0)$ $(\bar{k}h0)$ $(k\bar{h}0)$
2	a	2..	0, 0, z	0, 0, $\bar{z}$			Pinacoid or parallelohedron <i>Line segment through origin</i>	(001)	$(00\bar{1})$
1	o	$\bar{4}..$	0, 0, 0				Point in origin		
Symmetry of special projections									
			Along [001]	Along [100]	Along [110]				
			4	m	m				
$4/m$			C_{4h}						
8	c	1	x, y, z	$\bar{x}, \bar{y}, z$	$\bar{y}, x, z$	$y, \bar{x}, z$	Tetragonal dipyramid <i>Tetragonal prism</i>	(hkl)	$(\bar{h}\bar{k}l)$ $(\bar{k}hl)$ $(k\bar{h}l)$
			$\bar{x}, \bar{y}, \bar{z}$	$x, y, \bar{z}$	$y, \bar{x}, \bar{z}$	$\bar{y}, x, \bar{z}$		$(\bar{h}\bar{k}\bar{l})$	(hkl) $(\bar{k}hl)$ $(k\bar{h}l)$
4	b	m..	$x, y, 0$	$\bar{x}, \bar{y}, 0$	$\bar{y}, x, 0$	$y, \bar{x}, 0$	Tetragonal prism <i>Square through origin</i>	$(hk0)$	$(\bar{h}\bar{k}0)$ $(\bar{k}h0)$ $(k\bar{h}0)$
2	a	4..	0, 0, z	0, 0, $\bar{z}$			Pinacoid or parallelohedron <i>Line segment through origin</i>	(001)	$(00\bar{1})$
1	o	$4/m..$	0, 0, 0				Point in origin		
Symmetry of special projections									
			Along [001]	Along [100]	Along [110]				
			4	2mm	2mm				

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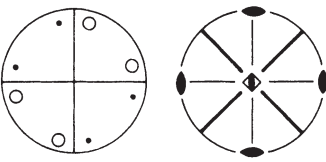
Table 3.2.3.2 (continued)

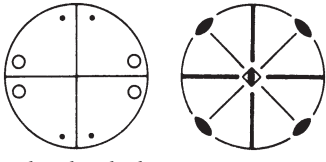
TETRAGONAL SYSTEM (<i>cont.</i>)									
422		D_4							
8	<i>d</i>	1	x, y, z $\bar{x}, y, \bar{z}$	$\bar{x}, \bar{y}, z$ $x, \bar{y}, \bar{z}$	$\bar{y}, x, z$ $y, x, \bar{z}$	$y, \bar{x}, z$ $\bar{y}, \bar{x}, \bar{z}$	Tetragonal trapezohedron <i>Twisted tetragonal antiprism</i>	(hkl) $(\bar{h}\bar{k}\bar{l})$	$(\bar{h}kl)$ $(h\bar{k}l)$
							Ditetragonal prism <i>Truncated square through origin</i>	$(hk0)$ $(\bar{h}\bar{k}0)$	$(\bar{k}h0)$ $(kh0)$
							Tetragonal dipyramid <i>Tetragonal prism</i>	$(h0l)$ $(\bar{h}0\bar{l})$	$(0hl)$ $(0h\bar{l})$
							Tetragonal dipyramid <i>Tetragonal prism</i>	(hhl) $(\bar{h}\bar{h}\bar{l})$	$(\bar{h}hl)$ (hhl)
4	<i>c</i>	.2.	$x, 0, 0$	$\bar{x}, 0, 0$	$0, x, 0$	$0, \bar{x}, 0$	Tetragonal prism <i>Square through origin</i>	(100)	$(\bar{1}00)$
4	<i>b</i>	..2	$x, x, 0$	$\bar{x}, \bar{x}, 0$	$\bar{x}, x, 0$	$x, \bar{x}, 0$	Tetragonal prism <i>Square through origin</i>	(110)	$(\bar{1}\bar{1}0)$
2	<i>a</i>	4..	$0, 0, z$	$0, 0, \bar{z}$			Pinacoid or parallelohedron <i>Line segment through origin</i>	(001)	$(00\bar{1})$
1	<i>o</i>	422	$0, 0, 0$				Point in origin		
Symmetry of special projections									
			Along [001]	Along [100]	Along [110]				
			4mm	2mm	2mm				

4mm		C_{4v}							
8	<i>d</i>	1	x, y, z $x, \bar{y}, z$	$\bar{x}, \bar{y}, z$ $\bar{x}, y, z$	$\bar{y}, x, z$ $\bar{y}, \bar{x}, z$	$y, \bar{x}, z$ y, x, z	Ditetragonal pyramid <i>Truncated square</i>	(hkl) $(\bar{h}\bar{k}\bar{l})$	$(\bar{k}hl)$ (khl)
							Ditetragonal prism <i>Truncated square through origin</i>	$(hk0)$ $(\bar{h}\bar{k}0)$	$(\bar{k}h0)$ $(kh0)$
4	<i>c</i>	.m.	$x, 0, z$	$\bar{x}, 0, z$	$0, x, z$	$0, \bar{x}, z$	Tetragonal pyramid <i>Square</i>	$(h0l)$ $(\bar{h}0\bar{l})$	$(0hl)$ $(0h\bar{l})$
							Tetragonal prism <i>Square through origin</i>	(100)	$(\bar{1}00)$
4	<i>b</i>	..m	x, x, z	$\bar{x}, \bar{x}, z$	$\bar{x}, x, z$	$x, \bar{x}, z$	Tetragonal pyramid <i>Square</i>	(hhl) $(\bar{h}\bar{h}\bar{l})$	$(\bar{h}hl)$ (hhl)
							Tetragonal prism <i>Square through origin</i>	(110)	$(\bar{1}\bar{1}0)$
1	<i>a</i>	4mm	$0, 0, z$				Pedion or monohedron <i>Single point</i>	(001) or $(00\bar{1})$	
Symmetry of special projections									
			Along [001]	Along [100]	Along [110]				
			4mm	m	m				

3.2. POINT GROUPS AND CRYSTAL CLASSES

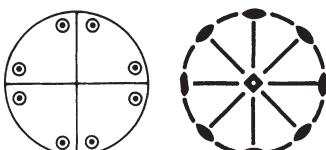
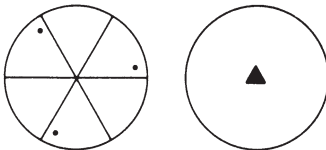
Table 3.2.3.2 (continued)

TETRAGONAL SYSTEM (<i>cont.</i>)									
$\bar{4}2m$		D_{2d}							
8	d	1	x, y, z $\bar{x}, y, \bar{z}$	$\bar{x}, \bar{y}, z$ $x, \bar{y}, \bar{z}$	$y, \bar{x}, \bar{z}$ $\bar{y}, \bar{x}, z$	$\bar{y}, x, \bar{z}$ y, x, z	Tetragonal scalenohedron <i>Tetragonal tetrahedron cut off by pinacoid</i> Ditetragonal prism <i>Truncated square through origin</i> Tetragonal dipyramid <i>Tetragonal prism</i>	(hkl) $(\bar{h}\bar{k}l)$ $(k\bar{h}l)$ $(\bar{k}hl)$ $(\bar{h}kl)$ $(h\bar{k}l)$ $(\bar{k}hl)$ (khl) $(hk0)$ $(\bar{h}\bar{k}0)$ $(k\bar{h}0)$ $(\bar{k}h0)$ $(\bar{h}k0)$ $(h\bar{k}0)$ $(\bar{k}h0)$ $(kh0)$ $(h0l)$ $(\bar{h}0l)$ $(0\bar{h}l)$ $(0hl)$ $(\bar{h}0l)$ $(h0l)$ $(0hl)$ $(0\bar{h}l)$	
4	c	$\dots m$	x, x, z	$\bar{x}, \bar{x}, z$	$x, \bar{x}, \bar{z}$	$\bar{x}, x, \bar{z}$	Tetragonal disphenoid or tetragonal tetrahedron <i>Tetragonal tetrahedron</i> Tetragonal prism <i>Square through origin</i>	(hhl) $(\bar{h}\bar{h}l)$ $(h\bar{h}l)$ $(\bar{h}hl)$ (110) $(\bar{1}\bar{1}0)$ $(1\bar{1}0)$ $(\bar{1}10)$	
4	b	$\dots 2$	$x, 0, 0$	$\bar{x}, 0, 0$	$0, \bar{x}, 0$	$0, x, 0$	Tetragonal prism <i>Square through origin</i>	(100) $(\bar{1}00)$ $(0\bar{1}0)$ (010)	
2	a	$2mm$	$0, 0, z$	$0, 0, \bar{z}$			Pinacoid or parallelohedron <i>Line segment through origin</i>	(001) $(00\bar{1})$	
1	o	$\bar{4}2m$	$0, 0, 0$				<i>Point in origin</i>		
Symmetry of special projections									
			Along [001]	Along [100]	Along [110]				
			$4mm$	$2mm$	m				

$\bar{4}m2$		D_{2d}							
8	d	1	x, y, z $x, \bar{y}, z$	$\bar{x}, \bar{y}, z$ $\bar{x}, y, z$	$y, \bar{x}, \bar{z}$ $y, x, \bar{z}$	$\bar{y}, x, \bar{z}$ $\bar{y}, \bar{x}, z$	Tetragonal scalenohedron <i>Tetragonal tetrahedron cut off by pinacoid</i> Ditetragonal prism <i>Truncated square through origin</i> Tetragonal dipyramid <i>Tetragonal prism</i>	(hkl) $(\bar{h}\bar{k}l)$ $(k\bar{h}l)$ $(\bar{k}hl)$ $(\bar{h}kl)$ $(h\bar{k}l)$ $(\bar{k}hl)$ (khl) $(hk0)$ $(\bar{h}\bar{k}0)$ $(k\bar{h}0)$ $(\bar{k}h0)$ $(\bar{h}k0)$ $(h\bar{k}0)$ $(\bar{k}h0)$ $(kh0)$ (hhl) $(\bar{h}\bar{h}l)$ $(h\bar{h}l)$ $(\bar{h}hl)$ $(\bar{h}hl)$ (hhl) (hhl) $(\bar{h}\bar{h}l)$	
4	c	$\dots m$	$x, 0, z$	$\bar{x}, 0, z$	$0, \bar{x}, \bar{z}$	$0, x, \bar{z}$	Tetragonal disphenoid or tetragonal tetrahedron <i>Tetragonal tetrahedron</i> Tetragonal prism <i>Square through origin</i>	$(h0l)$ $(\bar{h}0l)$ $(0\bar{h}l)$ $(0hl)$ (100) $(\bar{1}00)$ $(0\bar{1}0)$ (010)	
4	b	$\dots 2$	$x, x, 0$	$\bar{x}, \bar{x}, 0$	$x, \bar{x}, 0$	$\bar{x}, x, 0$	Tetragonal prism <i>Square through origin</i>	(110) $(\bar{1}\bar{1}0)$ $(1\bar{1}0)$ $(\bar{1}10)$	
2	a	$2mm$	$0, 0, z$	$0, 0, \bar{z}$			Pinacoid or parallelohedron <i>Line segment through origin</i>	(001) $(00\bar{1})$	
1	o	$\bar{4}m2$	$0, 0, 0$				<i>Point in origin</i>		
Symmetry of special projections									
			Along [001]	Along [100]	Along [110]				
			$4mm$	m	$2mm$				

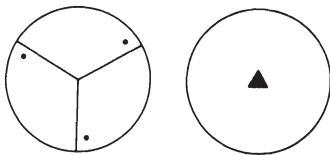
3. ADVANCED TOPICS ON SPACE-GROUP SYMMETRY

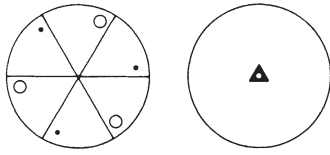
Table 3.2.3.2 (continued)

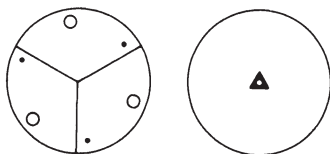
TETRAGONAL SYSTEM (<i>cont.</i>)									
$4/mmm$		D_{4h}							
$\begin{smallmatrix} 4 & 2 & 2 \\ \hline m & m & m \end{smallmatrix}$									
16	<i>g</i>	1	x, y, z $\bar{x}, y, \bar{z}$ $\bar{x}, \bar{y}, \bar{z}$ $x, \bar{y}, z$	$\bar{x}, \bar{y}, z$ $x, \bar{y}, \bar{z}$ $x, y, \bar{z}$ $\bar{x}, y, z$	$\bar{y}, x, z$ $y, x, \bar{z}$ $y, \bar{x}, \bar{z}$ $\bar{y}, x, \bar{z}$	$y, \bar{x}, z$ $\bar{y}, \bar{x}, \bar{z}$ $\bar{y}, x, \bar{z}$ y, x, z	Ditetragonal dipyramid <i>Edge-truncated tetragonal prism</i>		$(hkl) (\bar{h}\bar{k}l) (\bar{k}hl) (k\bar{h}l)$ $(\bar{h}k\bar{l}) (\bar{h}k\bar{l}) (khl) (\bar{k}h\bar{l})$ $(\bar{h}k\bar{l}) (\bar{h}k\bar{l}) (khl) (\bar{k}h\bar{l})$ $(h\bar{k}l) (\bar{h}kl) (\bar{k}hl) (khl)$
8	<i>f</i>	<i>.m.</i>	$x, 0, z$ $\bar{x}, 0, \bar{z}$	$\bar{x}, 0, z$ $x, 0, \bar{z}$	$0, x, z$ $0, x, \bar{z}$	$0, \bar{x}, z$ $0, \bar{x}, \bar{z}$	Tetragonal dipyramid <i>Tetragonal prism</i>		$(h0l) (\bar{h}0l) (0hl) (0\bar{h}l)$ $(\bar{h}0\bar{l}) (\bar{h}0\bar{l}) (0h\bar{l}) (0\bar{h}\bar{l})$
8	<i>e</i>	<i>..m</i>	x, x, z $\bar{x}, x, \bar{z}$	$\bar{x}, \bar{x}, z$ $x, \bar{x}, \bar{z}$	$\bar{x}, x, z$ $x, x, \bar{z}$	$x, \bar{x}, z$ $\bar{x}, \bar{x}, \bar{z}$	Tetragonal dipyramid <i>Tetragonal prism</i>		$(hhl) (\bar{h}\bar{h}l) (\bar{h}hl) (h\bar{h}l)$ $(\bar{h}h\bar{l}) (\bar{h}h\bar{l}) (hhl) (\bar{h}\bar{h}\bar{l})$
8	<i>d</i>	<i>m..</i>	$x, y, 0$ $\bar{x}, y, 0$	$\bar{x}, \bar{y}, 0$ $x, \bar{y}, 0$	$\bar{y}, x, 0$ $y, x, 0$	$y, \bar{x}, 0$ $\bar{y}, \bar{x}, 0$	Ditetragonal prism <i>Truncated square through origin</i>		$(hk0) (\bar{h}\bar{k}0) (\bar{k}h0) (k\bar{h}0)$ $(\bar{h}k0) (\bar{h}k0) (kh0) (\bar{k}h0)$
4	<i>c</i>	<i>m2m.</i>	$x, 0, 0$	$\bar{x}, 0, 0$	$0, x, 0$	$0, \bar{x}, 0$	Tetragonal prism <i>Square through origin</i>		$(100) (\bar{1}00) (010) (0\bar{1}0)$
4	<i>b</i>	<i>m.m2</i>	$x, x, 0$	$\bar{x}, \bar{x}, 0$	$\bar{x}, x, 0$	$x, \bar{x}, 0$	Tetragonal prism <i>Square through origin</i>		$(110) (\bar{1}\bar{1}0) (\bar{1}10) (1\bar{1}0)$
2	<i>a</i>	<i>4mm</i>	$0, 0, z$	$0, 0, \bar{z}$	Pinacoid or parallelohedron <i>Line segment through origin</i>		$(001) (00\bar{1})$		
1	<i>o</i>	<i>4/mmm</i>	$0, 0, 0$	Point in origin					
Symmetry of special projections									
			Along [001]	Along [100]	Along [110]				
			<i>4mm</i>	<i>2mm</i>	<i>2mm</i>				
TRIGONAL SYSTEM									
3		C_3							
HEXAGONAL AXES									
3	<i>b</i>	1	x, y, z	$\bar{y}, x - y, z$	$\bar{x} + y, \bar{x}, z$	Trigonal pyramid <i>Trigon</i>		$(hki\bar{l}) (\bar{i}hkl) (kihl)$	
					Trigonal prism <i>Trigon through origin</i>		$(hki0) (\bar{i}hk0) (kih0)$		
1	<i>a</i>	3..	$0, 0, z$	Pedion or monohedron <i>Single point</i>		$(0001) \text{ or } (000\bar{1})$			
Symmetry of special projections									
			Along [001]	Along [100]	Along [210]				
			3	1	1				

3.2. POINT GROUPS AND CRYSTAL CLASSES

Table 3.2.3.2 (continued)

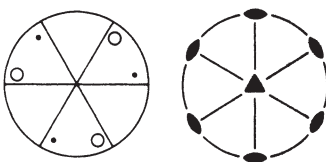
TRIGONAL SYSTEM (cont.)									
$\bar{3}$		C_3							
RHOMBOHEDRAL AXES									
3	b	1	x, y, z	z, x, y	y, z, x	Trigonal pyramid Trigon	(hkl)	(lhk)	(klh)
						Trigonal prism Trigon through origin	(hk(h+k))	((h+k)hk)	(k(h+k)h)
1	a	3.	x, x, x			Pedion or monohedron Single point	(111) or (111)		
Symmetry of special projections									
			Along [111]	Along [110]	Along [211]				
			3	1	1				

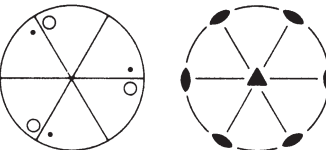
$\bar{3}$		C_{3i}							
HEXAGONAL AXES									
6	b	1	x, y, z	y, x - y, z	x + y, x, z	Rhombohedral Trigonal antiprism	(hkl)	(ihkl)	(kihl)
			x, y, z	y, x + y, z	x - y, x, z	Hexagonal prism Hexagon through origin	(hkl)	(ihkl)	(kihl)
						Hexagonal prism Hexagon through origin	(hki0)	(ihk0)	(kih0)
							(hki0)	(ihk0)	(kih0)
2	a	3..	0, 0, z	0, 0, z		Pinacoid or parallelohedron Line segment through origin	(0001)	(0001)	
1	o	3..	0, 0, 0			Point in origin			
Symmetry of special projections									
			Along [001]	Along [100]	Along [210]				
			6	2	2				

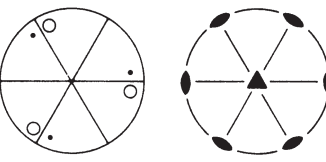
$\bar{3}$		C_{3i}							
RHOMBOHEDRAL AXES									
6	b	1	x, y, z	z, x, y	y, z, x	Rhombohedral Trigonal antiprism	(hkl)	(lhk)	(klh)
			x, y, z	z, x, y	y, z, x	Hexagonal prism Hexagon through origin	(hkl)	(lhk)	(klh)
						Hexagonal prism Hexagon through origin	(hk(h+k))	((h+k)hk)	(k(h+k)h)
							(hk(h+k))	((h+k)hk)	(k(h+k)h)
2	a	3.	x, x, x	x, x, x		Pinacoid or parallelohedron Line segment through origin	(111)	(111)	
1	o	3.	0, 0, 0			Point in origin			
Symmetry of special projections									
			Along [111]	Along [110]	Along [211]				
			6	2	2				

3. ADVANCED TOPICS ON SPACE-GROUP SYMMETRY

Table 3.2.3.2 (continued)

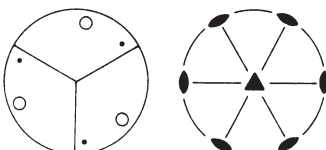
TRIGONAL SYSTEM (<i>cont.</i>)									
312		D_3							
HEXAGONAL AXES									
									
6	c	1	x, y, z $\bar{y}, \bar{x}, \bar{z}$	$\bar{y}, x - y, z$ $\bar{x} + y, y, \bar{z}$	$\bar{x} + y, \bar{x}, z$ $x, x - y, \bar{z}$	Trigonal trapezohedron <i>Twisted trigonal antiprism</i>	$(hki\bar{l})$ $(\bar{k}\bar{h}\bar{i}l)$	$(ihk\bar{l})$ $(\bar{h}\bar{i}\bar{k}l)$	$(kih\bar{l})$ $(\bar{i}\bar{k}\bar{h}l)$
						Ditrigonal prism <i>Truncated trigon through origin</i>	$(hki0)$ $(\bar{k}\bar{h}\bar{i}0)$	$(ihk0)$ $(\bar{h}\bar{i}\bar{k}0)$	$(kih0)$ $(\bar{i}\bar{k}\bar{h}0)$
						Trigonal dipyramid <i>Trigonal prism</i>	$(h0\bar{h}l)$ $(0\bar{h}hl)$	$(\bar{h}h0l)$ $(\bar{h}h0\bar{l})$	$(0\bar{h}hl)$ $(h0\bar{h}\bar{l})$
						Rhombohedron <i>Trigonal antiprism</i>	$(hh2\bar{h}l)$ $(\bar{h}\bar{h}2h\bar{l})$	$(2\bar{h}hhl)$ $(\bar{h}2h\bar{h}\bar{l})$	$(h2\bar{h}hl)$ $(2h\bar{h}\bar{h}\bar{l})$
						Hexagonal prism <i>Hexagon through origin</i>	$(11\bar{2}0)$ $(\bar{1}\bar{1}20)$	$(\bar{2}110)$ $(\bar{1}2\bar{1}0)$	$(1\bar{2}10)$ $(2\bar{1}\bar{1}0)$
3	b	..2	$x, \bar{x}, 0$	$x, 2x, 0$	$2\bar{x}, \bar{x}, 0$	Trigonal prism <i>Trigon through origin</i>	$(10\bar{1}0)$ or $(\bar{1}010)$	$(\bar{1}100)$ $(\bar{1}\bar{1}00)$	$(0\bar{1}10)$ $(01\bar{1}0)$
2	a	3..	$0, 0, z$	$0, 0, \bar{z}$		Pinacoid or parallelohedron <i>Line segment through origin</i>	(0001)	$(000\bar{1})$	
1	o	3.2	$0, 0, 0$			<i>Point in origin</i>			
Symmetry of special projections									
			Along [001]	Along [100]	Along [210]				
			3m	1	2				

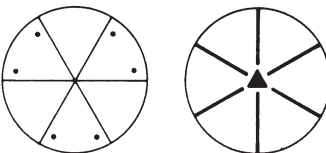
321									
D_3									
HEXAGONAL AXES									
									
6	c	1	x, y, z $y, x, \bar{z}$	$\bar{y}, x - y, z$ $x - y, \bar{y}, \bar{z}$	$\bar{x} + y, \bar{x}, z$ $\bar{x}, \bar{x} + y, \bar{z}$	Trigonal trapezohedron <i>Twisted trigonal antiprism</i>	$(hki\bar{l})$ $(khi\bar{l})$	$(ihk\bar{l})$ $(hik\bar{l})$	$(kih\bar{l})$ $(ikh\bar{l})$
						Ditrigonal prism <i>Truncated trigon through origin</i>	$(hki0)$ $(khi0)$	$(ihk0)$ $(hik0)$	$(kih0)$ $(ikh0)$
						Trigonal dipyramid <i>Trigonal prism</i>	$(hh2\bar{h}l)$ $(hh2h\bar{l})$	$(2\bar{h}hhl)$ $(h2h\bar{h}\bar{l})$	$(h2\bar{h}hl)$ $(2h\bar{h}\bar{h}\bar{l})$
						Rhombohedron <i>Trigonal antiprism</i>	$(h0\bar{h}l)$ $(0\bar{h}hl)$	$(\bar{h}h0l)$ $(h\bar{h}0\bar{l})$	$(0\bar{h}hl)$ $(\bar{h}0h\bar{l})$
						Hexagonal prism <i>Hexagon through origin</i>	$(10\bar{1}0)$ $(01\bar{1}0)$	$(\bar{1}100)$ $(1\bar{1}00)$	$(0\bar{1}10)$ $(\bar{1}010)$
3	b	..2.	$x, 0, 0$	$0, x, 0$	$\bar{x}, \bar{x}, 0$	Trigonal prism <i>Trigon through origin</i>	$(11\bar{2}0)$ or $(\bar{1}\bar{1}20)$	$(\bar{2}110)$ $(2\bar{1}\bar{1}0)$	$(1\bar{2}10)$ $(\bar{1}2\bar{1}0)$
2	a	3..	$0, 0, z$	$0, 0, \bar{z}$		Pinacoid or parallelohedron <i>Line segment through origin</i>	(0001)	$(000\bar{1})$	
1	o	32.	$0, 0, 0$			<i>Point in origin</i>			
Symmetry of special projections									
			Along [001]	Along [100]	Along [210]				
			3m	2	1				

321			D_3								
HEXAGONAL AXES											
											
6	c	1	x, y, z $y, x, \bar{z}$	$\bar{y}, x - y, z$ $x - y, \bar{y}, \bar{z}$	$\bar{x} + y, \bar{x}, z$ $\bar{x}, \bar{x} + y, \bar{z}$	Trigonal trapezohedron <i>Twisted trigonal antiprism</i>	(hkl) $(k\bar{h}\bar{l})$	$(ihkl)$ $(h\bar{i}\bar{k}\bar{l})$	$(kihl)$ $(i\bar{k}\bar{h}\bar{l})$		
						Ditrigonal prism <i>Truncated trigon through origin</i>	$(hki0)$ $(k\bar{h}\bar{i}0)$	$(ihk0)$ $(h\bar{i}\bar{k}0)$	$(kih0)$ $(i\bar{k}\bar{h}0)$		
						Trigonal dipyramid <i>Trigonal prism</i>	$(hh2\bar{h}l)$ $(hh2h\bar{l})$	$(\bar{2}h\bar{h}hl)$ $(h2h\bar{h}\bar{l})$	$(h2\bar{h}hl)$ $(2h\bar{h}\bar{h}\bar{l})$		
						Rhombohedron <i>Trigonal antiprism</i>	$(h0\bar{h}l)$ $(0\bar{h}hl)$	$(\bar{h}h0l)$ $(h\bar{h}0\bar{l})$	$(0\bar{h}hl)$ $(h0\bar{h}\bar{l})$		
						Hexagonal prism <i>Hexagon through origin</i>	$(10\bar{1}0)$ $(01\bar{1}0)$	$(\bar{1}100)$ $(\bar{1}100)$	$(0\bar{1}10)$ $(01\bar{1}0)$		
3	b	.2.	$x, 0, 0$	$0, x, 0$	$\bar{x}, \bar{x}, 0$	Trigonal prism <i>Trigon through origin</i>	$(11\bar{2}0)$ or $(\bar{1}\bar{1}20)$	$(\bar{2}110)$ $(2\bar{1}\bar{1}0)$	$(1\bar{2}10)$ $(\bar{1}2\bar{1}0)$		
2	a	3..	$0, 0, z$	$0, 0, \bar{z}$		Pinacoid or parallelohedron <i>Line segment through origin</i>	(0001)	$(000\bar{1})$			
1	o	32.	$0, 0, 0$			Point in origin					
Symmetry of special projections											
			Along [001]	Along [100]	Along [210]						
			3m	2	1						

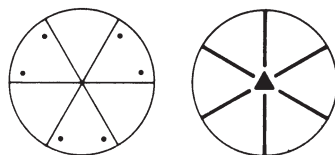
3.2. POINT GROUPS AND CRYSTAL CLASSES

Table 3.2.3.2 (continued)

TRIGONAL SYSTEM (<i>cont.</i>)									
32		D_3							
RHOMBOHEDRAL AXES									
6	<i>c</i>	1	x, y, z $\bar{z}, \bar{y}, \bar{x}$	z, x, y $\bar{y}, \bar{x}, \bar{z}$	y, z, x $\bar{x}, \bar{z}, \bar{y}$	Trigonal trapezohedron <i>Twisted trigonal antiprism</i>	(hkl) $(\bar{k}\bar{h}\bar{l})$	(lhk) $(\bar{h}\bar{l}\bar{k})$	(klh) $(\bar{l}\bar{k}\bar{h})$
Ditrigonal prism <i>Truncated trigon through origin</i>							$(hk(\bar{h}+\bar{k}))$ $(\bar{k}\bar{h}(h+k))$	$((\bar{h}+\bar{k})hk)$ $(\bar{h}(h+k)\bar{k})$	$(k(\bar{h}+\bar{k})h)$ $((h+k)\bar{k}\bar{h})$
Trigonal dipyramid <i>Trigonal prism</i>							$(hk(2k-h))$ $(\bar{k}\bar{h}(h-2k))$	$((2k-h)hk)$ $(\bar{h}(h-2k)\bar{k})$	$(k(2k-h)h)$ $((h-2k)\bar{k}\bar{h})$
Rhombohedron <i>Trigonal antiprism</i>							(hhl) $(\bar{h}\bar{h}\bar{l})$	(lhh) $(\bar{h}\bar{l}\bar{h})$	(hll) $(\bar{l}\bar{h}\bar{h})$
Hexagonal prism <i>Hexagon through origin</i>							$(11\bar{2})$ $(\bar{1}\bar{1}2)$	$(\bar{2}11)$ $(1\bar{2}\bar{1})$	$(1\bar{2}1)$ $(2\bar{1}\bar{1})$
3	<i>b</i>	2	$x, \bar{x}, 0$	$0, x, \bar{x}$	$\bar{x}, 0, x$	Trigonal prism <i>Trigon through origin</i>	$(01\bar{1})$ or $(0\bar{1}1)$	$(\bar{1}01)$ $(10\bar{1})$	$(1\bar{1}0)$ $(\bar{1}10)$
2	<i>a</i>	3	x, x, x	$\bar{x}, \bar{x}, \bar{x}$		Pinacoid or parallelohedron <i>Line segment through origin</i>	(111)	$(\bar{1}\bar{1}\bar{1})$	
1	<i>o</i>	32	$0, 0, 0$			Point in origin			
Symmetry of special projections									
			Along $[111]$ 3 <i>m</i>	Along $[\bar{1}\bar{1}0]$ 2	Along $[2\bar{1}\bar{1}]$ 1				

3m1		C_{3v}							
HEXAGONAL AXES									
6	<i>c</i>	1	x, y, z $\bar{y}, \bar{x}, z$	$\bar{y}, x-y, z$ $\bar{x}+y, y, z$	$\bar{x}+y, \bar{x}, z$ $x, x-y, z$	Ditrigonal pyramid <i>Truncated trigon</i>	$(hkil)$ $(\bar{k}\bar{h}\bar{i}\bar{l})$	$(ihkl)$ $(\bar{h}\bar{i}\bar{k}\bar{l})$	$(kihl)$ $(\bar{i}\bar{k}\bar{h}\bar{l})$
Ditrigonal prism <i>Truncated trigon through origin</i>							$(hki0)$ $(\bar{k}\bar{h}\bar{i}0)$	$(ihk0)$ $(\bar{h}\bar{i}\bar{k}0)$	$(kih0)$ $(\bar{i}\bar{k}\bar{h}0)$
Hexagonal pyramid <i>Hexagon</i>							$(hh2\bar{h}l)$ $(\bar{h}\bar{h}2hl)$	$(\bar{2}h\bar{h}hl)$ $(\bar{h}2h\bar{h}\bar{l})$	$(h2\bar{h}hl)$ $(2h\bar{h}\bar{h}\bar{l})$
Hexagonal prism <i>Hexagon through origin</i>							$(11\bar{2}0)$ $(\bar{1}\bar{1}20)$	$(\bar{2}110)$ $(1\bar{2}\bar{1}0)$	$(1\bar{2}10)$ $(2\bar{1}\bar{1}0)$
3	<i>b</i>	3 <i>m</i> .	$x, \bar{x}, z$	$x, 2x, z$	$2\bar{x}, \bar{x}, z$	Trigonal pyramid <i>Trigon</i>	$(h0\bar{h}l)$	$(\bar{h}h0l)$	$(0\bar{h}hl)$
Trigonal prism <i>Trigon through origin</i>							$(10\bar{1}0)$ or $(\bar{1}010)$	$(\bar{1}100)$ $(1\bar{1}00)$	$(0\bar{1}10)$ $(01\bar{1}0)$
1	<i>a</i>	3 <i>m</i> .	$0, 0, z$			Pedion or monohedron <i>Single point</i>	(0001) or $(000\bar{1})$		
Symmetry of special projections									
			Along $[001]$ 3 <i>m</i>	Along $[100]$ 1	Along $[210]$ <i>m</i>				

3m1 **C_{3v}**
HEXAGONAL AXES



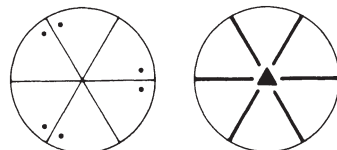
6	<i>c</i>	1	x, y, z $\bar{y}, \bar{x}, z$	$\bar{y}, x-y, z$ $\bar{x}+y, y, z$	$\bar{x}+y, \bar{x}, z$ $x, x-y, z$	Ditrigonal pyramid <i>Truncated trigon</i>	$(hki\bar{l})$ $(\bar{k}\bar{h}\bar{i}\bar{l})$	$(ihkl)$ $(\bar{h}\bar{i}kl)$	$(kihl)$ $(\bar{i}\bar{k}\bar{h}\bar{l})$
Ditrigonal prism <i>Truncated trigon through origin</i>							$(hki0)$ $(\bar{k}\bar{h}\bar{i}0)$	$(ihk0)$ $(\bar{h}\bar{i}k0)$	$(kih0)$ $(\bar{i}\bar{k}\bar{h}0)$
Hexagonal pyramid <i>Hexagon</i>							$(hh2\bar{h}l)$ $(\bar{h}\bar{h}2hl)$	$(\bar{2}h\bar{h}hl)$ $(h2h\bar{h}l)$	$(h2\bar{h}hl)$ $(\bar{h}2h\bar{h}l)$
Hexagonal prism <i>Hexagon through origin</i>							$(11\bar{2}0)$ $(\bar{1}\bar{1}20)$	$(\bar{2}110)$ $(1\bar{2}\bar{1}0)$	$(1\bar{2}10)$ $(2\bar{1}\bar{1}0)$
3	<i>b</i>	3 <i>m</i>	$x, \bar{x}, z$	$x, 2x, z$	$2\bar{x}, \bar{x}, z$	Trigonal pyramid <i>Trigon</i>	$(h0\bar{h}l)$	$(\bar{h}h0l)$	$(0\bar{h}hl)$
Trigonal prism <i>Trigon through origin</i>							$(10\bar{1}0)$ or $(\bar{1}010)$	$(\bar{1}100)$ $(1\bar{1}00)$	$(0\bar{1}10)$ $(01\bar{1}0)$
1	<i>a</i>	3 <i>m</i>	$0, 0, z$			Pedion or monohedron <i>Single point</i>	(0001) or $(000\bar{1})$		

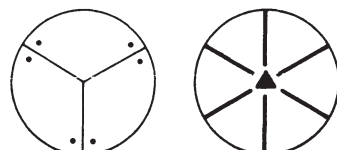
Symmetry of special projections

Along $[001]$	Along $[100]$	Along $[210]$
3 <i>m</i>	1	<i>m</i>

3. ADVANCED TOPICS ON SPACE-GROUP SYMMETRY

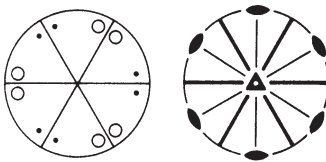
Table 3.2.3.2 (continued)

TRIGONAL SYSTEM (<i>cont.</i>)									
31m		C_{3v}							
HEXAGONAL AXES									
6	c	1	x, y, z y, x, z	$\bar{y}, x - y, z$ $x - y, \bar{y}, z$	$\bar{x} + y, \bar{x}, z$ $\bar{x}, \bar{x} + y, z$	Ditrigonal pyramid <i>Truncated trigon</i>	(hkl) (khl)	$(ihkl)$ $(hikl)$	$(kihl)$ $(ikhl)$
						Ditrigonal prism <i>Truncated trigon through origin</i>	$(hki0)$ $(khi0)$	$(ihk0)$ $(hik0)$	$(kih0)$ $(ikh0)$
						Hexagonal pyramid <i>Hexagon</i>	$(h0\bar{h}l)$ $(0h\bar{h}l)$	$(\bar{h}h0l)$ $(h\bar{h}0l)$	$(0\bar{h}hl)$ $(\bar{h}0hl)$
						Hexagonal prism <i>Hexagon through origin</i>	$(10\bar{1}0)$ $(01\bar{1}0)$	$(\bar{1}100)$ $(1\bar{1}00)$	$(0\bar{1}10)$ $(\bar{1}010)$
3	b	..m	$x, 0, z$ $0, x, z$	$\bar{x}, \bar{x}, z$		Trigonal pyramid <i>Trigon</i>	$(hh\bar{2}hl)$	$(\bar{2}h hhl)$	$(h\bar{2} hhl)$
						Trigonal prism <i>Trigon through origin</i>	$(11\bar{2}0)$ or $(\bar{1}\bar{1}20)$	$(\bar{2}110)$ $(2\bar{1}\bar{1}0)$	$(1\bar{2}10)$ $(\bar{1}2\bar{1}0)$
1	a	3.m	$0, 0, z$			Pedion or monohedron <i>Single point</i>	(0001) or $(000\bar{1})$		
Symmetry of special projections									
			Along [001]	Along [100]	Along [210]				
			3m	m	1				

3m									
		C_{3v}							
RHOMBOHEDRAL AXES									
6	c	1	x, y, z z, y, x	z, x, y y, x, z	y, z, x x, z, y	Ditrigonal pyramid <i>Truncated trigon</i>	(hkl) (khl)	(lhk) (hlk)	(klh) (lkh)
						Ditrigonal prism <i>Truncated trigon through origin</i>	$(hk(\bar{h}+\bar{k}))$ $(kh(\bar{h}+\bar{k}))$	$((\bar{h}+\bar{k})hk)$ $(h(\bar{h}+\bar{k})k)$	$(k(\bar{h}+\bar{k})h)$ $((\bar{h}+\bar{k})kh)$
						Hexagonal pyramid <i>Hexagon</i>	$(hk(2k-h))$ $(kh(2k-h))$	$((2k-h)hk)$ $(h(2k-h)k)$	$(k(2k-h)h)$ $((2k-h)kh)$
						Hexagonal prism <i>Hexagon through origin</i>	$(01\bar{1})$ $(10\bar{1})$	$(\bar{1}01)$ $(0\bar{1}1)$	$(\bar{1}10)$ $(1\bar{1}0)$
3	b	..m	x, y, x x, y, x	x, x, y y, x, x	y, x, x x, x, y	Trigonal pyramid <i>Trigon</i>	(hhl)	(lhh)	(hlh)
						Trigonal prism <i>Trigon through origin</i>	$(11\bar{2})$ or $(\bar{1}\bar{1}2)$	$(\bar{2}11)$ $(2\bar{1}\bar{1})$	$(1\bar{2}1)$ $(\bar{1}2\bar{1})$
1	a	3m	x, x, x			Pedion or monohedron <i>Single point</i>	(111) or $(\bar{1}\bar{1}\bar{1})$		
Symmetry of special projections									
			Along [111]	Along [110]	Along [211]				
			3m	1	m				

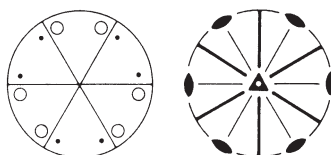
3.2. POINT GROUPS AND CRYSTAL CLASSES

Table 3.2.3.2 (continued)

TRIGONAL SYSTEM (<i>cont.</i>)														
$\bar{3}1m$			D_{3d}											
$\bar{3}1\frac{2}{m}$														
HEXAGONAL AXES														
12	<i>d</i>	1	x, y, z $\bar{y}, \bar{x}, \bar{z}$ $\bar{x}, \bar{y}, \bar{z}$ y, x, z	$\bar{y}, x - y, z$ $\bar{x} + y, y, \bar{z}$ $y, \bar{x} + y, \bar{z}$ $x - y, \bar{y}, z$	$\bar{x} + y, \bar{x}, z$ $x, x - y, \bar{z}$ $x - y, x, \bar{z}$ $\bar{x}, \bar{x} + y, z$	Ditrigonal scalenohedron or hexagonal scalenohedron <i>Trigonal antiprism sliced off by pinacoid</i>		(hkl) $(\bar{k}\bar{h}\bar{l})$ $(h\bar{k}\bar{l})$ (khl)	$(ihkl)$ $(\bar{h}\bar{i}\bar{k}\bar{l})$ $(\bar{i}\bar{h}\bar{k}\bar{l})$ $(hikl)$	$(kihl)$ $(\bar{i}\bar{k}\bar{h}\bar{l})$ $(\bar{k}\bar{i}\bar{h}\bar{l})$ $(ikh\bar{l})$				
						Dihexagonal prism <i>Truncated hexagon through origin</i>	$(hki0)$ $(\bar{k}\bar{h}\bar{i}0)$ $(h\bar{k}\bar{i}0)$ $(khi0)$	$(ihk0)$ $(\bar{h}\bar{i}\bar{k}0)$ $(\bar{i}\bar{h}\bar{k}0)$ $(hik0)$	$(kih0)$ $(\bar{i}\bar{k}\bar{h}0)$ $(\bar{k}\bar{i}\bar{h}0)$ $(ikh0)$					
						Hexagonal dipyrmaid <i>Hexagonal prism</i>	$(h0\bar{h}l)$ $(0\bar{h}hl)$ $(h0hl)$ $(0hh\bar{l})$	$(\bar{h}h0l)$ $(\bar{h}h0\bar{l})$ $(h\bar{h}0l)$ $(h\bar{h}0\bar{l})$	$(0\bar{h}hl)$ $(h0\bar{h}\bar{l})$ $(0hhl)$ $(\bar{h}0hl)$					
6	<i>c</i>	$\bar{3}m$	$x, 0, z$ $0, \bar{x}, \bar{z}$	$0, x, z$ $\bar{x}, 0, \bar{z}$	$\bar{x}, \bar{x}, z$ $x, x, \bar{z}$	Rhombohedron <i>Trigonal antiprism</i>		$(hh2hl)$ $(\bar{h}\bar{h}2h\bar{l})$	$(\bar{2}hhhl)$ $(h2hhl)$	$(h2\bar{h}hl)$ $(2hh\bar{h}\bar{l})$				
						Hexagonal prism <i>Hexagon through origin</i>	$(11\bar{2}0)$ $(\bar{1}\bar{1}20)$	$(\bar{2}110)$ $(\bar{1}2\bar{1}0)$	$(1\bar{2}10)$ $(2\bar{1}\bar{1}0)$					
6	<i>b</i>	$\bar{3}2$	$x, \bar{x}, 0$ $\bar{x}, x, 0$	$x, 2x, 0$ $\bar{x}, 2\bar{x}, 0$	$2\bar{x}, \bar{x}, 0$ $2x, x, 0$	Hexagonal prism <i>Hexagon through origin</i>		$(10\bar{1}0)$ $(\bar{1}010)$	$(\bar{1}100)$ $(\bar{1}\bar{1}00)$	$(0\bar{1}10)$ $(01\bar{1}0)$				
2	<i>a</i>	$3m$	$0, 0, z$	$0, 0, \bar{z}$		Pinacoid or parallelohedron <i>Line segment through origin</i>		(0001)	$(000\bar{1})$					
1	<i>o</i>	$\bar{3}m$	$0, 0, 0$			Point in origin								
Symmetry of special projections														
			Along [001]	Along [100]	Along [210]									
			6mm	2mm	2									

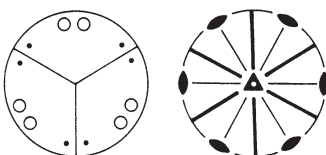
3. ADVANCED TOPICS ON SPACE-GROUP SYMMETRY

Table 3.2.3.2 (continued)

TRIGONAL SYSTEM (<i>cont.</i>)													
$\bar{3}m1$		D_{3d}											
$\bar{3}21$		m											
HEXAGONAL AXES													
12	<i>d</i>	1	x, y, z $y, x, \bar{z}$ $\bar{x}, \bar{y}, \bar{z}$ $\bar{y}, \bar{x}, z$	$\bar{y}, x - y, z$ $x - y, \bar{y}, \bar{z}$ $y, \bar{x} + y, \bar{z}$ $\bar{x} + y, y, z$	$\bar{x} + y, \bar{x}, z$ $\bar{x}, \bar{x} + y, \bar{z}$ $x - y, x, \bar{z}$ $x, x - y, z$	Ditrigonal scalenohedron or hexagonal scalenohedron <i>Trigonal antiprism sliced off by pinacoid</i>		(hkl) (khl) $(\bar{h}\bar{k}\bar{l})$ $(\bar{k}\bar{h}\bar{l})$	$(ihkl)$ $(hikl)$ $(\bar{i}\bar{h}\bar{k}\bar{l})$ $(\bar{h}\bar{i}\bar{k}\bar{l})$	$(kihl)$ $(ikh\bar{l})$ $(\bar{k}\bar{i}\bar{h}\bar{l})$ $(\bar{i}\bar{k}\bar{h}\bar{l})$			
						Dihexagonal prism <i>Truncated hexagon through origin</i>		$(hki0)$ $(khi0)$ $(\bar{h}\bar{k}\bar{i}0)$ $(\bar{k}\bar{h}\bar{i}0)$	$(ihk0)$ $(hik0)$ $(\bar{i}\bar{h}\bar{k}0)$ $(\bar{h}\bar{i}\bar{k}0)$	$(kih0)$ $(ikh0)$ $(\bar{k}\bar{i}\bar{h}0)$ $(\bar{i}\bar{k}\bar{h}0)$			
						Hexagonal dipyrmaid <i>Hexagonal prism</i>		$(hh2hl)$ $(hh2h\bar{l})$ $(\bar{h}\bar{h}2h\bar{l})$ $(\bar{h}\bar{h}2hl)$	$(\bar{2}h\bar{h}hl)$ $(h2\bar{h}h\bar{l})$ $(2h\bar{h}h\bar{l})$ $(\bar{h}2h\bar{h}l)$	$(h2\bar{h}hl)$ $(2h\bar{h}hl)$ $(\bar{h}2hhl)$ $(2h\bar{h}hl)$			
6	<i>c</i>	<i>.m.</i>	$x, \bar{x}, z$ $\bar{x}, x, \bar{z}$	$x, 2x, z$ $2x, x, \bar{z}$	$2\bar{x}, \bar{x}, z$ $\bar{x}, 2\bar{x}, \bar{z}$	Rhombohedron <i>Trigonal antiprism</i>		$(h0hl)$ $(0hhl)$	$(\bar{h}h0l)$ $(h\bar{h}0\bar{l})$	$(0\bar{h}hl)$ $(h0h\bar{l})$			
						Hexagonal prism <i>Hexagon through origin</i>		$(10\bar{1}0)$ $(01\bar{1}0)$	$(\bar{1}100)$ $(1\bar{1}00)$	$(0\bar{1}10)$ $(\bar{1}010)$			
6	<i>b</i>	<i>.2.</i>	$x, 0, 0$ $\bar{x}, 0, 0$	$0, x, 0$ $0, \bar{x}, 0$	$\bar{x}, \bar{x}, 0$ $x, x, 0$	Hexagonal prism <i>Hexagon through origin</i>		$(11\bar{2}0)$ $(\bar{1}\bar{1}20)$	$(\bar{2}110)$ $(12\bar{1}0)$	$(12\bar{1}0)$ $(\bar{2}\bar{1}10)$			
2	<i>a</i>	<i>3m.</i>	$0, 0, z$	$0, 0, \bar{z}$		Pinacoid or parallelohedron <i>Line segment through origin</i>		(0001)	$(000\bar{1})$				
1	<i>o</i>	<i>3m.</i>	$0, 0, 0$			Point in origin							
Symmetry of special projections													
			Along [001]	Along [100]	Along [210]								
			$6mm$	2	$2mm$								

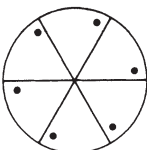
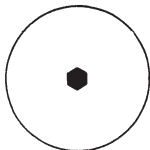
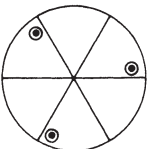
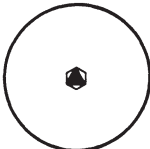
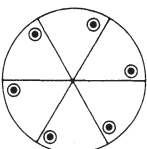
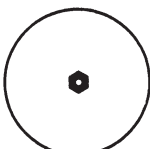
3.2. POINT GROUPS AND CRYSTAL CLASSES

Table 3.2.3.2 (continued)

TRIGONAL SYSTEM (<i>cont.</i>)									
$\bar{3}m$			D_{3d}						
$\bar{3}2$									
m									
RHOMBOHEDRAL AXES									
12	<i>d</i>	1	x, y, z $\bar{z}, \bar{y}, \bar{x}$ $\bar{x}, \bar{y}, \bar{z}$ z, y, x	z, x, y $\bar{y}, \bar{x}, \bar{z}$ $\bar{z}, \bar{x}, \bar{y}$ y, x, z	y, z, x $\bar{x}, \bar{z}, \bar{y}$ $\bar{y}, \bar{z}, \bar{x}$ x, z, y	Ditrigonal scalenohedron or hexagonal scalenohedron <i>Trigonal antiprism sliced off by pinacoid</i>	(hkl) $(\bar{k}\bar{h}\bar{l})$ $(\bar{h}\bar{k}\bar{l})$ (khl)	(lhk) $(\bar{h}\bar{l}\bar{k})$ $(\bar{l}\bar{h}\bar{k})$ (lkh)	(klh) $(\bar{l}\bar{k}\bar{h})$ $(\bar{k}\bar{l}\bar{h})$ (lkh)
						Dihexagonal prism <i>Truncated hexagon through origin</i>	$(hk(\bar{h}+\bar{k}))$ $(\bar{k}\bar{h}(h+k))$ $(\bar{h}\bar{k}(h+k))$ $(kh(\bar{h}+\bar{k}))$	$((\bar{h}+\bar{k})hk)$ $(\bar{h}(h+k)\bar{k})$ $((h+k)\bar{h}\bar{k})$ $(h(\bar{h}+\bar{k})k)$	$(k(\bar{h}+\bar{k})h)$ $((h+k)\bar{k}\bar{h})$ $(\bar{k}(h+k)\bar{h})$ $((\bar{h}+\bar{k})kh)$
						Hexagonal dipyramid <i>Hexagonal prism</i>	$(hk(2k-h))$ $(\bar{k}\bar{h}(h-2k))$ $(\bar{h}\bar{k}(h-2k))$ $(kh(2k-h))$	$((2k-h)hk)$ $(\bar{h}(h-2k)\bar{k})$ $((h-2k)\bar{h}\bar{k})$ $(h(2k-h)k)$	$(k(2k-h)h)$ $((h-2k)\bar{k}\bar{h})$ $(\bar{k}(h-2k)\bar{h})$ $((2k-h)kh)$
6	<i>c</i>	$.m$	x, y, x $\bar{x}, \bar{y}, \bar{x}$	x, x, y $\bar{y}, \bar{x}, \bar{x}$	y, x, x $\bar{x}, \bar{x}, \bar{y}$	Rhombohedron <i>Trigonal antiprism</i>	(hhl) $(\bar{h}\bar{h}\bar{l})$	(lhh) $(\bar{h}\bar{l}\bar{h})$	(hlh) $(\bar{l}\bar{h}\bar{h})$
						Hexagonal prism <i>Hexagon through origin</i>	$(11\bar{2})$ $(\bar{1}\bar{1}2)$	$(\bar{2}11)$ $(\bar{1}2\bar{1})$	$(\bar{1}21)$ $(2\bar{1}\bar{1})$
6	<i>b</i>	$.2$	$x, \bar{x}, 0$ $\bar{x}, x, 0$	$0, x, \bar{x}$ $0, \bar{x}, x$	$\bar{x}, 0, x$ $x, 0, \bar{x}$	Hexagonal prism <i>Hexagon through origin</i>	$(01\bar{1})$ $(0\bar{1}1)$	$(\bar{1}01)$ $(10\bar{1})$	$(\bar{1}10)$ (110)
2	<i>a</i>	$3m$	x, x, x	$\bar{x}, \bar{x}, \bar{x}$		Pinacoid or parallelohedron <i>Line segment through origin</i>	(111)	$(\bar{1}\bar{1}\bar{1})$	
1	<i>o</i>	$\bar{3}m$	$0, 0, 0$			<i>Point in origin</i>			
Symmetry of special projections									
			Along $[111]$	Along $[1\bar{1}0]$	Along $[2\bar{1}\bar{1}]$				
			$6mm$	2	$2mm$				

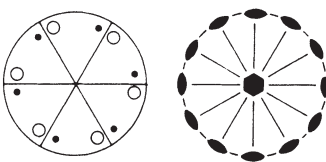
3. ADVANCED TOPICS ON SPACE-GROUP SYMMETRY

Table 3.2.3.2 (continued)

HEXAGONAL SYSTEM									
6	C_6								
									
6	b	1	x, y, z $\bar{x}, \bar{y}, z$	$\bar{y}, x - y, z$ $y, \bar{x} + y, z$	$\bar{x} + y, \bar{x}, z$ $x - y, x, z$	Hexagonal pyramid Hexagon		(hkl) $(\bar{h}\bar{k}l)$	$(ihkl)$ $(\bar{i}\bar{h}\bar{k}l)$ $(kihl)$ $(\bar{k}\bar{i}\bar{h}l)$
						Hexagonal prism Hexagon through origin		$(hki0)$ $(\bar{h}\bar{k}\bar{i}0)$	$(ihk0)$ $(\bar{i}\bar{h}\bar{k}0)$ $(kih0)$ $(\bar{k}\bar{i}\bar{h}0)$
1	a	6..	0, 0, z			Pedion or monohedron Single point		(0001) or (000 $\bar{1}$)	
Symmetry of special projections									
			Along [001]	Along [100]	Along [210]				
			6	m	m				
$\bar{6}$	C_{3h}								
									
6	c	1	x, y, z $x, y, \bar{z}$	$\bar{y}, x - y, z$ $\bar{y}, x - y, \bar{z}$	$\bar{x} + y, \bar{x}, z$ $\bar{x} + y, \bar{x}, \bar{z}$	Trigonal dipyramid Trigonal prism		(hkl) $(h\bar{k}l)$	$(ihkl)$ $(i\bar{h}\bar{k}l)$ $(kihl)$ $(k\bar{i}\bar{h}l)$
3	b	$m..$	$x, y, 0$	$\bar{y}, x - y, 0$	$\bar{x} + y, \bar{x}, 0$	Trigonal prism Trigon through origin		$(hki0)$	$(ihk0)$ $(kih0)$
2	a	3..	0, 0, z	0, 0, $\bar{z}$		Pinacoid or parallelohedron Line segment through origin		(0001)	(000 $\bar{1}$)
1	o	$\bar{6}..$	0, 0, 0			Point in origin			
Symmetry of special projections									
			Along [001]	Along [100]	Along [210]				
			3	m	m				
6/m	C_{6h}								
									
12	c	1	x, y, z $\bar{x}, \bar{y}, z$ $\bar{x}, \bar{y}, \bar{z}$ $x, y, \bar{z}$	$\bar{y}, x - y, z$ $y, \bar{x} + y, z$ $y, \bar{x} + y, \bar{z}$ $\bar{y}, x - y, \bar{z}$	$\bar{x} + y, \bar{x}, z$ $x - y, x, z$ $x - y, x, \bar{z}$ $\bar{x} + y, \bar{x}, \bar{z}$	Hexagonal dipyramid Hexagonal prism		(hkl) $(\bar{h}\bar{k}l)$ $(h\bar{k}l)$ $(\bar{h}kl)$	$(ihkl)$ $(\bar{i}\bar{h}\bar{k}l)$ $(ih\bar{k}l)$ $(\bar{i}hkl)$ $(kihl)$ $(\bar{k}\bar{i}\bar{h}l)$
6	b	$m..$	$x, y, 0$ $\bar{x}, \bar{y}, 0$	$\bar{y}, x - y, 0$ $y, \bar{x} + y, 0$	$\bar{x} + y, \bar{x}, 0$ $x - y, x, 0$	Hexagonal prism Hexagon through origin		$(hki0)$ $(\bar{h}\bar{k}\bar{i}0)$	$(ihk0)$ $(\bar{i}\bar{h}\bar{k}0)$ $(kih0)$ $(\bar{k}\bar{i}\bar{h}0)$
2	a	6..	0, 0, z	0, 0, $\bar{z}$		Pinacoid or parallelohedron Line segment through origin		(0001)	(000 $\bar{1}$)
1	o	6/m..	0, 0, 0			Point in origin			
Symmetry of special projections									
			Along [001]	Along [100]	Along [210]				
			6	$2mm$	$2mm$				

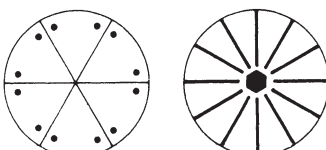
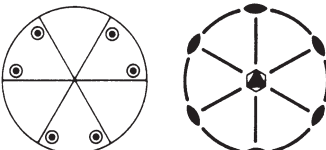
3.2. POINT GROUPS AND CRYSTAL CLASSES

Table 3.2.3.2 (continued)

HEXAGONAL SYSTEM (<i>cont.</i>)									
622		D_6							
12	d	1	x, y, z $\bar{x}, \bar{y}, z$ $y, x, \bar{z}$ $\bar{y}, \bar{x}, \bar{z}$	$\bar{y}, x - y, z$ $y, \bar{x} + y, z$ $x - y, \bar{y}, \bar{z}$ $\bar{x} + y, y, \bar{z}$	$\bar{x} + y, \bar{x}, z$ $x - y, x, z$ $\bar{x}, \bar{x} + y, \bar{z}$ $x, x - y, \bar{z}$	Hexagonal trapezohedron <i>Twisted hexagonal antiprism</i>	$(h\bar{k}l)$ $(\bar{h}k\bar{l})$ $(k\bar{h}l)$ $(\bar{k}h\bar{l})$	$(ihkl)$ $(\bar{i}\bar{h}k\bar{l})$ $(hik\bar{l})$ $(\bar{h}i\bar{k}l)$	$(kihl)$ $(\bar{k}\bar{i}h\bar{l})$ $(ikh\bar{l})$ $(\bar{i}k\bar{h}l)$
						Dihexagonal prism <i>Truncated hexagon through origin</i>	$(h\bar{k}i0)$ $(\bar{h}k\bar{i}0)$ $(k\bar{h}i0)$ $(\bar{k}h\bar{i}0)$	$(ihk0)$ $(\bar{i}\bar{h}k0)$ $(hik0)$ $(\bar{h}i\bar{k}0)$	$(kih0)$ $(\bar{k}\bar{i}h0)$ $(ikh0)$ $(\bar{i}k\bar{h}0)$
						Hexagonal dipyramid <i>Hexagonal prism</i>	$(h0\bar{h}l)$ $(\bar{h}0hl)$ $(0h\bar{h}l)$ $(0\bar{h}hl)$	$(\bar{h}h0l)$ $(h\bar{h}0l)$ $(h\bar{h}0l)$ $(\bar{h}h0l)$	$(0\bar{h}hl)$ $(h0\bar{h}l)$ $(h0\bar{h}l)$ $(\bar{h}0hl)$
						Hexagonal dipyramid <i>Hexagonal prism</i>	$(hh2\bar{h}l)$ $(\bar{h}\bar{h}2hl)$ $(hh2\bar{h}l)$ $(\bar{h}\bar{h}2hl)$	$(2\bar{h}hhl)$ $(2h\bar{h}\bar{h}l)$ $(h2\bar{h}hl)$ $(\bar{h}2h\bar{h}l)$	$(h2\bar{h}hl)$ $(\bar{h}2h\bar{h}l)$ $(2h\bar{h}hl)$ $(2h\bar{h}hl)$
6	c	..2	$x, \bar{x}, 0$ $\bar{x}, x, 0$	$x, 2x, 0$ $\bar{x}, 2\bar{x}, 0$	$2\bar{x}, \bar{x}, 0$ $2x, x, 0$	Hexagonal prism <i>Hexagon through origin</i>	$(10\bar{1}0)$ $(\bar{1}010)$	$(\bar{1}100)$ $(1\bar{1}00)$	$(0\bar{1}10)$ $(01\bar{1}0)$
6	b	.2.	$x, 0, 0$ $\bar{x}, 0, 0$	$0, x, 0$ $0, \bar{x}, 0$	$\bar{x}, \bar{x}, 0$ $x, x, 0$	Hexagonal prism <i>Hexagon through origin</i>	$(11\bar{2}0)$ $(\bar{1}\bar{1}20)$	$(\bar{2}110)$ $(2\bar{1}\bar{1}0)$	$(1\bar{2}10)$ $(\bar{1}2\bar{1}0)$
2	a	6..	$0, 0, z$	$0, 0, \bar{z}$		Pinacoid or parallelohedron <i>Line segment through origin</i>	(0001)	$(000\bar{1})$	
1	o	622	$0, 0, 0$			Point in origin			
Symmetry of special projections									
			Along [001]	Along [100]	Along [210]				
			6mm	2mm	2mm				

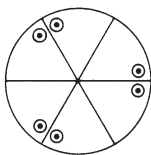
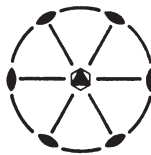
3. ADVANCED TOPICS ON SPACE-GROUP SYMMETRY

Table 3.2.3.2 (continued)

HEXAGONAL SYSTEM (<i>cont.</i>)									
$6mm$		C_{6v}							
12	d	1	x, y, z $\bar{y}, x - y, z$ $\bar{x} + y, \bar{x}, z$ $\bar{x}, \bar{y}, z$ $y, \bar{x} + y, z$ $x - y, x, z$ $\bar{y}, \bar{x}, z$ $\bar{x} + y, y, z$ $x, x - y, z$ y, x, z $x - y, \bar{y}, z$ $\bar{x}, \bar{x} + y, z$	Dihexagonal pyramid <i>Truncated hexagon</i>		$(hki\bar{l})$ $(ihk\bar{l})$ $(kih\bar{l})$ $(\bar{h}\bar{k}\bar{i}l)$ $(\bar{i}\bar{h}\bar{k}l)$ $(\bar{k}\bar{i}\bar{h}l)$ $(khi\bar{l})$ $(hik\bar{l})$ $(ikh\bar{l})$ $(\bar{k}\bar{h}\bar{i}l)$ $(\bar{h}\bar{i}\bar{k}l)$ $(\bar{i}\bar{k}\bar{h}l)$			
				Dihexagonal prism <i>Truncated hexagon through origin</i>		$(hki0)$ $(ihk0)$ $(kih0)$ $(\bar{h}\bar{k}\bar{i}0)$ $(\bar{i}\bar{h}\bar{k}0)$ $(\bar{k}\bar{i}\bar{h}0)$ $(khi0)$ $(hik0)$ $(ikh0)$ $(\bar{k}\bar{h}\bar{i}0)$ $(\bar{h}\bar{i}\bar{k}0)$ $(\bar{i}\bar{k}\bar{h}0)$			
6	c	$.m.$	$x, \bar{x}, z$ $x, 2x, z$ $2\bar{x}, \bar{x}, z$ $\bar{x}, x, z$ $\bar{x}, 2\bar{x}, z$ $2x, x, z$	Hexagonal pyramid <i>Hexagon</i>		$(h0\bar{h}l)$ $(\bar{h}h0l)$ $(0\bar{h}hl)$ $(\bar{h}0hl)$ $(h\bar{h}0l)$ $(0hhl)$			
				Hexagonal prism <i>Hexagon through origin</i>		$(10\bar{1}0)$ $(\bar{1}100)$ $(0\bar{1}10)$ $(\bar{1}010)$ $(1\bar{1}00)$ $(01\bar{1}0)$			
6	b	$.m$	$x, 0, z$ $0, x, z$ $\bar{x}, \bar{x}, z$ $\bar{x}, 0, z$ $0, \bar{x}, z$ x, x, z	Hexagonal pyramid <i>Hexagon</i>		$(hh2\bar{h}l)$ $(\bar{2}hhhl)$ $(h\bar{2}hhl)$ $(\bar{h}h2hl)$ $(2h\bar{h}hl)$ $(\bar{h}2hhl)$			
				Hexagonal prism <i>Hexagon through origin</i>		$(11\bar{2}0)$ $(\bar{2}110)$ $(1\bar{2}10)$ $(\bar{1}\bar{1}20)$ $(2\bar{1}\bar{1}0)$ $(12\bar{1}0)$			
1	a	$6mm$	$0, 0, z$	Pedion or monohedron <i>Single point</i>		(0001) or $(000\bar{1})$			
Symmetry of special projections									
Along [001]		Along [100]		Along [210]					
$6mm$		m		m					
$\bar{6}m2$		D_{3h}							
12	e	1	x, y, z $\bar{y}, x - y, z$ $\bar{x} + y, \bar{x}, z$ $x, y, \bar{z}$ $\bar{y}, x - y, \bar{z}$ $\bar{x} + y, \bar{x}, \bar{z}$ $\bar{y}, \bar{x}, z$ $\bar{x} + y, y, z$ $x, x - y, z$ $\bar{y}, \bar{x}, \bar{z}$ $\bar{x} + y, y, \bar{z}$ $x, x - y, \bar{z}$	Ditrigonal dipyramid <i>Edge-truncated trigonal prism</i>		$(hki\bar{l})$ $(ihk\bar{l})$ $(kih\bar{l})$ $(\bar{h}\bar{k}\bar{i}l)$ $(\bar{i}\bar{h}\bar{k}l)$ $(\bar{k}\bar{i}\bar{h}l)$ $(khi\bar{l})$ $(hik\bar{l})$ $(ikh\bar{l})$ $(\bar{k}\bar{h}\bar{i}l)$ $(\bar{h}\bar{i}\bar{k}l)$ $(\bar{i}\bar{k}\bar{h}l)$			
				Hexagonal dipyramid <i>Hexagonal prism</i>		$(hh2\bar{h}l)$ $(\bar{2}hhhl)$ $(h\bar{2}hhl)$ $(\bar{h}h2\bar{h}l)$ $(2h\bar{h}hl)$ $(\bar{h}2hhl)$ $(\bar{h}h2hl)$ $(h2hhl)$ $(2h\bar{h}hl)$ $(\bar{h}h2hl)$ $(h2hhl)$ $(2h\bar{h}hl)$			
6	d	$.m.$	$x, y, 0$ $\bar{y}, x - y, 0$ $\bar{x} + y, \bar{x}, 0$ $\bar{y}, \bar{x}, 0$ $\bar{x} + y, y, 0$ $x, x - y, 0$	Ditrigonal prism <i>Truncated trigon through origin</i>		$(hki0)$ $(ihk0)$ $(kih0)$ $(\bar{k}\bar{h}\bar{i}0)$ $(\bar{h}\bar{i}\bar{k}0)$ $(\bar{i}\bar{k}\bar{h}0)$			
				Hexagonal prism <i>Hexagon through origin</i>		$(11\bar{2}0)$ $(\bar{2}110)$ $(1\bar{2}10)$ $(\bar{1}\bar{1}20)$ $(2\bar{1}\bar{1}0)$ $(12\bar{1}0)$			
6	c	$.m.$	$x, \bar{x}, z$ $x, 2x, z$ $2\bar{x}, \bar{x}, z$ $x, \bar{x}, \bar{z}$ $x, 2x, \bar{z}$ $2\bar{x}, \bar{x}, \bar{z}$	Trigonal dipyramid <i>Trigonal prism</i>		$(h0\bar{h}l)$ $(\bar{h}h0l)$ $(0\bar{h}hl)$ $(\bar{h}0hl)$ $(h\bar{h}0l)$ $(0hhl)$			
3	b	$mm2$	$x, \bar{x}, 0$ $x, 2x, 0$ $2\bar{x}, \bar{x}, 0$	Trigonal prism <i>Trigon through origin</i>		$(10\bar{1}0)$ $(\bar{1}100)$ $(0\bar{1}10)$ or (1010) $(1\bar{1}00)$ $(01\bar{1}0)$			
2	a	$3m.$	$0, 0, z$ $0, 0, \bar{z}$	Pinacoid or parallelohedron <i>Line segment through origin</i>		(0001) $(000\bar{1})$			
1	o	$\bar{6}m2$	$0, 0, 0$	Point in origin					
Symmetry of special projections									
Along [001]		Along [100]		Along [210]					
$3m$		m		$2mm$					

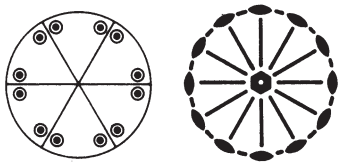
3.2. POINT GROUPS AND CRYSTAL CLASSES

Table 3.2.3.2 (continued)

HEXAGONAL SYSTEM (<i>cont.</i>)									
$\bar{6}2m$		D_{3h}							
12	e	1	x, y, z $\bar{y}, x - y, z$ $\bar{x} + y, \bar{x}, z$ $x, y, \bar{z}$ $\bar{y}, x - y, \bar{z}$ $\bar{x} + y, \bar{x}, \bar{z}$ $y, x, \bar{z}$ $x - y, \bar{y}, \bar{z}$ $\bar{x}, \bar{x} + y, \bar{z}$ y, x, z $x - y, \bar{y}, z$ $\bar{x}, \bar{x} + y, z$	Ditrigonal dipyramid <i>Edge-truncated trigonal prism</i>		$(hki\bar{l})$ $(ihk\bar{l})$ $(kih\bar{l})$ $(hk\bar{i}l)$ $(ihk\bar{l})$ $(kih\bar{l})$ $(khi\bar{l})$ $(hik\bar{l})$ $(ikh\bar{l})$ $(khil)$ $(hikl)$ $(ikh\bar{l})$			
				Hexagonal dipyramid <i>Hexagonal prism</i>		$(h0\bar{h}l)$ $(\bar{h}h0l)$ $(0\bar{h}hl)$ $(h0\bar{h}l)$ $(\bar{h}h0l)$ $(0\bar{h}hl)$ $(0h\bar{h}l)$ $(h\bar{h}0l)$ $(\bar{h}0hl)$ $(0h\bar{h}l)$ $(h\bar{h}0l)$ $(\bar{h}0hl)$			
6	d	$m..$	$x, y, 0$ $\bar{y}, x - y, 0$ $\bar{x} + y, \bar{x}, 0$ $y, x, 0$ $x - y, \bar{y}, 0$ $\bar{x}, \bar{x} + y, 0$	Ditrigonal prism <i>Truncated trigon through origin</i>		$(hki0)$ $(ihk0)$ $(kih0)$ $(khi0)$ $(hik0)$ $(ikh0)$			
				Hexagonal prism <i>Hexagon through origin</i>		$(10\bar{1}0)$ $(\bar{1}100)$ $(0\bar{1}10)$ $(01\bar{1}0)$ $(1\bar{1}00)$ $(\bar{1}010)$			
6	c	$..m$	$x, 0, z$ $0, x, z$ $\bar{x}, \bar{x}, z$ $x, 0, \bar{z}$ $0, x, \bar{z}$ $\bar{x}, \bar{x}, \bar{z}$	Trigonal dipyramid <i>Trigonal prism</i>		$(hh2\bar{h}l)$ $(\bar{2}h\bar{h}hl)$ $(h2\bar{h}hl)$ $(hh2\bar{h}\bar{l})$ $(\bar{2}h\bar{h}h\bar{l})$ $(h2\bar{h}h\bar{l})$			
3	b	$m2m$	$x, 0, 0$ $0, x, 0$ $\bar{x}, \bar{x}, 0$	Trigonal prism <i>Trigon through origin</i>		$(11\bar{2}0)$ $(\bar{2}110)$ $(1\bar{2}10)$ or $(\bar{1}\bar{1}20)$ $(2\bar{1}\bar{1}0)$ $(\bar{1}2\bar{1}0)$			
2	a	$3.m$	$0, 0, z$ $0, 0, \bar{z}$	Pinacoid or parallelohedron <i>Line segment through origin</i>		(0001) $(000\bar{1})$			
1	o	$\bar{6}2m$	$0, 0, 0$	Point in origin					
Symmetry of special projections									
Along [001] Along [100] Along [210]									
$3m$ $2mm$ m									

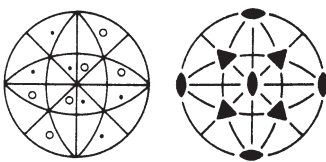
3. ADVANCED TOPICS ON SPACE-GROUP SYMMETRY

Table 3.2.3.2 (continued)

HEXAGONAL SYSTEM (<i>cont.</i>)									
$6/mmm$			D_{6h}						
$\frac{6}{m} \frac{2}{m} \frac{2}{m}$									
24	<i>g</i>	1	x, y, z $\bar{x}, \bar{y}, z$ $y, x, \bar{z}$ $\bar{y}, \bar{x}, \bar{z}$	$\bar{y}, x - y, z$ $y, \bar{x} + y, z$ $x - y, \bar{y}, \bar{z}$ $\bar{x} + y, y, \bar{z}$	$\bar{x} + y, \bar{x}, z$ $x - y, x, z$ $\bar{x}, \bar{x} + y, \bar{z}$ $x, x - y, \bar{z}$	Dihexagonal dipyramid Edge-truncated hexagonal prism	(hkl) $(\bar{h}k\bar{l})$ $(kh\bar{l})$ $(\bar{k}h\bar{l})$	$(ihkl)$ $(\bar{i}h\bar{k}l)$ $(hik\bar{l})$ $(\bar{h}i\bar{k}l)$	$(kihl)$ $(\bar{k}i\bar{h}l)$ $(ikh\bar{l})$ $(\bar{i}k\bar{h}l)$
12	<i>f</i>	<i>m..</i>	$x, y, 0$ $\bar{x}, \bar{y}, 0$ $y, x, 0$ $\bar{y}, \bar{x}, 0$	$\bar{y}, x - y, 0$ $y, \bar{x} + y, 0$ $x - y, \bar{y}, 0$ $\bar{x} + y, y, 0$	$\bar{x} + y, \bar{x}, 0$ $x - y, x, 0$ $\bar{x}, \bar{x} + y, 0$ $x, x - y, 0$	Dihexagonal prism Truncated hexagon through origin	$(hki0)$ $(\bar{h}\bar{k}\bar{i}0)$ $(khi0)$ $(\bar{k}\bar{h}\bar{i}0)$	$(ihk0)$ $(\bar{i}h\bar{k}0)$ $(hik0)$ $(\bar{h}i\bar{k}0)$	$(kih0)$ $(\bar{k}i\bar{h}0)$ $(ikh0)$ $(\bar{i}k\bar{h}0)$
12	<i>e</i>	<i>.m.</i>	$x, 2x, z$ $\bar{x}, 2\bar{x}, z$ $2x, x, \bar{z}$ $2\bar{x}, \bar{x}, \bar{z}$	$2\bar{x}, \bar{x}, z$ $2x, x, z$ $\bar{x}, 2\bar{x}, \bar{z}$ $x, 2x, \bar{z}$	$x, \bar{x}, z$ $\bar{x}, x, z$ $\bar{x}, x, \bar{z}$ $x, \bar{x}, \bar{z}$	Hexagonal dipyramid Hexagonal prism	$(h0\bar{h}l)$ $(h0hl)$ $(0h\bar{h}l)$ $(0hhl)$	$(\bar{h}h0l)$ $(h\bar{h}0l)$ $(h\bar{h}0\bar{l})$ $(hh0\bar{l})$	$(0\bar{h}hl)$ $(0hhl)$ $(h0\bar{h}l)$ $(h0hl)$
12	<i>d</i>	<i>.m</i>	$x, 0, z$ $\bar{x}, 0, z$ $0, x, \bar{z}$ $0, \bar{x}, \bar{z}$	$0, x, z$ $0, \bar{x}, z$ $x, 0, \bar{z}$ $\bar{x}, 0, \bar{z}$	$\bar{x}, \bar{x}, z$ x, x, z $\bar{x}, \bar{x}, \bar{z}$ $x, x, \bar{z}$	Hexagonal dipyramid Hexagonal prism	$(hh2\bar{h}l)$ $(\bar{h}h2hl)$ $(hh2\bar{h}l)$ $(\bar{h}h2hl)$	$(\bar{2}hhhl)$ $(2hhhl)$ $(h2\bar{h}hl)$ $(\bar{h}2hhl)$	$(h\bar{2}hhl)$ $(\bar{h}2hhl)$ $(2hhhl)$ $(\bar{2}hhhl)$
6	<i>c</i>	<i>mm2</i>	$x, 2x, 0$ $\bar{x}, 2\bar{x}, 0$	$2\bar{x}, \bar{x}, 0$ $2x, x, 0$	$x, \bar{x}, 0$ $\bar{x}, x, 0$	Hexagonal prism Hexagon through origin	$(10\bar{1}0)$ $(\bar{1}010)$	$(\bar{1}100)$ $(1\bar{1}00)$	$(0\bar{1}10)$ $(01\bar{1}0)$
6	<i>b</i>	<i>m2m</i>	$x, 0, 0$ $\bar{x}, 0, 0$	$0, x, 0$ $0, \bar{x}, 0$	$\bar{x}, \bar{x}, 0$ $x, x, 0$	Hexagonal prism Hexagon through origin	$(11\bar{2}0)$ $(\bar{1}120)$	$(\bar{2}110)$ $(2\bar{1}\bar{1}0)$	$(1\bar{2}10)$ $(\bar{1}2\bar{1}0)$
2	<i>a</i>	<i>6mm</i>	$0, 0, z$	$0, 0, \bar{z}$		Pinacoid or parallelohedron Line segment through origin	(0001)	$(000\bar{1})$	
1	<i>o</i>	<i>6mmm</i>	$0, 0, 0$			Point in origin			
Symmetry of special projections Along [001] Along [100] Along [210] 6mm 2mm 2mm									

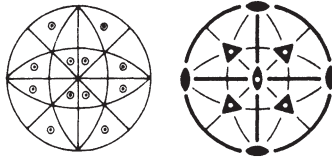
3.2. POINT GROUPS AND CRYSTAL CLASSES

Table 3.2.3.2 (continued)

CUBIC SYSTEM									
23			<i>T</i>						
12	<i>c</i>	1	x, y, z z, x, y y, z, x	$\bar{x}, \bar{y}, \bar{z}$ $\bar{z}, \bar{x}, \bar{y}$ $\bar{y}, \bar{z}, \bar{x}$	$\bar{x}, y, \bar{z}$ $\bar{z}, \bar{x}, y$ $y, \bar{z}, \bar{x}$	$x, \bar{y}, \bar{z}$ $\bar{z}, x, \bar{y}$ $\bar{y}, \bar{z}, x$	Pentagon-tritetrahedron or tetartoid or tetrahedral pentagon-dodecahedron <i>Snub tetrahedron</i> (= <i>pentagon-tritetra-</i> <i>hedron</i> + <i>two tetrahedra</i>)	(hkl) (lhk) (klh)	$(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$
Trigon-tritetrahedron or tristetrahedron (for $h < l$) <i>Tetrahedron truncated by tetrahedron</i> (for $x < z$) Tetragon-tritetrahedron or deltohedron or deltoid-dodecahedron (for $h > l$) <i>Cube & two tetrahedra</i> (for $x > z$)							(hhl) (lhh) (hlh)	$(\bar{h}\bar{h}\bar{l})$ $(\bar{l}\bar{h}\bar{h})$ $(\bar{h}\bar{l}\bar{h})$	$(\bar{h}\bar{h}\bar{l})$ $(\bar{l}\bar{h}\bar{h})$ $(\bar{h}\bar{l}\bar{h})$
							$(0kl)$ $(l0k)$ $(k l 0)$	$(0\bar{k}\bar{l})$ $(l0\bar{k})$ $(\bar{k}\bar{l}0)$	$(0k\bar{l})$ $(\bar{l}0k)$ $(k\bar{l}0)$
Pentagon-dodecahedron or dihexahedron or pyritohedron <i>Irregular icosahedron</i> (= <i>pentagon-dodecahedron</i> + <i>octahedron</i>)							$(0kl)$ $(l0k)$ $(k l 0)$	$(0\bar{k}\bar{l})$ $(l0\bar{k})$ $(\bar{k}\bar{l}0)$	$(0k\bar{l})$ $(\bar{l}0k)$ $(k\bar{l}0)$
Rhomb-dodecahedron <i>Cuboctahedron</i>							(011) (101) (110)	$(0\bar{1}\bar{1})$ $(\bar{1}01)$ $(\bar{1}\bar{1}0)$	$(01\bar{1})$ $(10\bar{1})$ $(1\bar{1}0)$
6	<i>b</i>	2..	$x, 0, 0$ $0, x, 0$ $0, 0, x$	$\bar{x}, 0, 0$ $0, \bar{x}, 0$ $0, 0, \bar{x}$	Cube or hexahedron <i>Octahedron</i>		(100) (010) (001)	$(\bar{1}00)$ $(0\bar{1}0)$ $(00\bar{1})$	
4	<i>a</i>	.3.	x, x, x $\bar{x}, \bar{x}, \bar{x}$	$\bar{x}, x, \bar{x}$ $x, \bar{x}, \bar{x}$	Tetrahedron <i>Tetrahedron</i>		(111) or $(\bar{1}\bar{1}\bar{1})$	$(\bar{1}\bar{1}1)$ $(11\bar{1})$	$(\bar{1}\bar{1}\bar{1})$ (111)
1	<i>o</i>	23.	0, 0, 0	Point in origin					
Symmetry of special projections Along [001] Along [111] Along [110] 2mm 3 m									

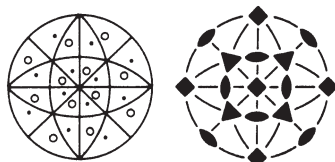
3. ADVANCED TOPICS ON SPACE-GROUP SYMMETRY

Table 3.2.3.2 (continued)

CUBIC SYSTEM (<i>cont.</i>)									
$m\bar{3}$ $\frac{2}{m}\bar{3}$ m			T_h						
24	<i>d</i>	1	x, y, z z, x, y y, z, x $\bar{x}, \bar{y}, \bar{z}$ $\bar{z}, \bar{x}, \bar{y}$ $\bar{y}, \bar{z}, \bar{x}$	$\bar{x}, \bar{y}, z$ $\bar{z}, \bar{x}, y$ $y, \bar{z}, x$	$\bar{x}, y, \bar{z}$ $\bar{z}, \bar{x}, y$ $y, \bar{z}, x$	$x, \bar{y}, \bar{z}$ $z, x, \bar{y}$ $y, z, \bar{x}$	Didodecahedron or diploid or dyakisdodecahedron <i>Cube & octahedron & pentagon-dodecahedron</i>	(hkl) (lhk) (klh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$	$(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ $(h\bar{k}\bar{l})$ $(l\bar{h}\bar{k})$ $(k\bar{l}\bar{h})$
			$\bar{x}, \bar{y}, \bar{z}$ $\bar{z}, \bar{x}, \bar{y}$ $\bar{y}, \bar{z}, \bar{x}$	$x, y, \bar{z}$ $\bar{z}, x, y$ $y, \bar{z}, x$	$x, \bar{y}, z$ $z, x, \bar{y}$ $y, z, \bar{x}$	$\bar{x}, y, z$ $z, \bar{x}, y$ $y, z, \bar{x}$		$(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$	$(hk\bar{l})$ $(l\bar{h}k)$ $(k\bar{l}h)$
			$\left\{ \begin{array}{l} \text{Tetragon-trioctahedron or trapezohedron} \\ \text{or deltoid-icositetrahedron} \\ \text{(for } h < l) \\ \text{Cube \& octahedron \& rhomb-dodecahedron} \\ \text{(for } x < z) \end{array} \right\}$ $\left\{ \begin{array}{l} \text{Trigon-trioctahedron or trisoctahedron} \\ \text{(for } h > l) \\ \text{Cube truncated by octahedron} \\ \text{(for } x > z) \end{array} \right\}$				(hhl) (lhh) (hll)	$(\bar{h}\bar{h}\bar{l})$ $(\bar{l}\bar{h}\bar{h})$ $(\bar{h}\bar{l}\bar{h})$	
							$(\bar{h}\bar{h}\bar{l})$ $(\bar{l}\bar{h}\bar{h})$ $(\bar{h}\bar{l}\bar{h})$	(hhl) (lhh) (hll)	$(\bar{h}\bar{h}\bar{l})$ $(\bar{l}\bar{h}\bar{h})$ $(\bar{h}\bar{l}\bar{h})$
12	<i>c</i>	$m..$	$0, y, z$ $z, 0, y$ $y, z, 0$	$0, \bar{y}, z$ $\bar{z}, 0, y$ $y, \bar{z}, 0$	$0, y, \bar{z}$ $\bar{z}, 0, y$ $y, \bar{z}, 0$	$0, \bar{y}, \bar{z}$ $\bar{z}, 0, \bar{y}$ $\bar{y}, \bar{z}, 0$	Pentagon-dodecahedron or dihexahedron or pyritohedron <i>Irregular icosahedron</i> (= <i>pentagon-dodecahedron + octahedron</i>)	$(0kl)$ $(l0k)$ $(k\bar{l}0)$	$(0\bar{k}l)$ $(\bar{l}0\bar{k})$ $(\bar{k}l0)$
			Rhomb-dodecahedron <i>Cuboctahedron</i>				(011) (101) (110)	$(0\bar{1}\bar{1})$ $(\bar{1}0\bar{1})$ $(\bar{1}\bar{1}0)$	
8	<i>b</i>	$.3.$	x, x, x $\bar{x}, \bar{x}, \bar{x}$	$\bar{x}, \bar{x}, x$ $x, x, \bar{x}$	$\bar{x}, x, \bar{x}$ $x, \bar{x}, x$	$x, \bar{x}, \bar{x}$ $\bar{x}, x, x$	Octahedron <i>Cube</i>	(111) $(\bar{1}\bar{1}\bar{1})$	$(\bar{1}\bar{1}1)$ $(11\bar{1})$
6	<i>a</i>	$2mm..$	$x, 0, 0$ $0, x, 0$ $0, 0, x$	$\bar{x}, 0, 0$ $0, \bar{x}, 0$ $0, 0, \bar{x}$			Cube or hexahedron <i>Octahedron</i>	(100) (010) (001)	$(\bar{1}00)$ $(0\bar{1}0)$ $(00\bar{1})$
1	<i>o</i>	$m\bar{3}$	$0, 0, 0$			Point in origin			
Symmetry of special projections									
			Along [001]	Along [111]	Along [110]				
			2mm	6	2mm				

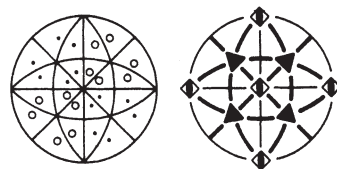
3.2. POINT GROUPS AND CRYSTAL CLASSES

Table 3.2.3.2 (continued)

CUBIC SYSTEM (<i>cont.</i>)										
432		<i>O</i>								
24	<i>d</i>	1	x, y, z z, x, y y, z, x $y, x, \bar{z}$ $x, z, \bar{y}$ $z, y, \bar{x}$	$\bar{x}, \bar{y}, z$ $\bar{z}, \bar{x}, \bar{y}$ $\bar{y}, z, \bar{x}$ $\bar{y}, \bar{x}, \bar{z}$ $\bar{x}, z, \bar{y}$ $\bar{z}, y, \bar{x}$	$\bar{x}, y, \bar{z}$ $\bar{z}, x, \bar{y}$ $\bar{y}, \bar{z}, x$ $y, \bar{x}, z$ $x, \bar{z}, y$ $\bar{z}, y, x$	$x, \bar{y}, \bar{z}$ $\bar{z}, x, \bar{y}$ $\bar{y}, \bar{z}, x$ $y, \bar{x}, z$ $x, \bar{z}, y$ $\bar{z}, y, x$	Pentagon-trioctahedron or gyroid or pentagon-icositetrahedron <i>Snub cube</i> (= <i>cube</i> + <i>octahedron</i> + <i>pentagon-</i> <i>trioctahedron</i>)	(hkl) (lhk) (klh) $(kh\bar{l})$ $(h\bar{l}k)$ $(l\bar{k}h)$	$(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(k\bar{h}\bar{l})$ $(h\bar{l}\bar{k})$ $(l\bar{k}\bar{h})$	
			{ Tetragon-trioctahedron or trapezohedron or deltoid-icositetrahedron (for $ h < l $) <i>Cube & octahedron &</i> <i>rhomb-dodecahedron</i> (for $ x < z $) Trigon-trioctahedron or trisoctahedron (for $ h > l $) <i>Cube truncated by octahedron</i> (for $ x > z $) }				(hhl) (lhh) (hlh)	$(\bar{h}\bar{h}\bar{l})$ $(\bar{l}\bar{h}\bar{h})$ $(\bar{h}\bar{l}\bar{h})$	$(h\bar{h}\bar{l})$ $(h\bar{l}\bar{h})$ $(l\bar{h}\bar{h})$	$(\bar{h}\bar{h}\bar{l})$ $(\bar{l}\bar{h}\bar{h})$ $(\bar{h}\bar{l}\bar{h})$
							(hhl) (lhh) (hlh)	$(\bar{h}\bar{h}\bar{l})$ $(\bar{l}\bar{h}\bar{h})$ $(\bar{h}\bar{l}\bar{h})$	$(h\bar{h}\bar{l})$ $(h\bar{l}\bar{h})$ $(l\bar{h}\bar{h})$	$(\bar{h}\bar{h}\bar{l})$ $(\bar{l}\bar{h}\bar{h})$ $(\bar{h}\bar{l}\bar{h})$
			Tetrahexahedron or tetrakisshexahedron <i>Octahedron truncated by cube</i>				$(0kl)$ $(l0k)$ $(k\bar{l}0)$	$(0\bar{k}\bar{l})$ $(l0\bar{k})$ $(\bar{k}\bar{l}0)$	$(0k\bar{l})$ $(0l\bar{k})$ $(l\bar{k}0)$	
12	<i>c</i>	..2	$0, y, y$ $y, 0, y$ $y, y, 0$	$0, \bar{y}, y$ $y, 0, \bar{y}$ $\bar{y}, y, 0$	$0, y, \bar{y}$ $\bar{y}, 0, y$ $y, \bar{y}, 0$	$0, \bar{y}, \bar{y}$ $\bar{y}, 0, \bar{y}$ $\bar{y}, \bar{y}, 0$	Rhomb-dodecahedron <i>Cuboctahedron</i>	(011) (101) (110)	$(0\bar{1}\bar{1})$ $(10\bar{1})$ $(1\bar{1}0)$	
8	<i>b</i>	.3.	x, x, x $x, x, \bar{x}$	$\bar{x}, \bar{x}, x$ $\bar{x}, \bar{x}, \bar{x}$	$\bar{x}, x, \bar{x}$ $x, \bar{x}, x$	$x, \bar{x}, \bar{x}$ $\bar{x}, x, x$	Octahedron <i>Cube</i>	(111) $(\bar{1}\bar{1}\bar{1})$	$(\bar{1}\bar{1}\bar{1})$ (111)	
6	<i>a</i>	4..	$x, 0, 0$ $0, x, 0$ $0, 0, x$	$\bar{x}, 0, 0$ $0, \bar{x}, 0$ $0, 0, \bar{x}$			Cube or hexahedron <i>Octahedron</i>	(100) (010) (001)	$(\bar{1}00)$ $(0\bar{1}0)$ $(00\bar{1})$	
1	<i>o</i>	432	$0, 0, 0$				Point in origin			
Symmetry of special projections Along [001] Along [111] Along [110] 4mm 3m 2mm										

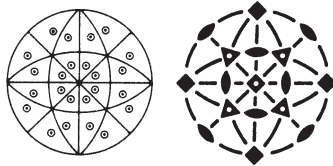
3. ADVANCED TOPICS ON SPACE-GROUP SYMMETRY

Table 3.2.3.2 (continued)

CUBIC SYSTEM (<i>cont.</i>)									
$\bar{4}3m$		T_d							
24	d	1	x, y, z z, x, y y, z, x y, x, z x, z, y z, y, x	$\bar{x}, \bar{y}, \bar{z}$ $\bar{z}, \bar{x}, \bar{y}$ $\bar{y}, \bar{z}, \bar{x}$ $\bar{y}, \bar{x}, \bar{z}$ $\bar{x}, \bar{z}, \bar{y}$ $\bar{z}, \bar{y}, \bar{x}$	$\bar{x}, y, \bar{z}$ $\bar{z}, \bar{x}, y$ $y, \bar{z}, \bar{x}$ $y, \bar{x}, \bar{z}$ $\bar{x}, \bar{z}, y$ $\bar{z}, y, \bar{x}$	$x, \bar{y}, \bar{z}$ $\bar{z}, x, \bar{y}$ $\bar{y}, \bar{z}, x$ $\bar{y}, x, \bar{z}$ $x, \bar{z}, \bar{y}$ $\bar{z}, \bar{y}, x$	Hexatetrahedron or hexakistetrahedron <i>Cube truncated by two tetrahedra</i>	(hkl) (lhk) (klh) (khl) (hlk) (lkh)	$(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(\bar{k}\bar{h}\bar{l})$ $(\bar{h}\bar{l}\bar{k})$ $(\bar{l}\bar{k}\bar{h})$
							Tetrahexahedron or tetrakisshexahedron <i>Octahedron truncated by cube</i>	$(0kl)$ $(l0k)$ $(k0l)$ $(k0l)$ $(0lk)$ $(lk0)$	$(0\bar{k}\bar{l})$ $(\bar{l}0\bar{k})$ $(\bar{k}0\bar{l})$ $(\bar{k}0\bar{l})$ $(\bar{0}\bar{l}\bar{k})$ $(\bar{l}\bar{k}0)$
12	c	$.m$	x, x, z z, x, x x, z, x	$\bar{x}, \bar{x}, \bar{z}$ $\bar{z}, \bar{x}, \bar{x}$ $\bar{x}, \bar{z}, \bar{x}$	$\bar{x}, x, \bar{z}$ $\bar{z}, \bar{x}, x$ $x, \bar{z}, \bar{x}$	$x, \bar{x}, \bar{z}$ $\bar{z}, x, \bar{x}$ $\bar{x}, \bar{z}, x$	Trigon-tritetrahedron or tristetrahedron (for $ h < l $) <i>Tetrahedron truncated by tetrahedron</i> (for $ x < z $) Tetragon-tritetrahedron or deltohedron or deltoid-dodecahedron (for $ h > l $) <i>Cube & two tetrahedra</i> (for $ x > z $)	(hhl) (lhh) (hll)	$(\bar{h}\bar{h}\bar{l})$ $(\bar{l}\bar{h}\bar{h})$ $(\bar{h}\bar{l}\bar{h})$
									(110) (011) (101)
6	b	$2.mm$	$x, 0, 0$ $0, x, 0$ $0, 0, x$	$\bar{x}, 0, 0$ $0, \bar{x}, 0$ $0, 0, \bar{x}$			Cube or hexahedron <i>Octahedron</i>	(100) (010) (001)	$(\bar{1}00)$ $(0\bar{1}0)$ $(00\bar{1})$
4	a	$.3m$	x, x, x	$\bar{x}, \bar{x}, \bar{x}$	$\bar{x}, x, \bar{x}$	$x, \bar{x}, \bar{x}$	Tetrahedron <i>Tetrahedron</i>	(111) or $(\bar{1}\bar{1}\bar{1})$	$(\bar{1}\bar{1}\bar{1})$ (111)
1	o	$\bar{4}3m$	$0, 0, 0$				Point in origin		
Symmetry of special projections									
			Along [001]	Along [111]	Along [110]				
			$4mm$	$3m$	m				

3.2. POINT GROUPS AND CRYSTAL CLASSES

Table 3.2.3.2 (continued)

CUBIC SYSTEM (<i>cont.</i>)							
$m\bar{3}m$ $\frac{4}{m}\frac{3}{m}\frac{2}{m}$		O_h					
48	<i>f</i>	1	x, y, z z, x, y y, z, x $y, x, \bar{z}$ $x, z, \bar{y}$ $z, y, \bar{x}$ $\bar{x}, \bar{y}, \bar{z}$ $\bar{z}, \bar{x}, \bar{y}$ $\bar{y}, \bar{z}, \bar{x}$ $\bar{y}, \bar{x}, z$ $\bar{x}, \bar{z}, y$ $\bar{z}, \bar{y}, x$	$\bar{x}, \bar{y}, z$ $z, \bar{x}, \bar{y}$ $y, \bar{z}, \bar{x}$ $\bar{y}, \bar{x}, z$ $\bar{x}, \bar{z}, y$ $\bar{z}, y, x$ $x, y, \bar{z}$ $z, x, \bar{y}$ $y, z, \bar{x}$ $\bar{x}, \bar{y}, z$ $\bar{z}, \bar{x}, y$ $\bar{y}, \bar{z}, x$ $y, x, \bar{z}$ $x, z, \bar{y}$ $z, y, \bar{x}$	$x, \bar{y}, \bar{z}$ $\bar{z}, x, y$ $y, \bar{z}, \bar{x}$ $\bar{y}, x, z$ $x, \bar{z}, y$ $\bar{z}, \bar{y}, x$ $\bar{x}, y, z$ $z, \bar{x}, \bar{y}$ $y, z, \bar{x}$ $\bar{x}, y, z$ $z, x, \bar{y}$ $y, z, \bar{x}$ $\bar{x}, y, z$ $z, \bar{x}, \bar{y}$ $y, z, \bar{x}$ $\bar{x}, y, z$ $z, \bar{x}, \bar{y}$ $y, z, \bar{x}$	Hexaoctahedron or hexakisoctahedron <i>Cube truncated by octahedron and by rhomb-dodecahedron</i>	(hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ $(\bar{k}\bar{l}\bar{h})$ $(h\bar{k}\bar{l})$ $(\bar{h}l\bar{k})$ $(k\bar{h}\bar{l})$ $(\bar{h}kl)$ $(\bar{l}hk)$ $(\bar{k}lh)$ (hkl) (lhk) (klh) (khl) (hlk) (lkh) $(\bar{h}\bar{k}\bar{l})$ $(\bar{l}\bar{h}\bar{k})$ 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3. ADVANCED TOPICS ON SPACE-GROUP SYMMETRY

Table 3.2.3.3

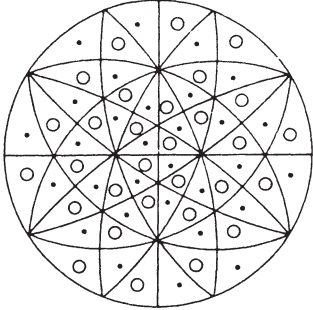
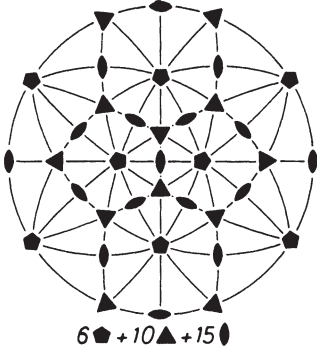
The two icosahedral point groups

Each point group is specified by its Hermann–Mauguin and Schoenflies symbol. For each point group, the stereographic projections show (on the left) the general position and (on the right) the symmetry elements.

The list of the Wyckoff positions includes:

Columns 1 to 3: multiplicity, Wyckoff letter, oriented site-symmetry symbol;

Under the left stereographic projection: face forms (in roman type) and point forms (in italics), corresponding to the values for the Miller indices and coordinates listed in the last column; only 'initial' Miller indices and coordinates are given (see text).

235	<i>I</i>					
						
60	<i>d</i>	1	Pentagon-hexecontahedron <i>Snub pentagon-dodecahedron</i> (= <i>pentagon-dodecahedron</i> + <i>icosahedron</i> + <i>pentagon-hexecontahedron</i>)	(<i>hkl</i>) <i>x, y, z</i>		
			Tricosahedron <i>Pentagon-dodecahedron truncated by icosahedron</i> (poles between axes 2 and 3)	(<i>0kl</i>) with $ l < 0.382 k $ <i>0, y, z</i> with $ z < 0.382 y $		
			Deltoid-hexecontahedron <i>Rhomb-triacontahedron</i> & <i>pentagon-dodecahedron</i> & <i>icosahedron</i> (poles between axes 3 and 5)	(<i>0kl</i>) with $0.382 k < l < 1.618 k $ <i>0, y, z</i> with $0.382 y < z < 1.618 y $		
			Pentakisidodecahedron <i>Icosahedron truncated by pentagon-dodecahedron</i> (poles between axes 5 and 2)	(<i>0kl</i>) with $ l > 1.618 k $ <i>0, y, z</i> with $ z > 1.618 y $		
30	<i>c</i>	2..	Rhomb-triacontahedron <i>Icosadodecahedron</i> (= <i>pentagon-dodecahedron</i> & <i>icosahedron</i>)	(100) <i>x, 0, 0</i>		
20	<i>b</i>	.3.	Regular icosahedron <i>Regular pentagon-dodecahedron</i>	(111) <i>x, x, x</i>		
12	<i>a</i>	..5	Regular pentagon-dodecahedron <i>Regular icosahedron</i>	(01 τ) <i>0, y, τy</i> } with $\tau = \frac{1}{2}(\sqrt{5} + 1) = 1.618$		
1	<i>o</i>	235	<i>Point in origin</i>	<i>0, 0, 0</i>		
Symmetry of special projections						
			Along [001]	Along [111]	Along [1 τ 0]	
			2mm	3m	5m	

3.2. POINT GROUPS AND CRYSTAL CLASSES

Table 3.2.3.3 (continued)

$m\bar{3}\bar{5}$						
$\frac{2}{m}\bar{3}\bar{5}$	I_h					
120	e	1	Hecaticosahedron or hexaicosahedron <i>Pentagon-dodecahedron truncated by icosahedron and by rhomb-triacontahedron</i>			
60	d	$m..$	Trisicosahedron <i>Pentagon-dodecahedron truncated by icosahedron (poles between axes 2 and $\bar{3}$)</i> Deltoid-hexacontahedron <i>Rhomb-triacontahedron & pentagon-dodecahedron & icosahedron (poles between axes $\bar{3}$ and $\bar{5}$)</i> Pentakisdodecahedron <i>Icosahedron truncated by pentagon-dodecahedron (poles between axes $\bar{5}$ and 2)</i>	(hkl) x, y, z $(0kl)$ with $l < 0.382 k$ $0, y, z$ with $z < 0.382 y$ $(0kl)$ with $0.382 k < l < 1.618 k$ $0, y, z$ with $0.382 y < z < 1.618 y$ $(0kl)$ with $l > 1.618 k$ $0, y, z$ with $z > 1.618 y$		
30	c	$2mm..$	Rhomb-triacontahedron <i>Icosadodecahedron (= pentagon-dodecahedron & icosahedron)</i>	(100) $x, 0, 0$		
20	b	$3m (m\bar{3}).$	Regular icosahedron <i>Regular pentagon-dodecahedron</i>	(111) x, x, x		
12	a	$5m (m\bar{5})$	Regular pentagon-dodecahedron <i>Regular icosahedron</i>	(01τ) $0, y, \tau y$	$\left. \vphantom{\begin{matrix} (01\tau) \\ 0, y, \tau y \end{matrix}} \right\}$ with $\tau = \frac{1}{2}(\sqrt{5} + 1) = 1.618$	
1	o	$2/m\bar{3}\bar{5}$	Point in origin	$0, 0, 0$		
Symmetry of special projections Along $[001]$ Along $[111]$ Along $[1\tau 0]$ $2mm$ $6mm$ $10mm$						

3.2.4. Molecular symmetry

By U. MÜLLER

3.2.4.1. Introduction

In this section, by ‘molecule’ we mean a separate set of closely bonded atoms; this definition includes molecular ions. The term ‘molecule’ as used here also includes molecular aggregates held together by hydrogen bonds or other important molecular interactions. We consider only the symmetry of isolated molecules, *i.e.* we act as if the molecule had no surroundings or a completely isotropic surrounding (even if it is a molecular ion that in reality has nearby counter ions), and we do not consider any vibrational distortions. Of course, in crystals the surroundings are not isotropic; their anisotropy is reflected in the site symmetry of the molecule. The site symmetry of the molecule in the crystal may not be higher than the molecular symmetry, and, in most cases, it is lower, *i.e.* it is a subgroup of the point group of the isolated molecule. From experience we know that the actual deviation of the molecular symmetry in crystals as compared to the symmetry in an isotropic surrounding is usually unimportant; generally, only conformation angles are affected significantly.

According to the statistical data for organic molecules (molecules having C—C or C—H bonds), the point group 1 is the most frequent (70.9%), followed by $\bar{1}$ (8.1%), 2 (7.5%), m (6.5%), $2/m$ (2.1%) and $mm2$ (1.7%); all others are <1% (Pidcock *et al.*, 2003). In crystals, the site symmetry of organic molecules is 1 with a frequency of 80.7%, followed by $\bar{1}$ (~11%). Centrosymmetric molecules ($\bar{1}$ and higher centrosymmetric point groups) almost always crystallize in centrosymmetric space groups (99%) and lie on centrosymmetric sites ($\bar{1}$ and higher) with a frequency of 97% (Pidcock *et al.*, 2003; Müller, 1978).

Since all real molecules are objects of three-dimensional space, we consider only three-dimensional symmetry groups. This choice also applies to so-called ‘zero-’, ‘one-’ and ‘two-dimensional’ molecules; these misleading terms are sometimes used for, respectively, finite molecules, polymeric chain molecules and polymeric layer molecules. For chemists who use these terms, the ‘dimensionality’ mostly refers to the linkage by covalent bonds; for physicists, ‘dimensionality’ may refer to some other anisotropic property, *e.g.* conductivity.

3.2.4.2. Definitions

The set of all isometries that map a molecule onto itself is its *molecular symmetry*. Its symmetry operations form a group which is the *point group* $\mathcal{P}_M$ of the molecule.

The group $\mathcal{P}_M$ is finite if the molecule consists of a finite number of atoms and is mapped onto itself by a finite number of isometries. However, the group is infinite for linear molecules like HCl and CO₂, because of the infinite order of the molecular axis. For the symmetry of polymeric molecules see below (Section 3.2.4.4).

All symmetry operations of a finite molecule leave its centre of mass unchanged. If this point is chosen as the origin, all symmetry operations can be represented by 3×3 matrices $\mathbf{W}_i$, referred to some predefined coordinate system.

According to their equivalence, point groups are classified into *point-group types*. Two point groups $\mathcal{P}_{M1}$ and $\mathcal{P}_{M2}$ belong to the same *point-group type* if, after selection of appropriate coordinate systems, the matrix groups of $\mathcal{P}_{M1}$ and $\mathcal{P}_{M2}$ coincide. Why point groups and point-group types have to be distinguished is shown in the following example of metaldehyde; however, for

most molecules the distinction is usually of only minor importance.

In Section 1.1.7 a general definition is given for an *orbit* under the action of a group. Applied to molecules, we formulate this definition in the following way:

The set of points symmetry-equivalent to a point X of a molecule is the *orbit* of X under $\mathcal{P}_M$.

A point X in a molecule has a definite *site symmetry* $\mathcal{S}_X$ (site-symmetry group). It consists of all those symmetry operations of the point group $\mathcal{P}_M$ which leave the point fixed. The site-symmetry group $\mathcal{S}_X$ is always a subgroup of the point group of the molecule: $\mathcal{S}_X \leq \mathcal{P}_M$.

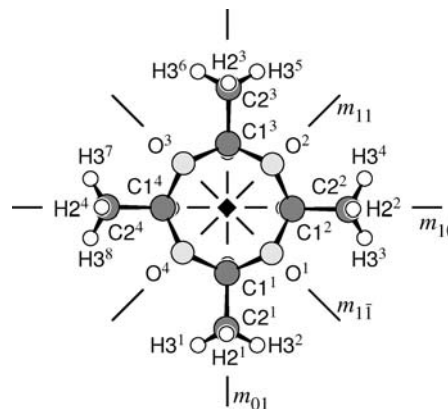
A point X is on a *general position* if its site symmetry $\mathcal{S}_X$ consists of nothing more than the identity, $\mathcal{S}_X = \mathcal{I}$. Otherwise, if $\mathcal{S}_X > \mathcal{I}$, the point is on a *special position*.

The *multiplicity* of a point X in a molecule is equal to the number of points which are symmetry-equivalent to this point, *i.e.* the number of points in its orbit. If the point is on a general position, its multiplicity is equal to the group order $|\mathcal{P}_M|$. If $|\mathcal{S}_X|$ is the order of the site symmetry of a point on a special position, the multiplicity of X is $|\mathcal{P}_M|/|\mathcal{S}_X|$.

Two points X and Y of a molecule belong to the same *Wyckoff position* if the site-symmetry groups $\mathcal{S}_X$ and $\mathcal{S}_Y$ are conjugate subgroups of $\mathcal{P}_M$, *i.e.* if $\mathcal{S}_X$ and $\mathcal{S}_Y$ are symmetry-equivalent under a symmetry operation of $\mathcal{P}_M$; expressed mathematically, this is $\mathcal{S}_Y = g\mathcal{S}_Xg^{-1}$ with $g \in \mathcal{P}_M$. The special case $\mathcal{S}_X = \mathcal{S}_Y$ is included.

Example

A molecule of metaldehyde [tetrameric acetaldehyde, 2,4,6,8-tetramethyl-1,3,5,7-tetraoxacyclooctane, (H₃CCHO)₄] has the point group $\mathcal{P}_M = 4mm$ (C_{4v}) of order $|\mathcal{P}_M| = 8$, consisting of the symmetry operations 1, 4^+ , 2 , 4^- , m_{10} , m_{01} , m_{11} and $m_{\bar{1}\bar{1}}$.



The atoms C¹...C⁴, the four H atoms bonded to them (not labelled in the figure and largely hidden), C²...C² and H²...H² make up four sets of symmetry-equivalent atoms and thus form four orbits. All these atoms are placed on the mirror planes m_{10} and m_{01} ; their site-symmetry groups are $\{1, m_{10}\}$ or $\{1, m_{01}\}$. $\{1, m_{10}\}$ and $\{1, m_{01}\}$ are conjugate subgroups of $4mm$ because they are equivalent by a fourfold rotation, *i.e.* $\{1, m_{01}\} = 4^+\{1, m_{10}\}4^-$. All of these atoms belong to *one* Wyckoff position, labelled 4c in Table 3.2.3.2. The symmetry-equivalent atoms O¹...O⁴ make up another orbit which belongs to a different Wyckoff position, labelled 4b, with the conjugate site symmetries $\{1, m_{11}\}$ and $\{1, m_{\bar{1}\bar{1}}\}$. Although the site symmetries of both 4b and 4c consist of the identity and one reflection, they are not conjugate because there is no symmetry operation of $4mm$ that relates them. The subgroups $\{1, m_{10}\}$ and $\{1, m_{11}\}$ are different point groups of the same point-group type.

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These four site-symmetry groups all have order 2. Accordingly, the multiplicity for each of the atoms discussed above is $8/2 = 4$. The atoms $\text{H}3^1 \dots \text{H}3^8$, however, are on a general position, their site symmetry is 1 and their multiplicity corresponds to the order 8 of the point group $4mm$. Their orbit belongs to the Wyckoff position $8d$. The Wyckoff position $1a$ on the fourfold rotation axis is not occupied.

3.2.4.3. Tables of the point groups

In Table 3.2.3.2 all crystallographic point groups are listed, *i.e.* all point groups with axes of orders 1, 2, 3, 4 and 6. As described in Section 3.2.1.2.3, the tables also contain the crystal forms which refer to the faces of crystals. As far as molecular symmetry is concerned, only the first four columns and the names of the polyhedra (point forms) in the fifth column given in italics are of interest.

The polyhedra listed in the fifth column result when the points of one orbit of the corresponding Wyckoff position are connected by lines. As for their names, see the remarks on 'point forms' in Sections 3.2.1.2.3 and 3.2.1.2.4.

In Table 3.2.3.2 each point group is headed by its short and, if it differs, full Hermann–Mauguin symbol and by its Schoenflies symbol. Each of the subsequent entries corresponds to one Wyckoff position, in order of increasing site symmetry. The last Wyckoff position of the list is always the one with the highest site symmetry and a multiplicity of 1. The sequence of the coordinate triplets follows the sequence as for the symmorphic space groups.

The columns of Table 3.2.3.2 contain the following data:

Column 1: The multiplicity, *i.e.* the number of equivalent points.

Column 2: The Wyckoff letter, which is an alphabetical label. For all point groups with a 'floating' origin, *i.e.*, where the origin is not fixed on a unique point of highest symmetry, the last entry is the Wyckoff letter *a*. For point groups with a fixed origin, the unique origin point has the highest site symmetry and has the letter *o* (for origin).

Column 3: The site symmetry, given as an 'oriented point-group symbol' (*cf.* Section 2.1.3.12).

Column 4: Coordinates x, y, z of equivalent points.

Column 5: The name of the polyhedron (point form) spanned by the set of symmetry-equivalent points with the coordinates from column 4 is given in italics. The names given in roman type refer to crystal forms described in Section 3.2.1.2.3 and have no importance for molecules. In some cases, a Wyckoff position has several entries in columns 5 and 6, with several polyhedron names. All but the first one correspond to orbits of the same Wyckoff position where some special values have been assigned to one or more of the variable coordinates, *e.g.* $z = 0$, with the consequence that the eigensymmetries of these orbits are higher than the symmetry of the generating point group (non-characteristic orbits, *cf.* Section 1.4.4.4).

Since there is no lattice, point groups of molecules, unlike point groups of crystals, have no 'natural' coordinate system. It is common for coordinates of atoms in molecules to refer to a Cartesian xyz coordinate system; with the exception of trigonal and hexagonal point groups, the coordinates given in Table 3.2.3.2 can be taken as Cartesian coordinates with arbitrary unit length. For point groups with a floating origin, the origin 0, 0, 0 has to be assigned to an arbitrarily chosen point of the Wyckoff position $1a$. The following conventions are used for the orientation of the coordinate axes:

Triclinic system: arbitrary orientation of the axes.

Monoclinic system: unique axis either y or z , as specified. The other two directions arbitrary but normal to the unique axis.

Orthorhombic system: Axes x, y and z along the symmetry directions.

Tetragonal system: z along the fourfold symmetry axis, x along the symmetry direction mentioned in the second position of the Hermann–Mauguin symbol. In the case of one-digit Hermann–Mauguin symbols the x direction is arbitrary within the plane normal to the unique axis. The axis y is normal to both x and z .

Trigonal and hexagonal systems: z along the unique symmetry axis, x along the symmetry direction mentioned second in the Hermann–Mauguin symbol. In the case of one-digit Hermann–Mauguin symbols the x direction is arbitrary within the plane normal to the unique axis. The axis y is also normal to z . *Note:* The coordinates given in Table 3.2.3.2 do not refer to a Cartesian, but to a symmetry-adapted coordinate system. For the hexagonal point groups and the trigonal point groups marked 'hexagonal axes' there is an angle of 120° between the x and y axes. These coordinates can be converted to a Cartesian system according to $x - \frac{1}{2}y, \frac{1}{2}\sqrt{3}y, z$. Coordinates that refer to 'rhombohedral axes' are not suited for separate molecules.

Cubic system: x, y and z along the symmetry directions described by the first position of the Hermann–Mauguin symbol.

For the trigonal and hexagonal systems, the following pairs of point-group types are distinguished in Table 3.2.3.2: 321 & 312 , $3m1$ & $31m$, $\bar{3}m1$ & $\bar{3}1m$ and $\bar{6}m2$ & $\bar{6}2m$. In fact, each of these pairs represents only one molecular point-group type, the difference being only the arbitrarily chosen direction of the x axis. The members of each pair can be interconverted by turning the coordinate system by 30° about the z axis (referred to hexagonal axes). The same applies to the tetragonal pair $\bar{4}2m$ & $\bar{4}m2$, the turning angle being then 45° .

The terms triclinic, monoclinic, orthorhombic, tetragonal, trigonal, hexagonal and cubic designate the crystal systems, *i.e.* the point groups of crystal lattices. As far as point groups of molecules are concerned, these terms only have the meaning of a classification by analogy with the crystal classes. For those who are accustomed to using Schoenflies symbols, it may seem odd to term the point groups $C_{3h} = S_3 = \bar{6}$ and $D_{3h} = \bar{6}2m$ hexagonal rather than trigonal, because the Schoenflies symbolism relies on rotoreflection operations rather than rotoinversion operations. In fact, some properties of C_{3h} and D_{3h} are rather more 'trigonal' than 'hexagonal'; for example, they have only one kind of degenerate irreducible representation (E), rather than two (E_1 and E_2) as do the other hexagonal point groups.

For the sake of treating molecular symmetries in the most uniform way, some recommendations for the preferred directions of the coordinate axes have been proposed (Joint Commission for Spectroscopy, 1955). These recommendations include:

- (1) If any specifications are being made that depend on the choice of the axes (*e.g.* certain irreducible representations), the choice should explicitly be stated.
- (2) For planar molecules with one three- or higher-fold axis (z axis): x axis through the greater number of atoms. For planar molecules with $mm2$ (C_{2v}) or mmm (D_{2h}) symmetry: x axis perpendicular to the plane of the molecule.

Molecular symmetry is not restricted to the crystallographic point groups. No detailed tables are given for noncrystallographic point groups, but the symbols for them can be found in Table 3.2.1.6. For the two icosahedral point groups see Section 3.2.1.4.2 and Table 3.2.3.3.

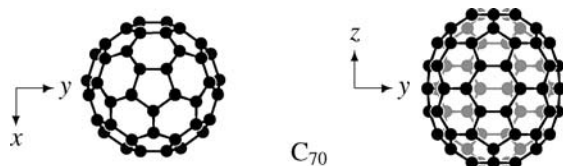
Certain physical properties depend on symmetry. In particular, a molecule can have a dipole moment only if it is polar, *i.e.* if it

has no rotoinversion axis ($\bar{2} = m$ excepted) and if it has at most one symmetry axis of order $N > 1$ and no reflection plane perpendicular to it. The polar point groups are N ($N = 1, 2, \dots$), m , Nm (N odd) and Nmm (N even). The corresponding crystallographic point groups are those marked with a + sign under the heading 'First-rank tensor' in Table 3.2.2.1.

Examples

For the example of metaldehyde mentioned in the preceding section, we find in Table 3.2.3.2, point group $4mm$, the Wyckoff positions $8d$, $4c$, $4b$ and $1a$. Looking at the image of the molecule, we choose a coordinate system with z along the fourfold axis and x and y perpendicular to the mirror planes marked m_{10} and m_{01} . Some value has to be assigned to the unit of the (Cartesian) coordinate system, say $1 \text{ \AA}$. The origin must be chosen at some arbitrary site on the Wyckoff position $1a$, say such that $z = 0$ for the C1 atoms. There is no atom on the Wyckoff position $1a$ ($0, 0, z$). The orbit consisting of the four oxygen atoms is on position $4b$ (x, x, z), $(-x, -x, z)$, $(-x, x, z)$, $(x, -x, z)$, with $x = 0.831$, $z = 0.586$; these atoms form a square and have the site symmetry $.m$, which in the image is on the mirror planes m_{11} and $m_{1\bar{1}}$. Position $4c$ ($x, 0, z$), $(-x, 0, z)$, $(0, x, z)$, $(0, -x, z)$ has three occupied orbits: C1 with $x = 1.710$ and (the arbitrary) $z = 0$, C2 with $x = 3.194$, $z = 0.294$ and H2; each of these orbits again forms a square and their site symmetry is $.m$ on the mirror planes m_{01} and m_{10} . The orbit of the eight H3 atoms is on the general position $8d$; they form a truncated square. (The numerical values have been adapted from the coordinates of Barnett *et al.*, 2005).

The fullerene C_{70} is an example of a molecule with noncrystallographic symmetry, $\bar{10}2m$ (D_{5h}). This point-group type is not listed in Table 3.2.3.2, but it can be found in Table 3.2.1.6.



3.2.4.4. Polymeric molecules

Polymeric molecules actually consist of a finite number of atoms, but it is more practical to treat them as parts of infinitely extended molecules in the same way as crystals are treated as parts of ideal infinite crystals. If an infinitely long ideal molecule has translational symmetry in one direction, then its symmetry group is a rod group. If the molecule forms a layer with translational symmetry in two dimensions, its symmetry group is a layer group (not to be confused with a plane group, where space is restricted to two dimensions, *i.e.* to an infinitely thin plane).

Rod groups can have noncrystallographic symmetries, with axes of any order, such as the 10_3 screw axes in polymeric sulfur or the 29_9 axes in polyoxymethylene. Symbols for rod groups correspond to Hermann–Mauguin space-group symbols, but begin with a script style $\mathcal{R}$. The unique direction c is the direction with the translational symmetry; lattice vectors exist in this direction only and are integral multiples of c . This is also the direction of any rotation or screw axis with order $N > 2$. There is no natural length for the a and b vectors. If there are (twofold) rotation axes perpendicular to c or normals of mirror or glide planes perpendicular to c , these directions serve to determine the directions of a and b ; otherwise, their directions are arbitrary.

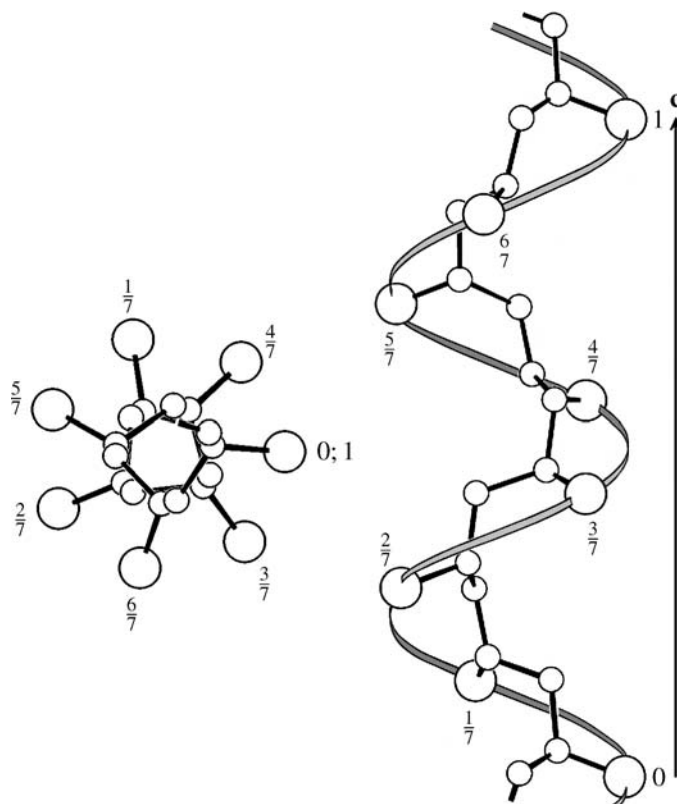


Figure 3.2.4.1

P-helix of isotactic poly-4-methyl-1-pentene (form I) with a $7/2$ helix which corresponds to a 7_4 helix in Hermann–Mauguin notation. Large circles represent isobutyl groups (adapted from Natta & Corradini, 1960).

Molecules of chain polymers tend to be entangled, in which case they have no overall symmetry; symmetry is then restricted to the local symmetry in the immediate surrounding of an atom. In crystalline polymers the chains are forced to align themselves and to adopt a symmetric conformation. This symmetry can be crystallographic, but often the symmetry of the single molecule within the crystal matrix is noncrystallographic.

Crystalline chain polymers often adopt helical molecular structures. In polymer science helices are designated according to the 'helical point net system' by N/r (' N/r helix') where N is the number of repeating units within one translation period and r is the corresponding number of helical coil turns along the molecular chain (Alexander, 1969; *Compendium of Polymer Terminology and Nomenclature* (2008); Spruiell & Clark, 1980).²⁵ Provided that all repeating units in the chain are symmetry-equivalent, the corresponding Hermann–Mauguin screw axis symbol N_q can be calculated from

$$Nn \pm 1 = rq,$$

where n ($= 0, 1, 2, \dots$) and q ($0 < q < N$) are integers to be chosen such that the equation is satisfied. The chemical handedness of the helix does not follow from the N/r symbol, but is specified by the letters M (minus; or L) and P (or R) for left and right, respectively.

Example

Isotactic poly-4-methyl-1-pentene (form I) consists of helical $7/2$ chains (7 monomers per two chain windings). The equation

²⁵ In the IUPAC Recommendations 2011, a different notation for polymeric helices is recommended, in that a $7/2$ helix be named 7_2 helix (Meille *et al.*, 2011). This is very unfortunate because it causes confusion with the Hermann–Mauguin symbols: the Hermann–Mauguin symbol of the $7/2$ helix is not 7_2 , but 7_4 or 7_3 .

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above is fulfilled either as $7 \times 1 + 1 = 2 \times 4$ or $7 \times 1 - 1 = 2 \times 3$, *i.e.* $q = 4$ or $q = 3$, and the corresponding Hermann–Mauguin screw operation symbol is either 7_4 or 7_3 , depending on chirality. The helix is shown in Fig. 3.2.4.1. The N/r symbol cannot be deduced uniquely from the Hermann–Mauguin symbol, because a $7/9$ or any other $7/(2 \text{ modulo } 7)$ P -helix and any $7/(5 \text{ modulo } 7)$ M -helix also has 7_4 symmetry.

Layer groups can have axes of orders of 1, 2, 3, 4 and 6 only. There exist 80 layer-group types. Layer-group symbols refer to a crystallographic coordinate system with **c** (of arbitrary length) perpendicular to the layer defined by the basis vectors **a** and **b**. Layer-group symbols are like space-group symbols, but begin with a lower-case *p* or *c* specifying the centring. Since plane-group symbols also begin by *p* or *c* and can therefore be confused with layer-group symbols, it is advisable to always mention what kind of group is meant.

Rod and layer groups with axis orders of 1, 2, 3, 4 and 6 are the subject of *International Tables for Crystallography* (2010), Volume E, *cf.* Section 1.7.3. Table 3.2.4.1 lists the rod-group types for any axis order.

3.2.4.5. Enantiomorphism and chirality

Definition: An object is *chiral* if it cannot be superposed by pure rotation and translation on its image formed by inversion through a point.

The symmetry group of a chiral object contains no symmetry operations of the second kind, *i.e.* no inversion, rotoinversion, reflection or glide reflection. As a consequence, a chiral object can occur in two different forms, which are related in the same way as are a right and a left hand. Further terms in this context are (Flack, 2003; Moss, 1996):

Absolute configuration	Spatial arrangement of atoms in a chiral molecule and its appropriate designation [<i>e.g.</i> by (<i>R</i>), (<i>S</i>) <i>etc.</i>]
Absolute (crystal) structure	Spatial arrangement of atoms in a chiral crystal and its description (lattice parameters, space group, atomic coordinates)
Enantiomorph	One out of a pair of objects of opposite chirality
Enantiomer	One molecule out of a pair of opposite chirality (special designation for enantiomorphic molecules)
Racemate	Equimolar mixture of a pair of enantiomers
Chirality sense (chirality)	Property which distinguishes enantiomorphs from one another; the two enantiomorphs of a pair have opposite chirality
Achiral	Refers to an object that is not chiral

Possible point groups for chiral molecules correspond to the crystal classes marked in Table 3.2.2.1 with a + sign under the heading ‘Enantiomorphism’. In addition, the noncrystallographic point groups N , $N2$ (N odd, $N \geq 5$), $N22$ (N even, $N \geq 8$) and 235 are possible. Chiral crystal structures are compatible only with space groups that have no inversion centres, rotoinversion axes, reflection or glide-reflection planes. For a chiral molecular compound, these symmetry elements would generate the opposite enantiomer and the compound would be a racemate. Chiral

Table 3.2.4.1

Classes of rod groups

If $N = 2$, the conventional symbols differ (the symmetry direction with the N must be shifted to the last position of the symbol and $\bar{2}$ is to be replaced by m). n is an arbitrary positive integer.

Short symbol	Full symbol	N	q
$\bar{N}$		$N = n$	$q = 1, \dots, N - 1$
$\bar{N}_q$		$N = n \geq 2$	
$\bar{N}$		$N = n$	
$\bar{N}/m$		$N = 2n$	$q = \frac{1}{2}N$
$\bar{N}_q/m$		$N = 2n$	
$\bar{N}2$		$N = 2n + 1$	
$\bar{N}22$		$N = 2n$	$q = 1, \dots, N - 1$
$\bar{N}_q2$		$N = 2n + 1$	
$\bar{N}_q22$		$N = 2n$	
$\bar{N}m$		$N = 2n + 1$	$q = 1, \dots, N - 1$
$\bar{N}mm$		$N = 2n$	
$\bar{N}c$		$N = 2n + 1$	
$\bar{N}cc$		$N = 2n$	$q = \frac{1}{2}N$
$\bar{N}_qmc$		$N = 2n$	
$\bar{N}m$	$\bar{N}2/m$	$N = 2n + 1$	
$\bar{N}2m$		$N = 2n$	$q = \frac{1}{2}N$
$\bar{N}c$	$\bar{N}2/c$	$N = 2n + 1$	
$\bar{N}2c$		$N = 2n$	
$\bar{N}/mmm$	$\bar{N}/m2/m2/m$	$N = 2n$	$q = \frac{1}{2}N$
$\bar{N}/mcc$	$\bar{N}/m2/c2/c$	$N = 2n$	
$\bar{N}_q/mmc$	$\bar{N}_q/m2/m2/c$	$N = 2n$	

crystals belong to one of the 11 crystal classes allowing enantiomorphism (Table 3.2.2.1); they can adopt one out of 65 space-group types. These are called the 65 *Sohncke space-group types* [after L. Sohncke who was the first to derive them; Flack (2003). Julian (2015) calls them ‘proper space groups’]. The Sohncke space-group types comprise the eleven pairs of enantiomorphic space-group types (*cf.* Section 1.3.4.1) and further 43 non-enantiomorphic space-group types. Whereas an enantiomorphic space group (*e.g.* $P3_1$ or $P4_122$) is chiral in itself, which means that its Euclidean normalizer is noncentrosymmetric, the space groups of the other 43 Sohncke space-group types (*e.g.* $P2_12_12_1$) are not chiral themselves; they have centrosymmetric Euclidean normalizers. Keep in mind that the symmetry of a molecule is specified by its point group; the symmetry of a crystal by its space group; and the symmetry of a space group by its Euclidean normalizer (*cf.* Section 3.5.1.2).²⁶ However, the arrangement of the atoms in any Sohncke space group is always chiral. In other words, an enantiomorphic space group is a sufficient but not a necessary condition for a chiral crystal structure. Chirality in a non-chiral space group results if the molecules or other building blocks of the crystal consist of only one kind of enantiomers or because the molecules or building blocks, without having to be chiral themselves, are arranged in a chiral manner in the crystal.

Examples

Trigonal selenium exhibits two enantiomorphic forms that consist of either right-handed or left-handed helical chain molecules (rod groups $\bar{3}_12$ and $\bar{3}_22$, respectively). The molecules as well as the space groups $P3_121$ and $P3_221$ are enantiomorphic and the helical molecules of right-handed

²⁶ Many structural researchers are not aware of the difference between chiral (enantiomorphic) space groups and Sohncke space groups; frequently, even in textbooks, the term ‘chiral space group’ is used although an achiral Sohncke space group is meant.

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selenium are incompatible with the space group of left-handed selenium.

The case of selenium can be contrasted to that of NaP (details for NaP are given at the end of Section 3.5.3.2 in the *Example for case 1*). Crystalline NaP contains one kind of enantiomeric, helical $(P^-)_\infty$ ions that are either 4_1 or 4_3 helices (rod groups $\#4_122$ and $\#4_322$, respectively). In this case the space group $P2_12_12_1$ is a non-enantiomorphic (achiral) Sohncke space group. It is therefore compatible with either of the two different NaP enantiomorphs.

The building blocks of NaClO_3 , namely spherical Na^+ ions and pyramidal ClO_3^- ions (point group $3m$), are not chiral, but their arrangement in the crystal is chiral. Both enantiomorphs adopt the same non-enantiomorphic Sohncke space group $P2_13$ (Abrahams & Bernstein, 1977; Bruke-Laing & Trueblood, 1977).

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